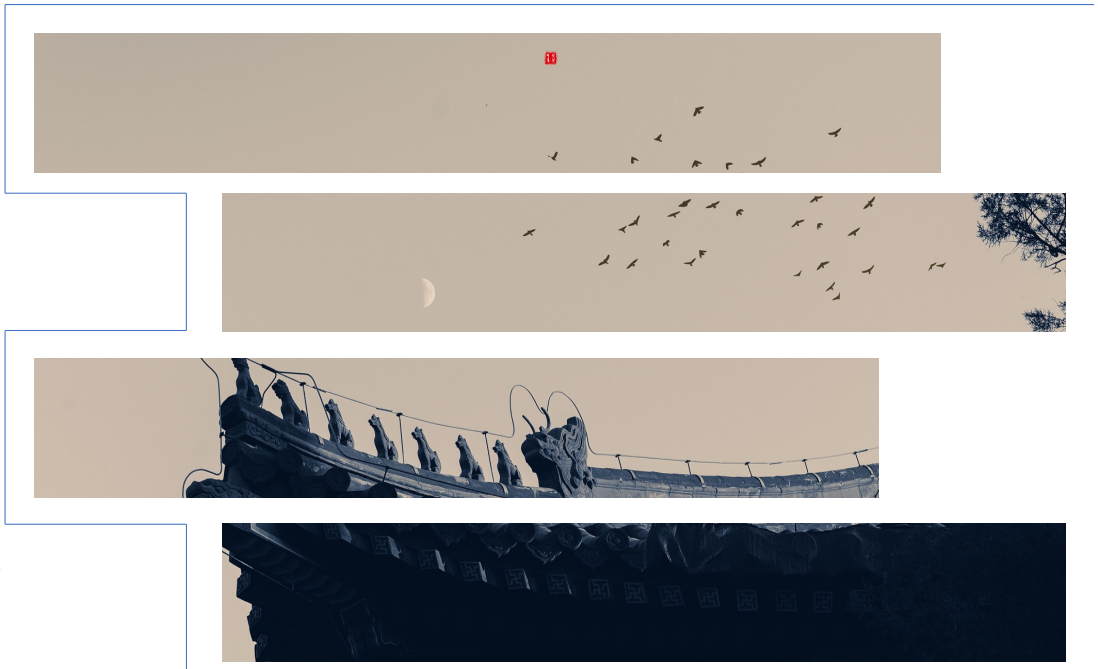
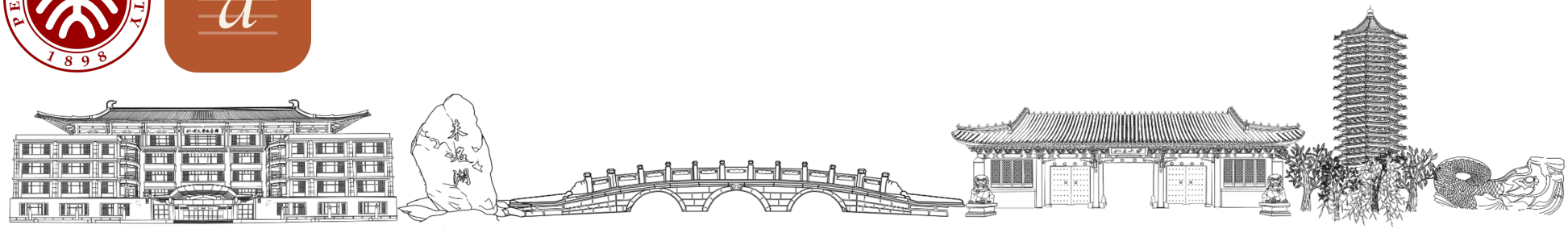
Cognitive Reasoning

Fall 2026

Lecture 01
Introduction

Instructor: Dr. Chi Zhang
School of Intelligence Science and Technology





规定时间: 07:15:00
入校
07:06:17
晚上考勤
规定时间: 21:15:00
出校
21:21:03

Interacting with the Team

Chat with us for **any** course and research questions

Make appointments **via email**

Assignment and project problems? Talk to the TAs

0. Disenchantment

What's so special about the human brain?

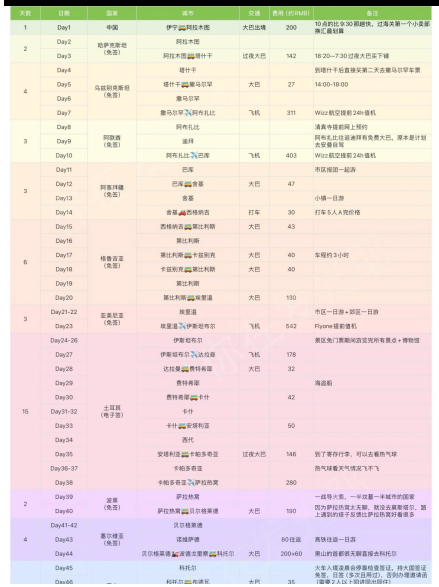
Torrents of data from cell atlases, brain organoids and other methods are finally delivering answers to an age-old question.

By Kerri Smith
Infographics by Nik Spencer

The Fundamental Questions

There must be something about the human brain that's different from the brains of other animals — something that:

- enables humans to plan



天数	日期	国家	城市	交通	费用（约RMB）	备注
1	Day1	中国	伊宁🇨🇳阿拉木图	大巴出境	200	10点的比9:30那趟快。过海关第一个小关那换汇最划算
2	Day2	哈萨克斯坦（免签）	阿拉木图			
	Day3		阿拉木图🇰🇿塔什干	过夜大巴	142	18:20—7:30过夜大巴买下铺
4	Day4	乌兹别克斯坦（免签）	塔什干			到塔什干后直接买第二天去撒马尔罕车票
	Day5		塔什干🇺🇿撒马尔罕	大巴	27	14:00-18:00
	Day6		撒马尔罕			
	Day7		撒马尔罕✈️阿布扎比	飞机	311	Wizz 航空提前 24h 值机
3	Day8	阿联酋（免签）	阿布扎比			清真寺提前网上预约
	Day9		迪拜			阿布扎比往迪迪拜有免费大巴，原本是计划去安曼自驾
	Day10		阿布扎比✈️巴库	飞机	403	Wizz 航空提前 24h 值机
3	Day11	阿塞拜疆（免签）	巴库			市区很团一起游
	Day12		巴库🇬🇪奇圣	大巴	47	
	Day13		奇圣			小镇一日游
	Day14		奇圣🇲🇦西格纳吉	打车	30	打车5人A完价格
6	Day15	格鲁吉亚（免签）	西格纳吉🇬🇪第比利斯	大巴	43	
	Day16		第比利斯			
	Day17		第比利斯🇬🇪卡兹别克	大巴	40	车程约3小时
	Day18		卡兹别克🇬🇪第比利斯	大巴	40	
	Day19		第比利斯			
	Day20		第比利斯🇬🇪库里逼	大巴	130	
3	Day21-22	亚美尼亚（免签）	库里逼			市区一日游+郊区一日游
	Day23		埃里温✈️伊斯坦布尔	飞机	542	Flyone 提前值机
15	Day24-26	土耳其（电子签）	伊斯坦布尔			景区免门票期间西亚买完所有景点+博物馆
	Day27		伊斯坦布尔✈️达拉斯	飞机	178	
	Day28		达拉斯🇬🇪费特希耶	大巴	32	
	Day29		费特希耶			海盗船
	Day30		费特希耶🇬🇪卡什		42	
	Day31-32		卡什			
	Day33		卡什🇬🇪安塔利亚		50	
	Day34		西代			
	Day35		安塔利亚🇬🇪卡帕多奇亚	过夜大巴	146	到了寄存行李，可以去看热气球
	Day36-37		卡帕多奇亚			热气球看天气情况飞不飞
	Day38		卡帕多奇亚✈️伊拉克		280	
2	Day39	波黑（免签）	伊拉克莫			一趟导火索，一半边基一半城市的国家
	Day60		伊拉克莫🇬🇪贝尔格莱德	大巴	190	巴尔伊拉克之无限，就没去塞维亚，就上面的搭子反馈过伊拉克没有看很多
4	Day41-42	塞尔维亚（免签）	贝尔格莱德			
	Day43		诺维萨德		60往返	高铁往返一日游
	Day44		贝尔格莱德🇽🇰普里兹里恩🇽🇰科托尔	大巴	200+60	黑山的首都都有天眼直接去科托尔
	Day45		科托尔			火车入境法奥合停靠检查签证，持大国签证免签，日签（多次往用过），否则办理普通单语（需要2人以上同团出团件）
	Day46		科托尔🇲🇪布德瓦	大巴	35	

大背包（背身后）	上衣	冲锋衣	有些地方比较冷，防风防寒，比如卡兹别克圣三一教堂那
		羽绒内胆	走完巴尔干半岛，去埃及之前我就把便宜、厚（占地方）的衣服丢了，给背包腾地方，所以不用带太贵的衣服
		保暖内衣	
		卫衣	
		羊绒针织衫	
		T恤	可以当睡衣
		吊带	我带了四五件，搭配穿
	裙子		连衣服、半身裙等主要为了拍照，泳衣、水母衣
	裤子		长裤*2、短裤*2、瑜伽裤*1（潜水时可以当水母衣穿）
	鞋	运动鞋	日常走路
		鞋子	为了搭衣服（一般挂在包外面不占地方）
		拖鞋	热带地区穿拖鞋还有洗澡
	贴身衣物		内衣*2、无肩带内衣*1、内裤*5、胸贴*3、一次性内裤*10
			短袜*3不同颜色、中筒袜*1、过膝袜*1
			各种搭配饰品、帽子、防晒口罩、墨镜等等
	洗护用品		电动牙刷、刷头、洗面奶、洗发水、沐浴露、卸妆油、一次性浴巾&洗面巾
			护肤脱化妆就带日常用的，润肤露跟面膜要带，有些地区很干燥缺水
	药品		感冒、过敏、晕车、肠胃、风油精、布洛芬、藿胶、维C、创口贴、碘伏棉签
	其他好物	电子产品收纳包	可以规整相机等电子产品收纳
		压缩袋	用来压缩衣物，放大背包里少位置
		消磁防盗刷包	随身携带
		钓鱼马甲	说是廉航必备，其实没用上两次
		折叠衣架	
		速干浴巾	非常实用！一定要带，而且不占什么空间
		便携餐具	
		防盗锁	住青旅时可以锁柜子、锁包
			眼罩、耳塞、颈枕（掏空内胆塞衣服）、指甲剪
			辣椒酱等各种调味品、辣椒粉，必备！这一路真的太难吃了
小背包（背前面）	证件		护照、身份证（伊宁出境坐大巴）、签证、国际学生证（学生票半价）
			现金（800美金➕300欧元➕500人民币）｜VISA➕AE➕银联（防止吞卡意外多带了）
	电子设备		相机、运动相机、转换插头、备用手机、充电宝、充电器、备用电池、降噪耳机

The Fundamental Questions

There must be something about the human brain that's different from the brains of other animals — something that:

- enables humans to plan or not

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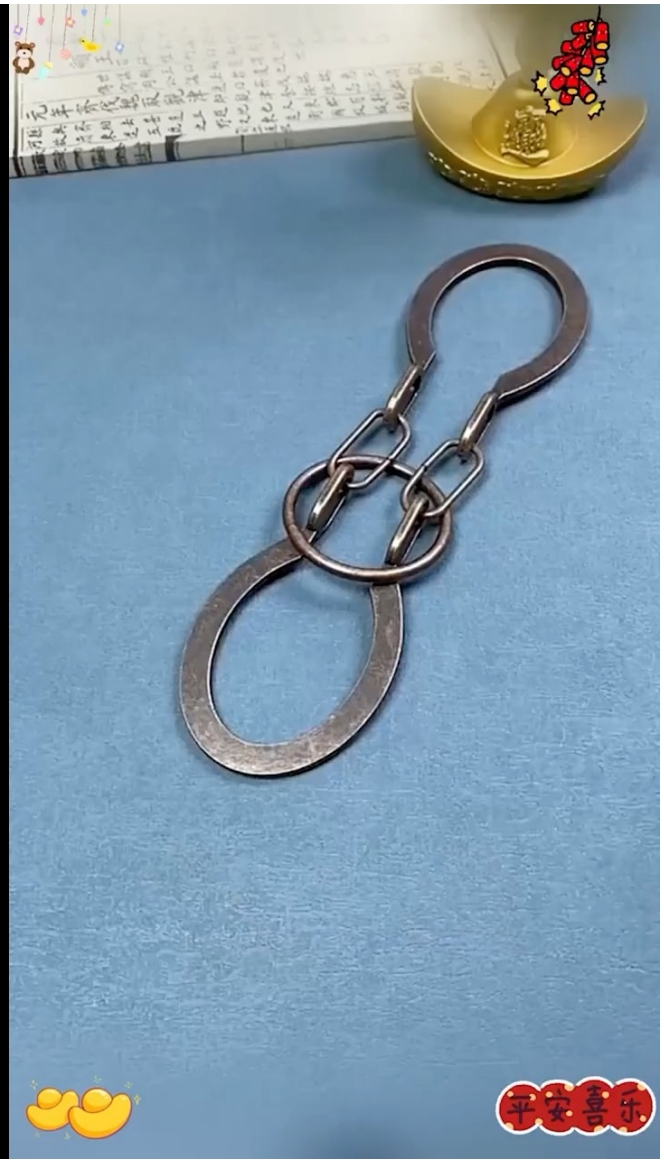
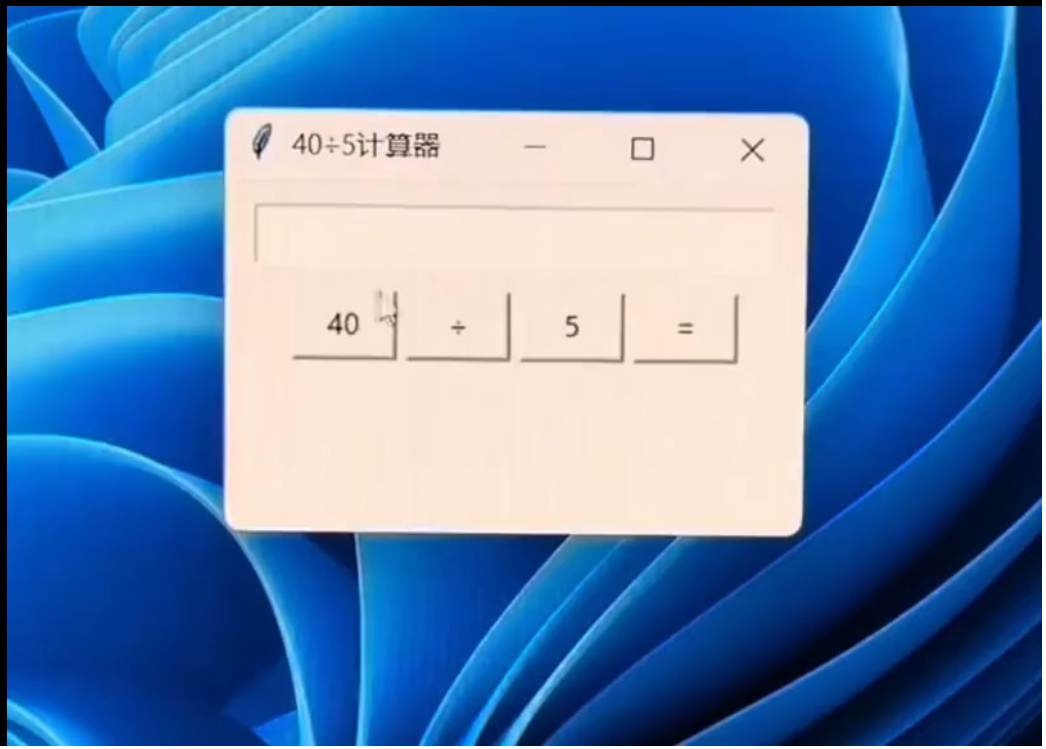


The Fundamental Questions

There must be something about the human brain that's different from the brains of other animals — something that:

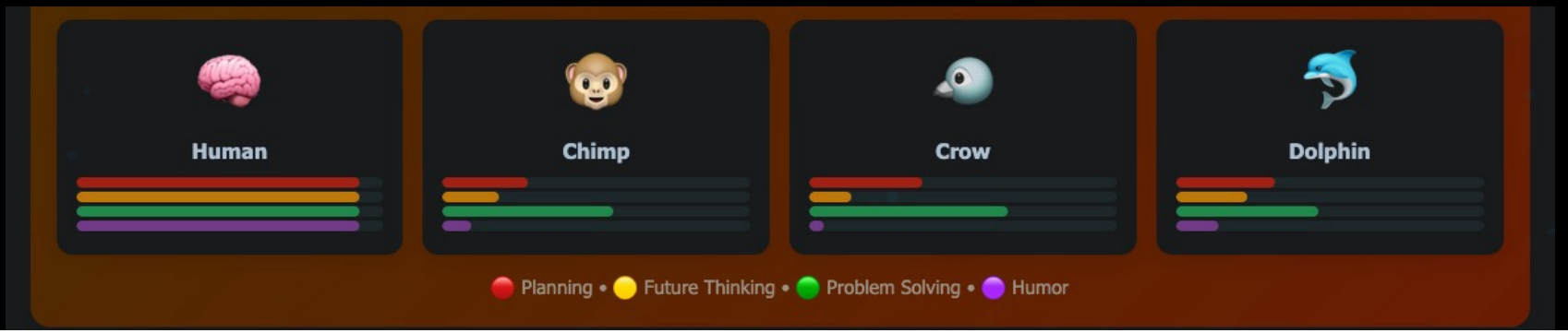
- enables humans to plan or not
- imagine the future or not
- solve puzzles or not
- understand sarcasm and humor



The Fundamental Questions

There must be something about the human brain that's different from the brains of other animals — something that:

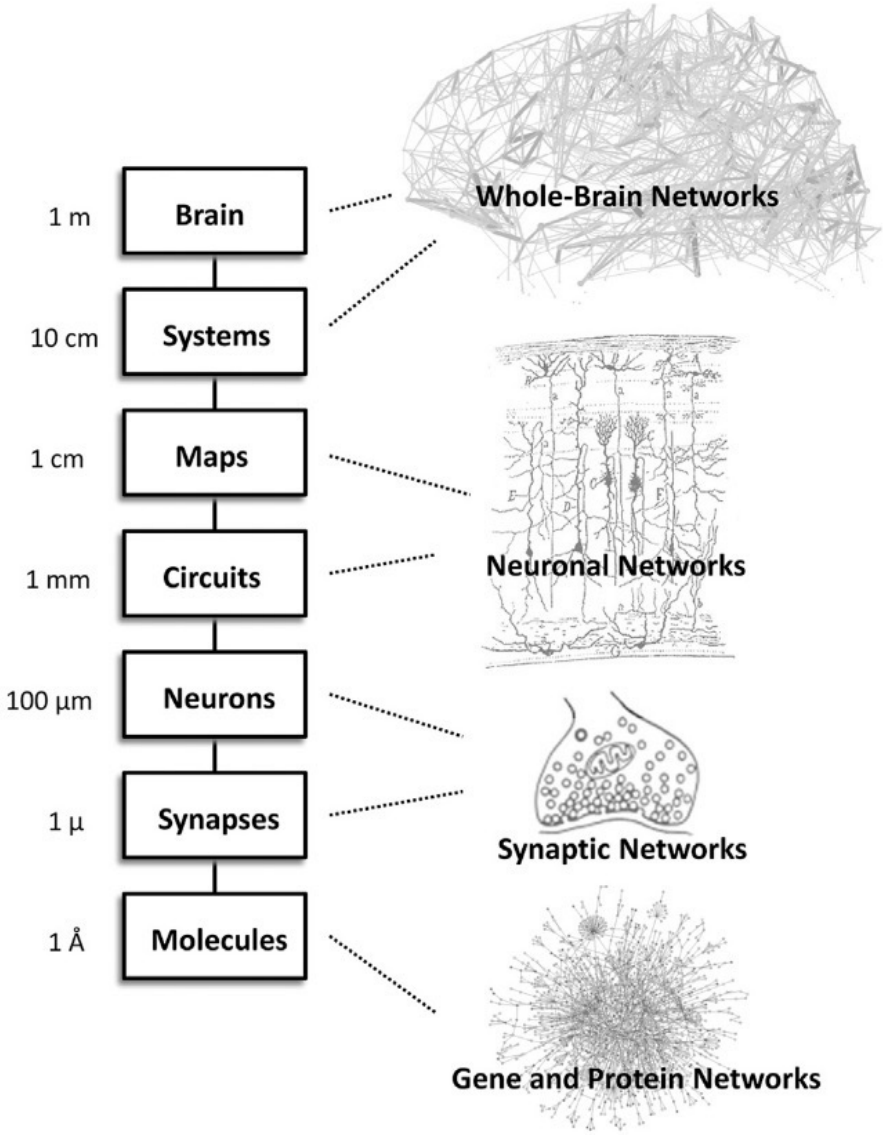
- enables humans to plan or not
- imagine the future or not
- solve puzzles or not
- understand sarcasm and humor
- advanced cognitive abilities



Why This Matters?

Understanding human information processing differences may explain:

- Our unique cognitive abilities
  what makes us human
- Why humans develop conditions like bipolar disorder and schizophrenia
  what makes you famous and rich
- The evolutionary advantages of our neural architecture – nature vs nurture
  how “god” thinks


1 m
Brain
Whole-Brain Networks
10 cm
Systems
1 cm
Maps
Neuronal Networks
1 mm
Circuits
100 µm
Neurons
1 µ
Synapses
Synaptic Networks
1 Å
Molecules
Gene and Protein Networks

Interesting (Counter-intuitive) Facts

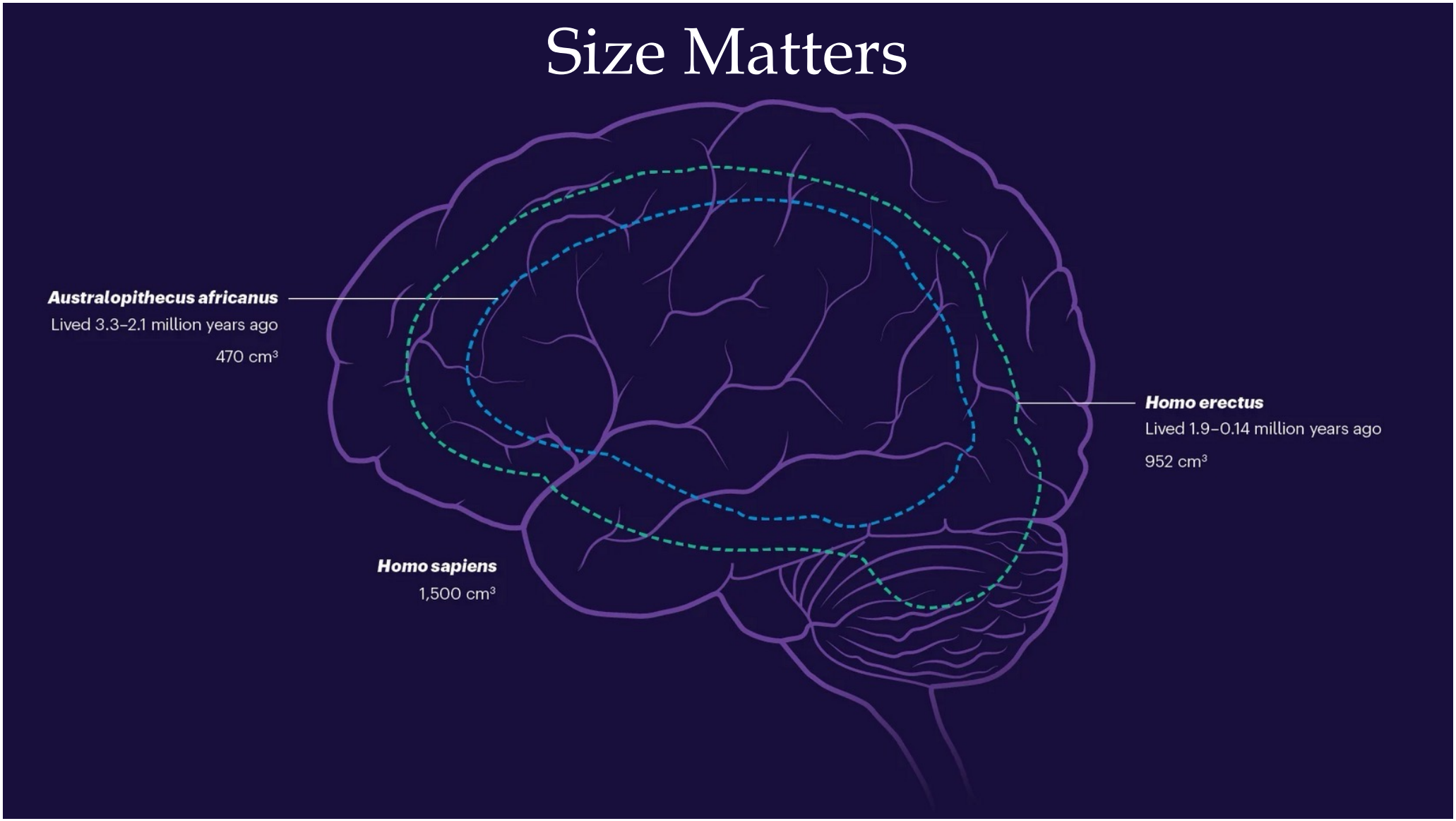
- The differences between human brain cells and those of other species are often subtle
- The human brain develops **slowly** compared with other animals

Interesting (Counter-intuitive) Facts

- The differences between human brain cells and those of other species are often subtle
- The human brain develops **slowly** compared with other animals
- How these small differences give rise to our superior cognitive skills is still a mystery

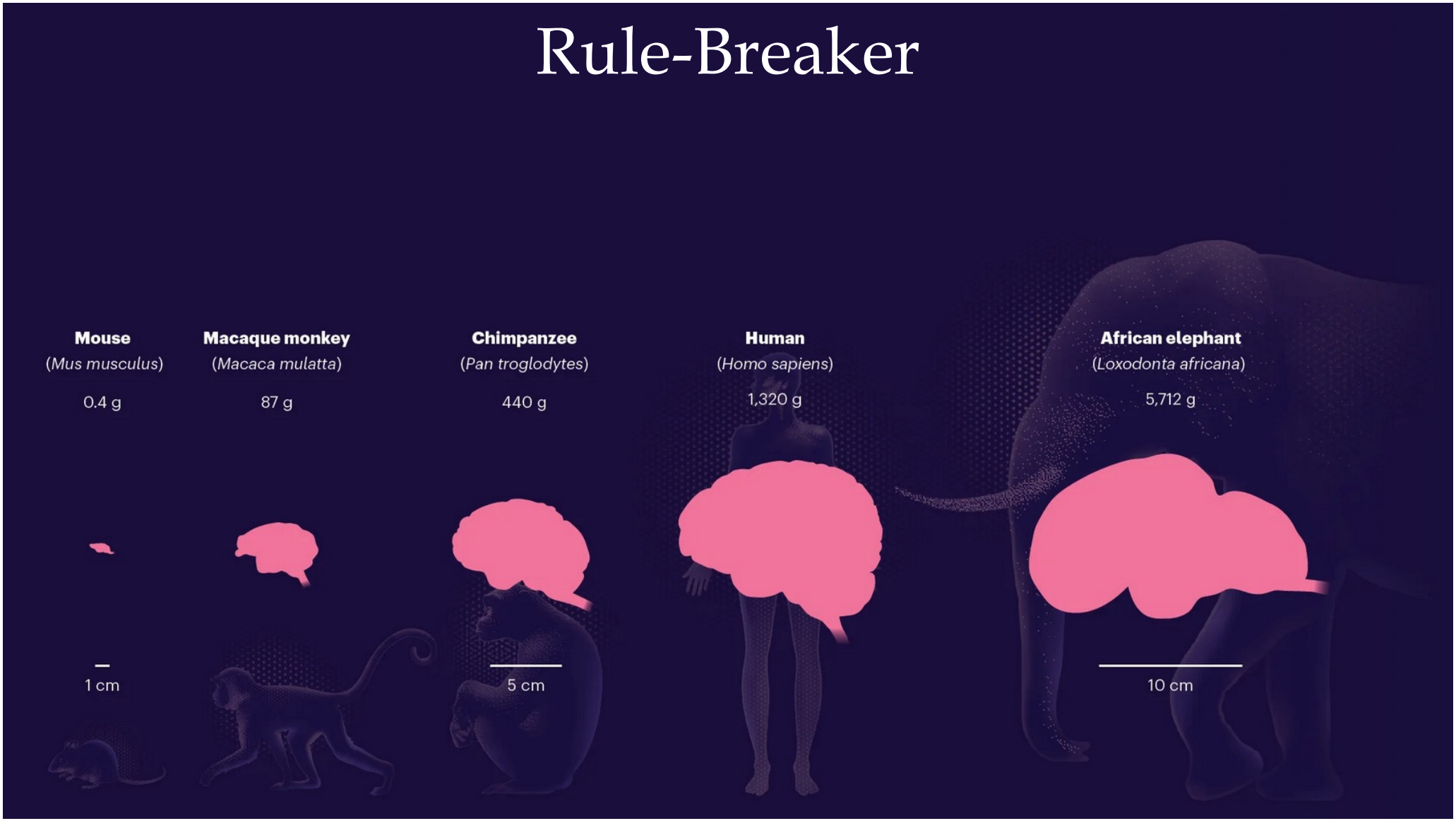
Size Matters?

Size Matters

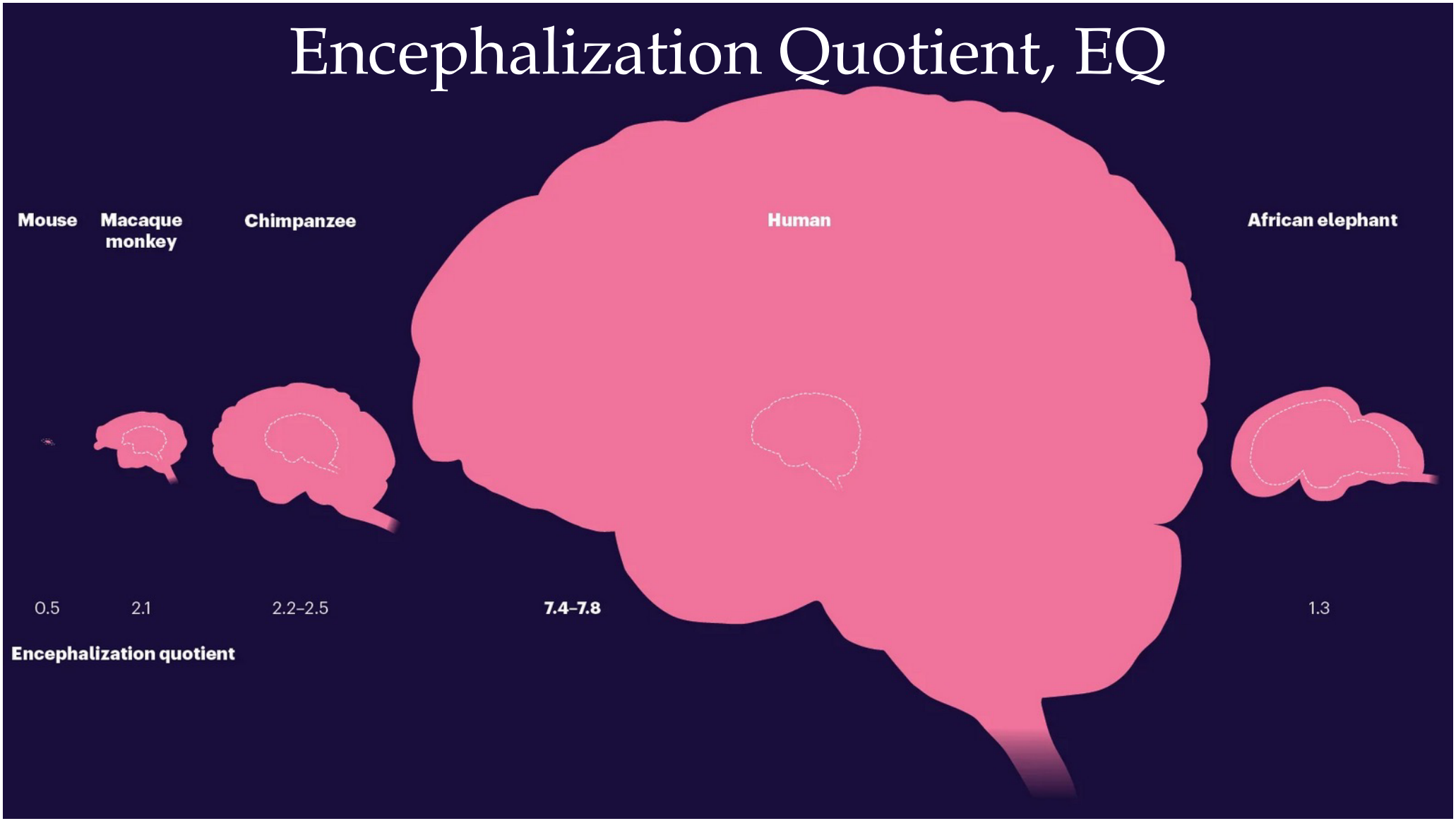


Size Matters

Rule-Breaker



Encephalization Quotient, EQ



Size Comparison

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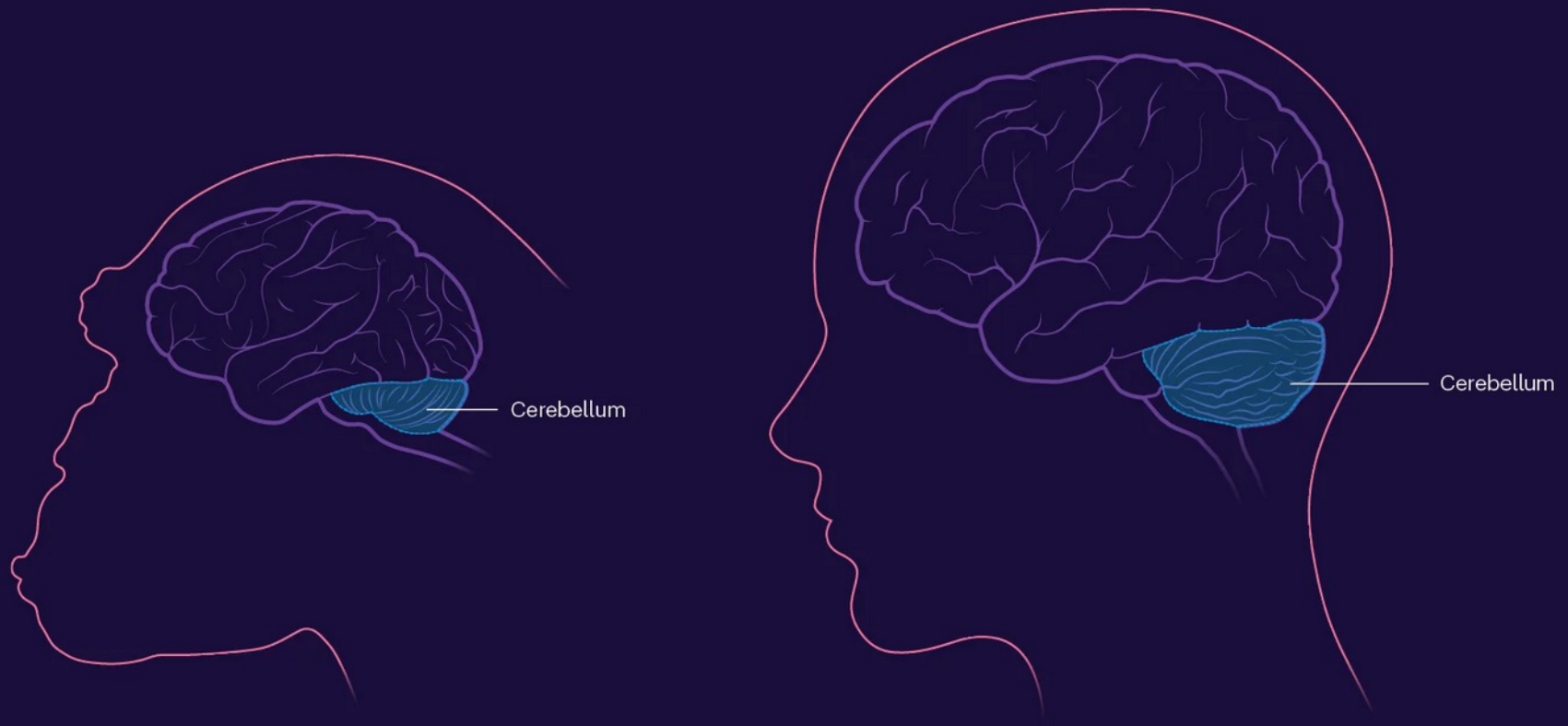

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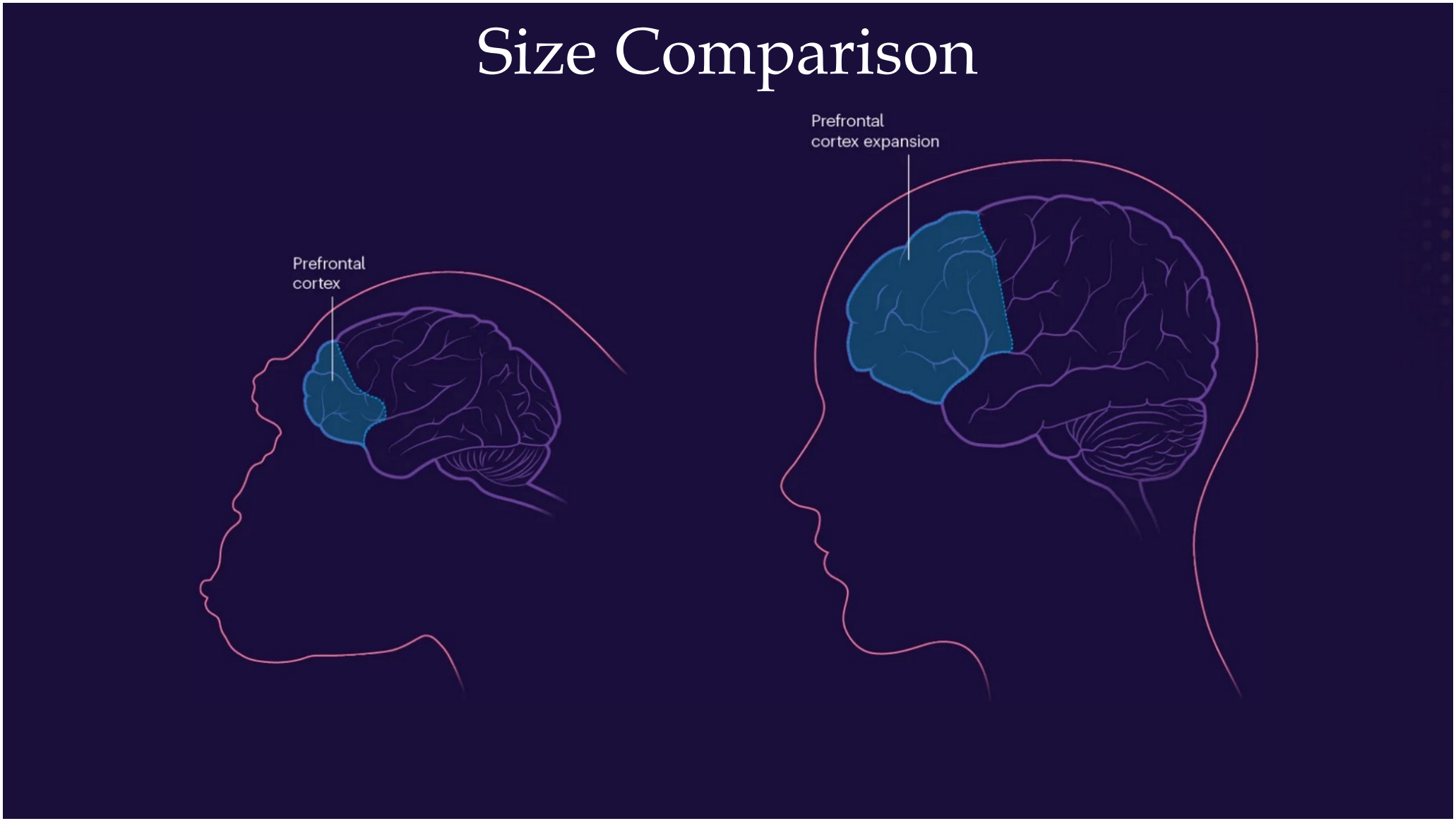


Is Size Everything?

Number vs Architecture

Mouse
71 million neurons

Raven
2.2 billion neurons

Human
86 billion neurons

Number vs Architecture

Macaque monkey
6.4 billion neurons

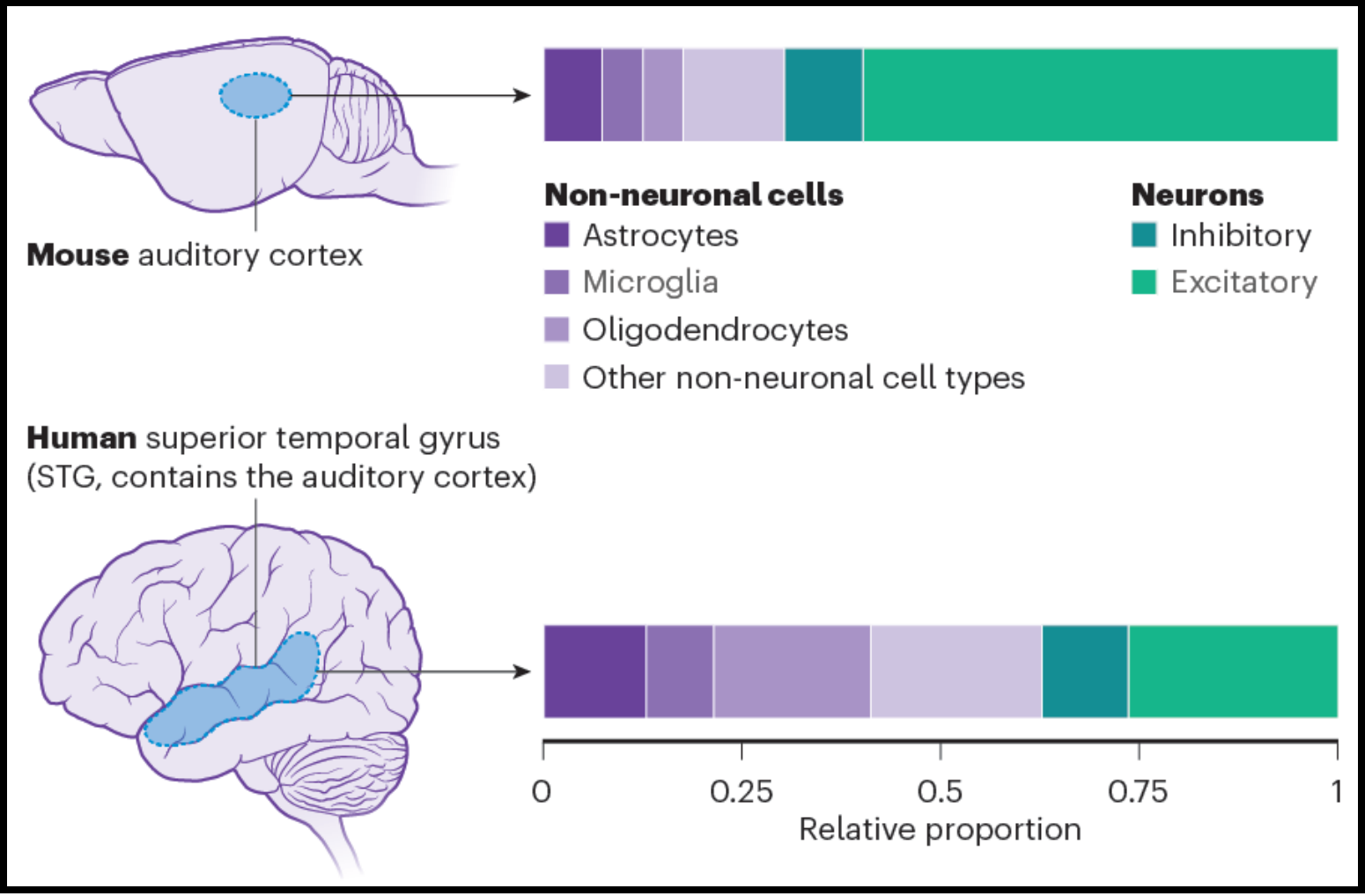
Human
86 billion neurons

Number vs Architecture

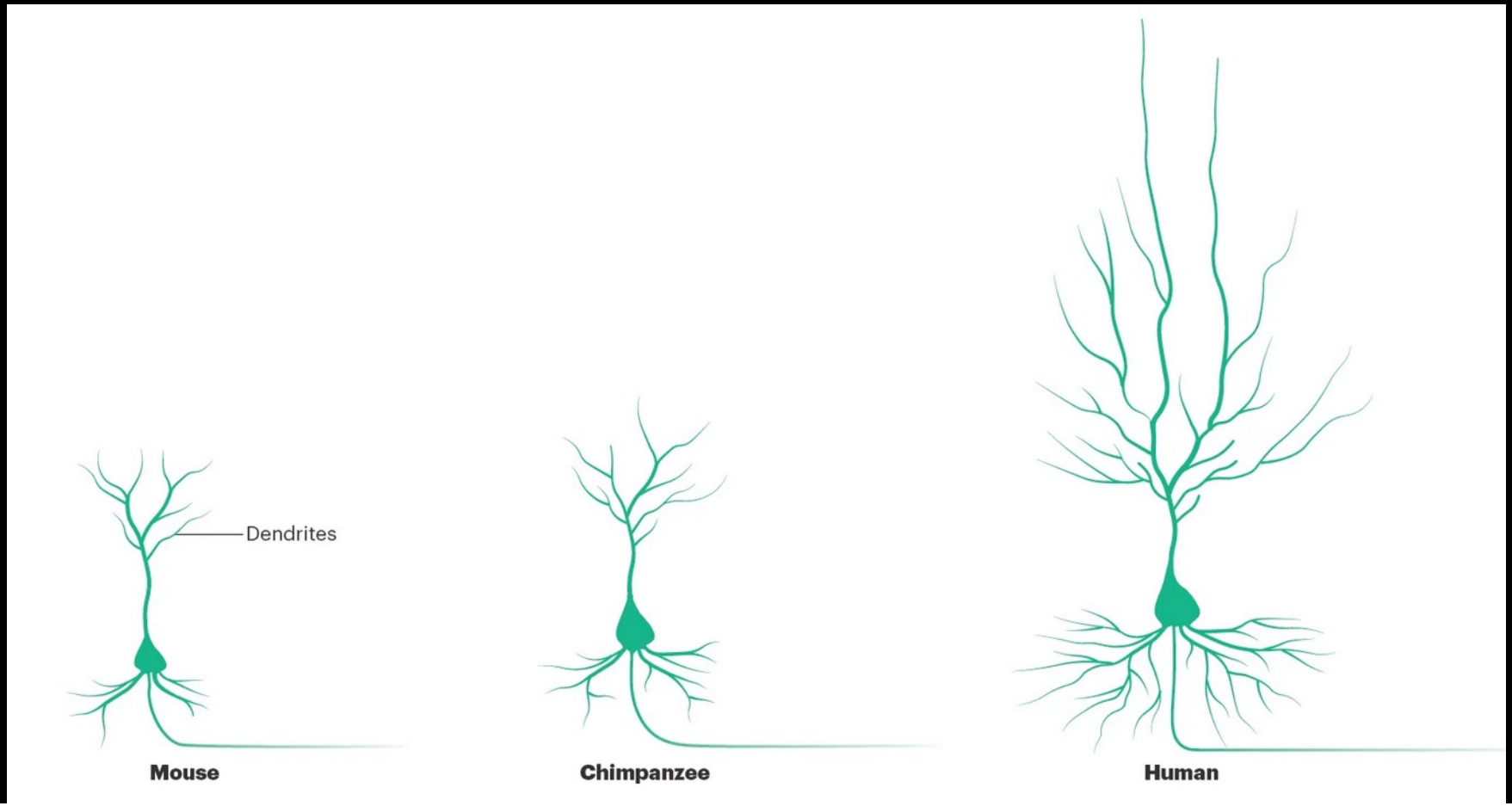
What Makes us Special?

Scaling?

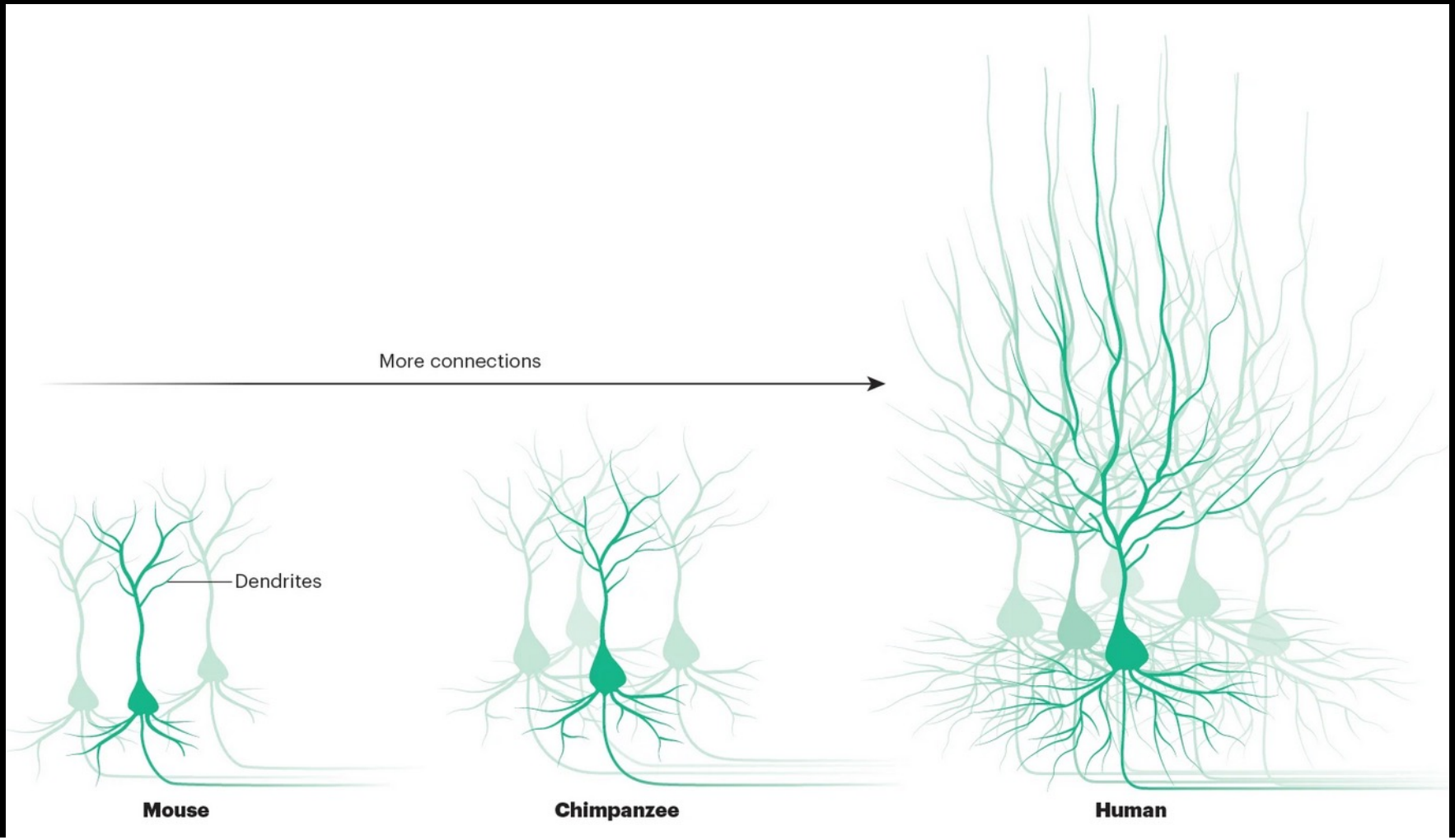
Special Recipe



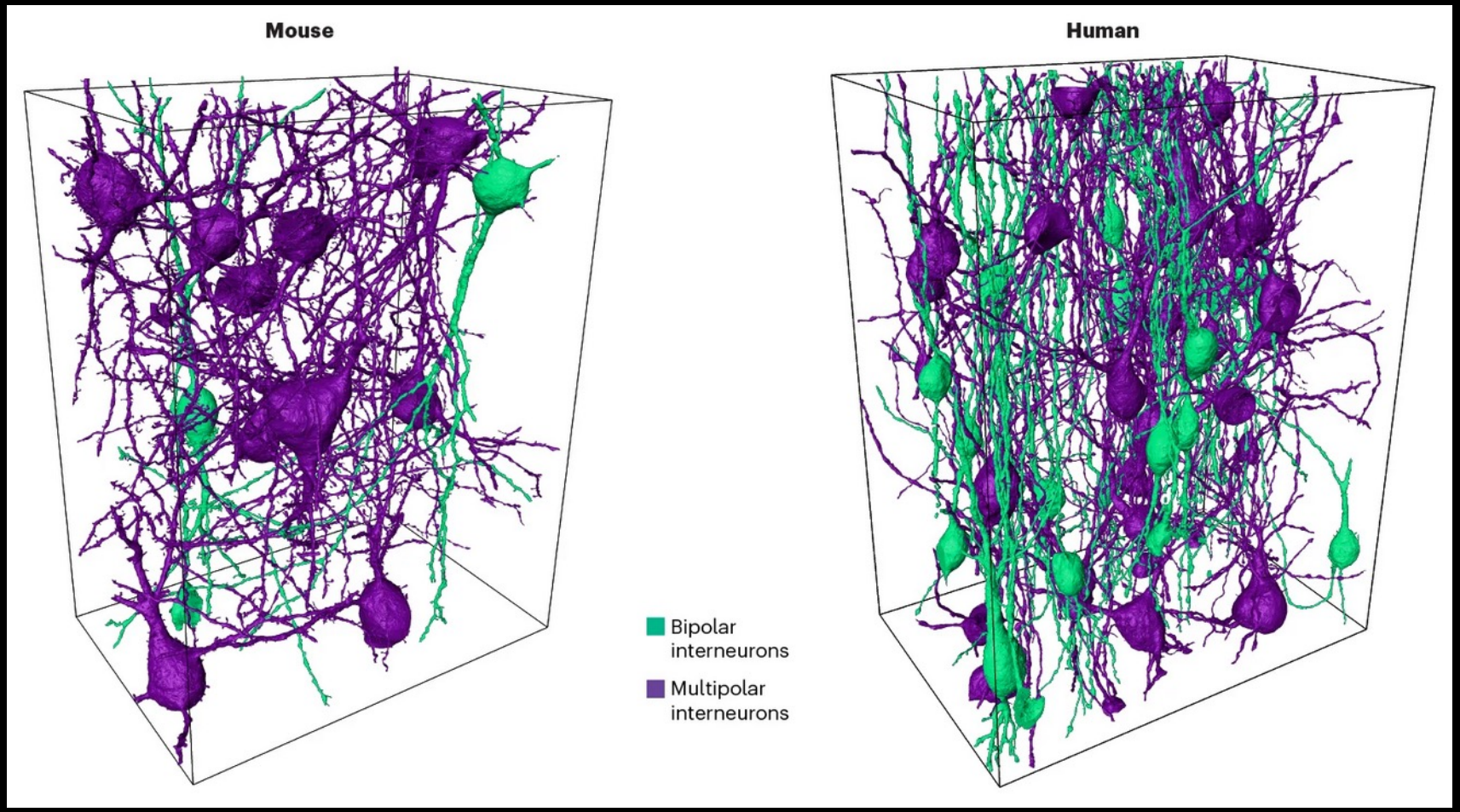
Special Recipe



Special Recipe



Making Connections



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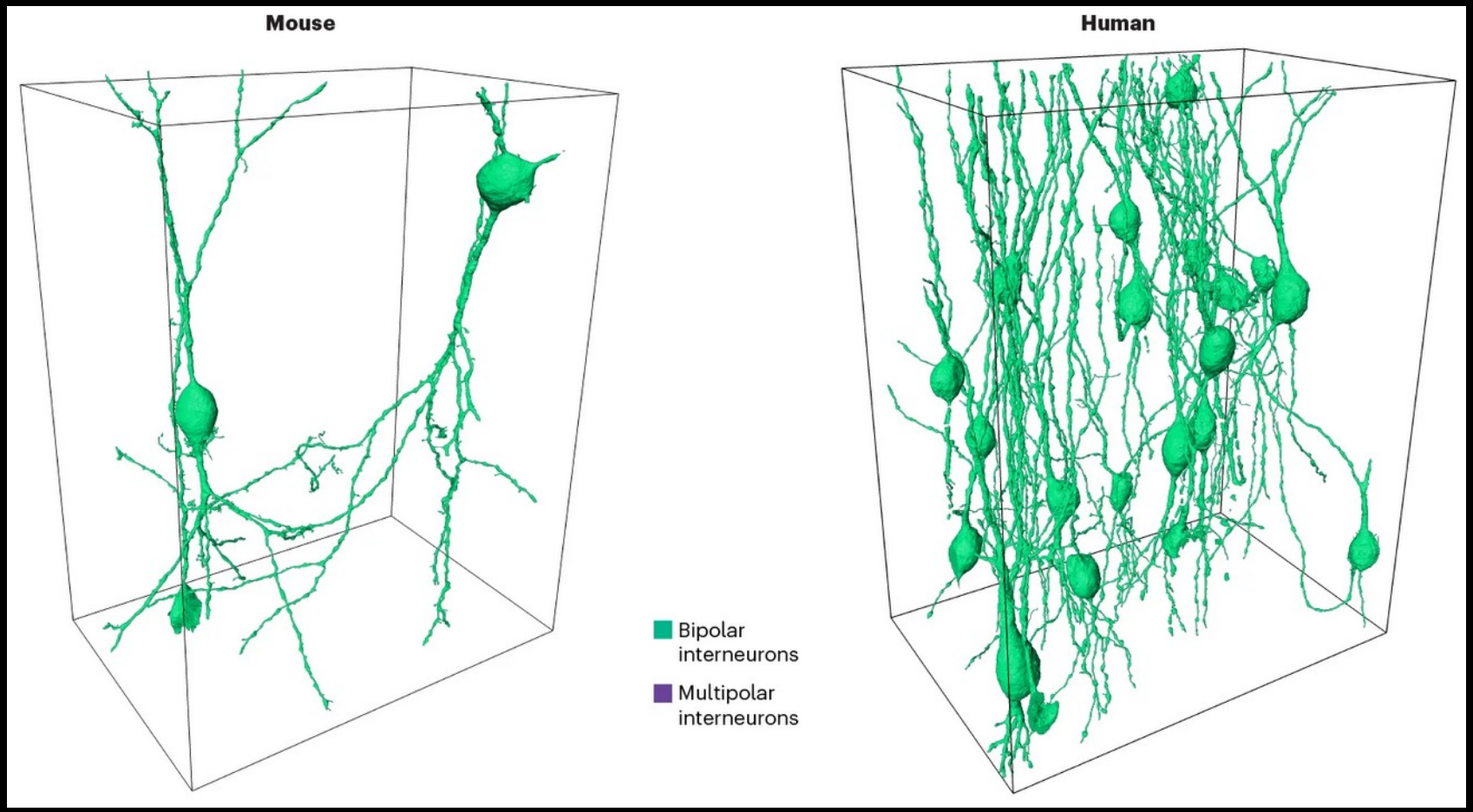
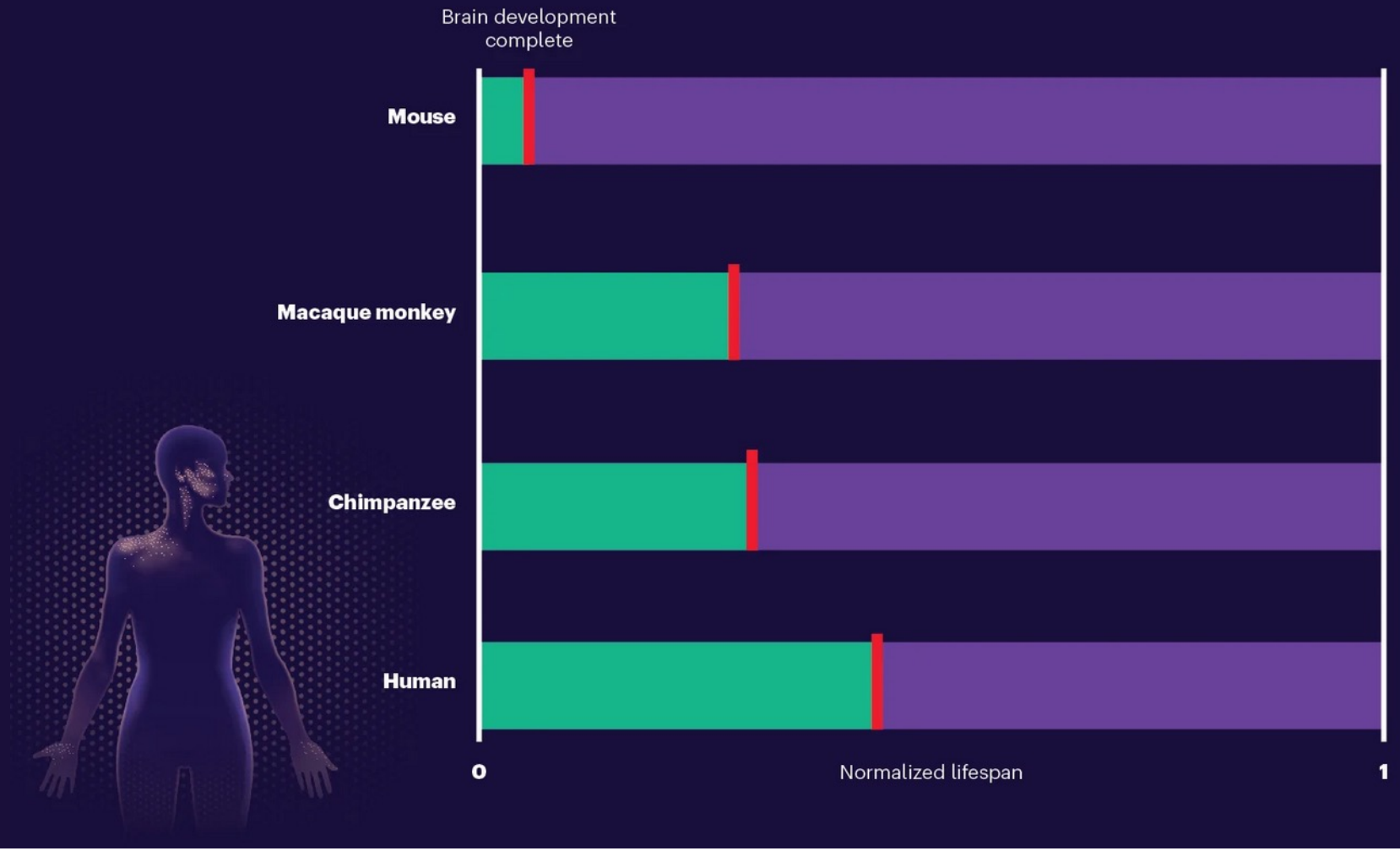


Slow is the New Fast



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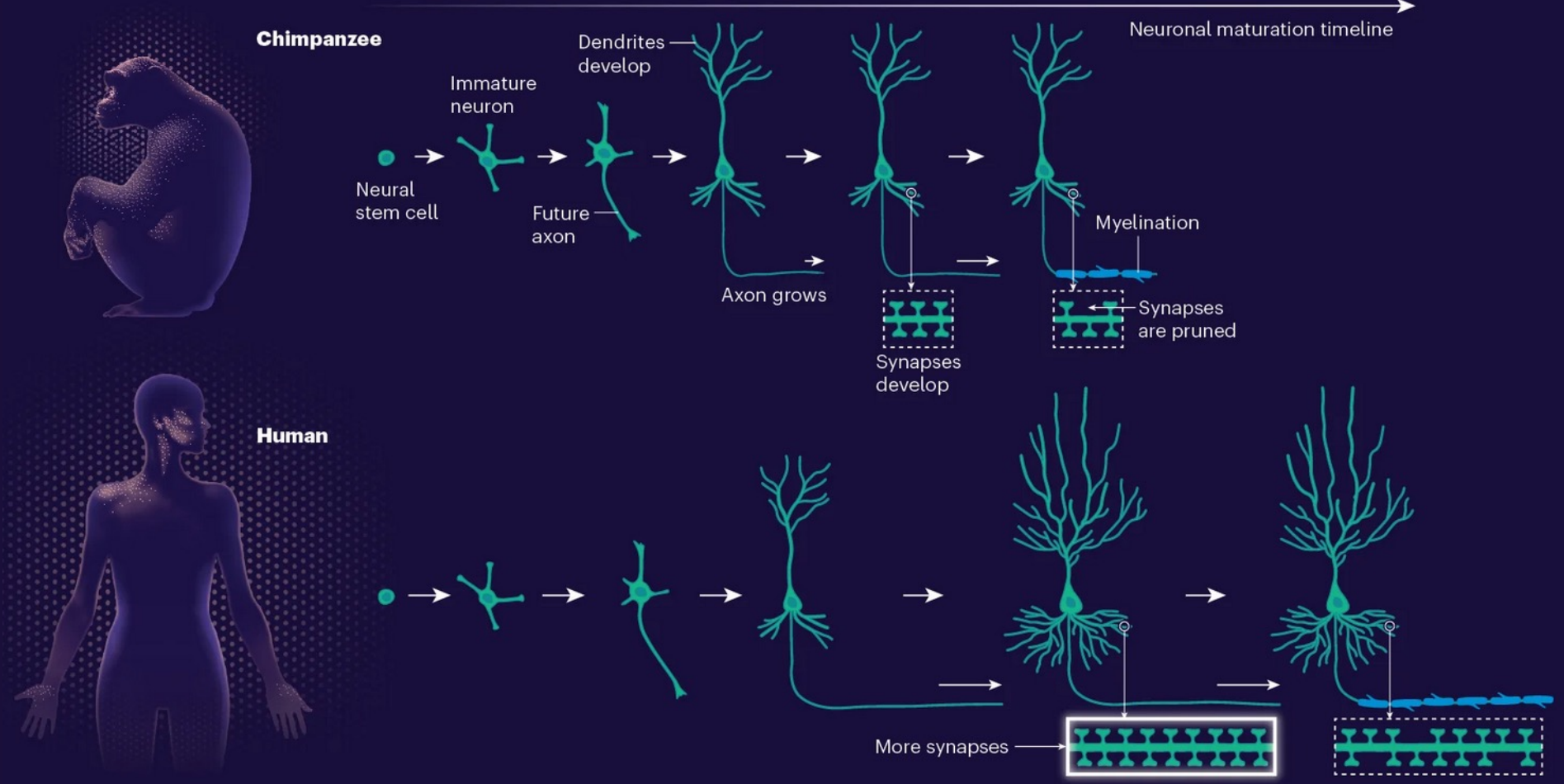


Science is Slow but Solid!

1. Herculano-Houzel, S. Front. Hum. Neurosci. 3, 31 (2009).
2. Yao, Z. et al. Nature 624, 317–332 (2023).
3. Siletti, K. et al. Science 382, eadd7046 (2023).
4. Jorstad, N. L. et al. Science 382, eadf6812 (2023).
5. Fang, R. et al. Science 377, 56–62 (2022).
6. Lindhout, F. W. et al. Nature 630, 596–608 (2024).
7. Loomba, S. et al. Science 377, eabo0924 (2022).
8. Jorstad, N. L. et al. Science 382, eade9516 (2023).
9. Otani, T. et al. Cell Stem Cell 18, 467–480 (2016).
10. Delgado, R. N. et al. Nature 601, 397–403 (2022).
11. Schmidt, E. R. E. et al. Nature 599, 640–644 (2021).
12. Fiddes, I. T. et al. Cell 173, 1356–1369 (2018).
13. Suzuki, I. K. et al. Cell 173, 1370–1384 (2018).

Course Logistics

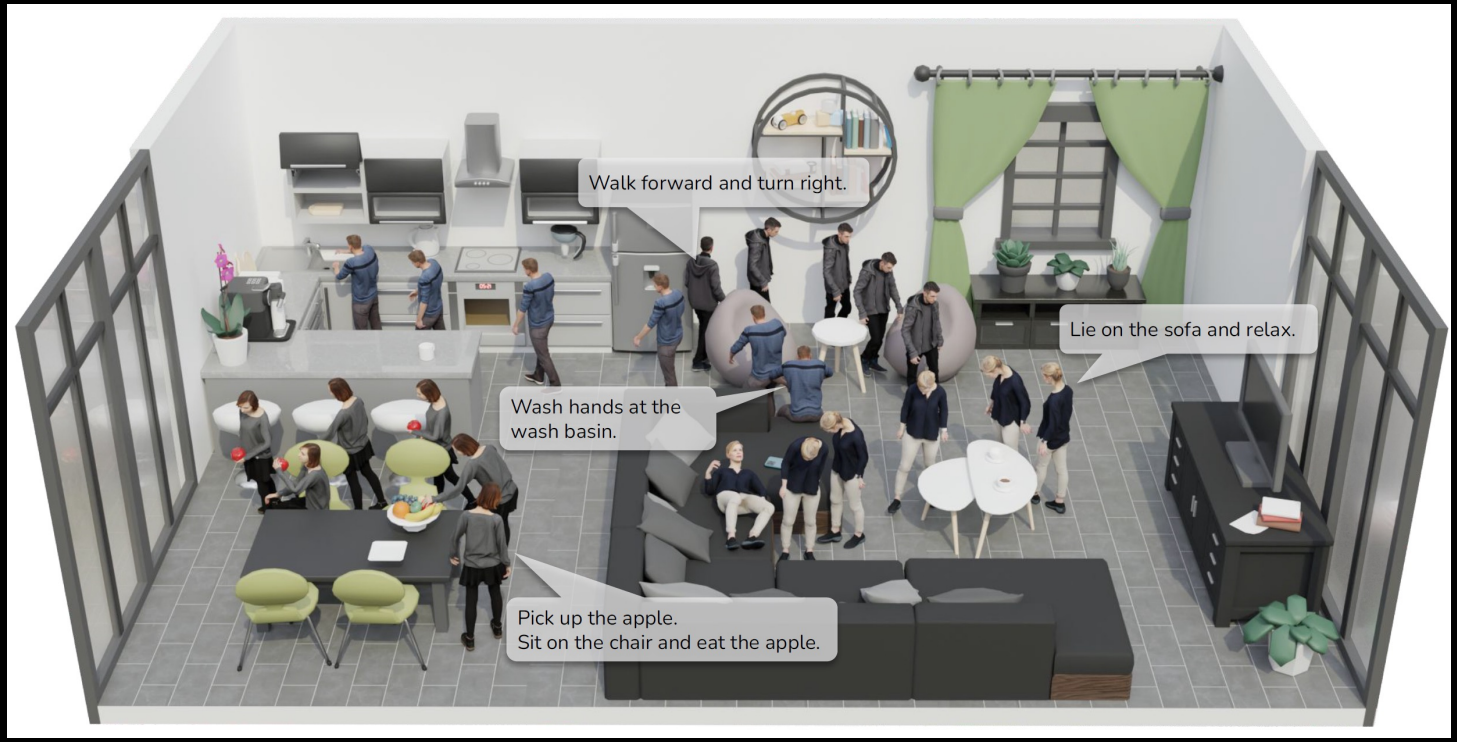
What this class is about?

- An attempt to see how recent work in cognitive science can inform some of the core questions in AI
- Developing formal accounts of cognition: how people think and learn, and how we can make machines do the same thing
- Using tools from: AI, ML, statistics, cognitive science, neuroscience, robotics

What you should expect from this class

- Tools (concepts, observations, computational methods) for both engineering and reverse engineering intelligence.
- Experience building models from both the cognitive science and AI perspectives, working with different kinds of data and modeling goals.
- Exposure to ways of thinking about intelligence you won't get in other classes (cogsci or AI): tools for studying mental models, how they are used, and how they are learned.

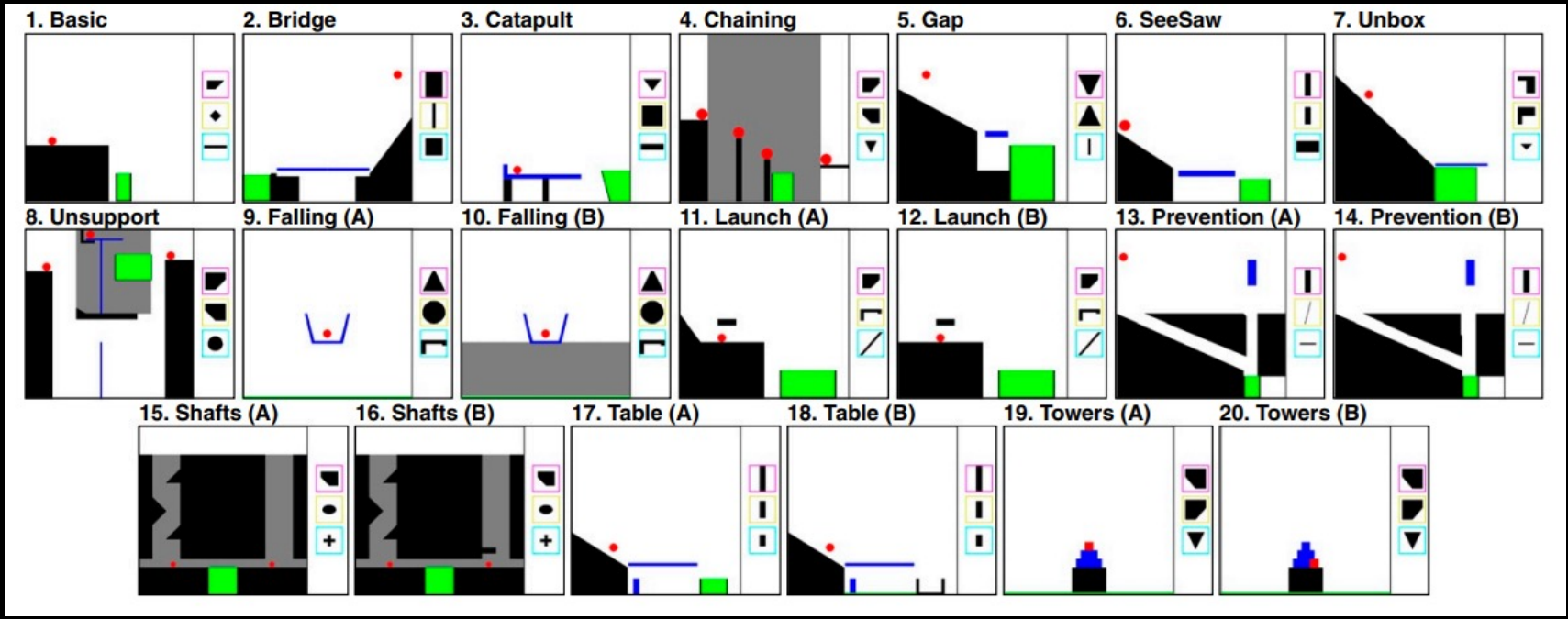
Topic 1: Affordance/Functionality/HOIs



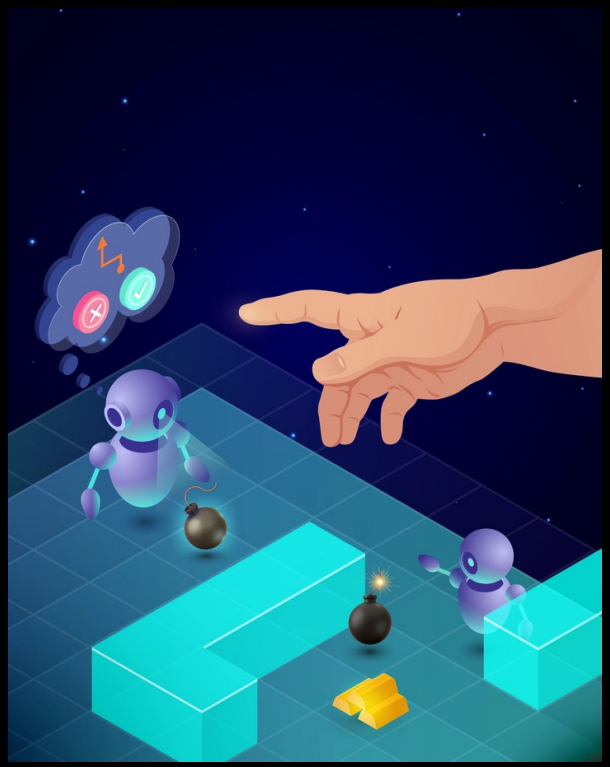
Topic 2: Intuitive Physics

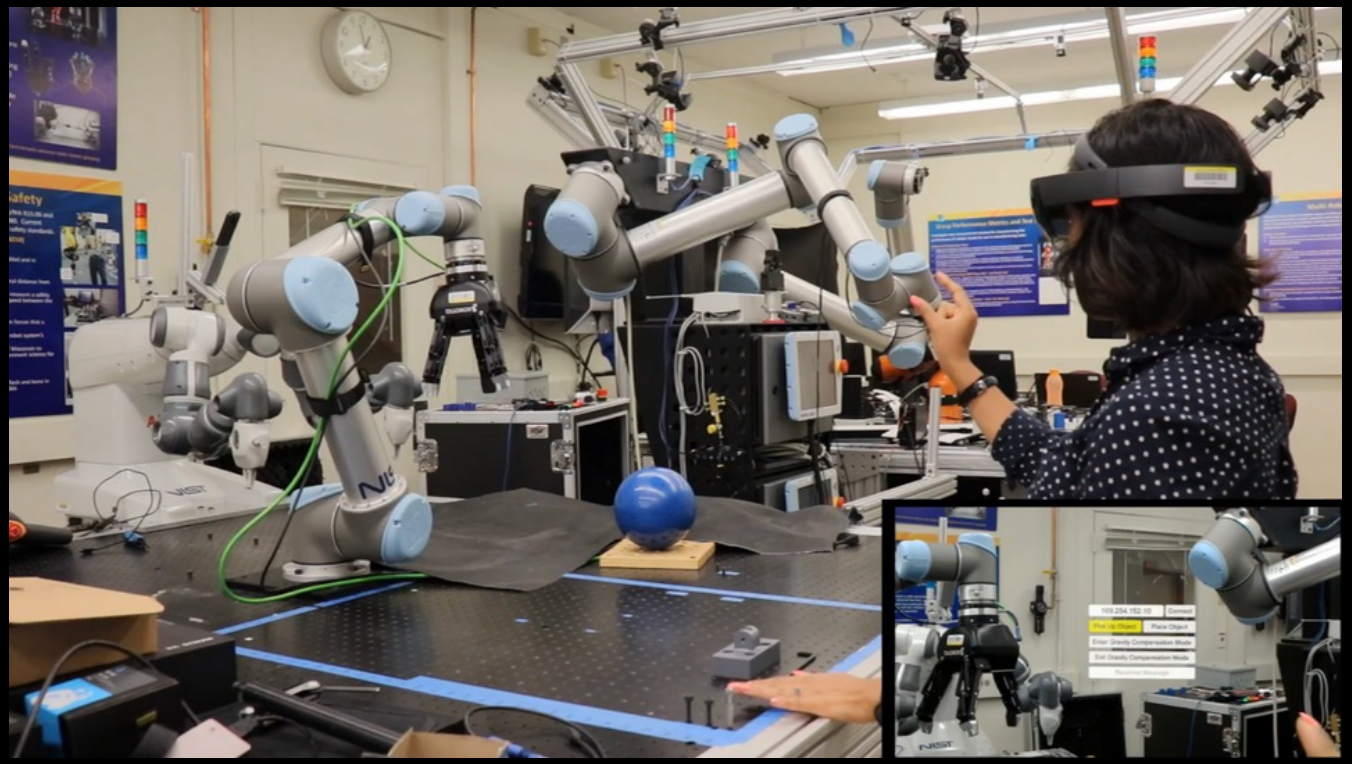
Topic 3: Causality

Topic 4: Tool, Mirroring, and Imitation

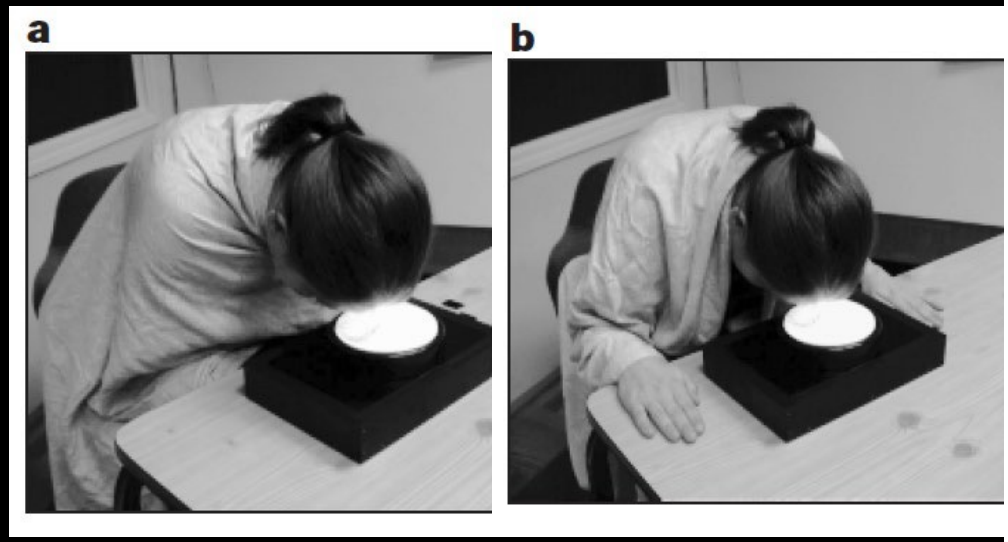


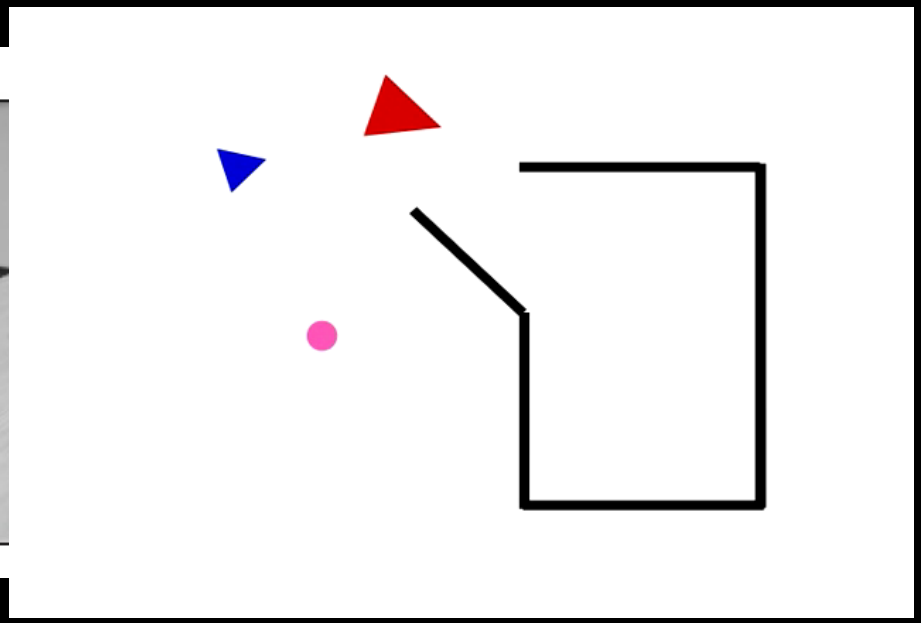
Topic 5: Social Intelligence



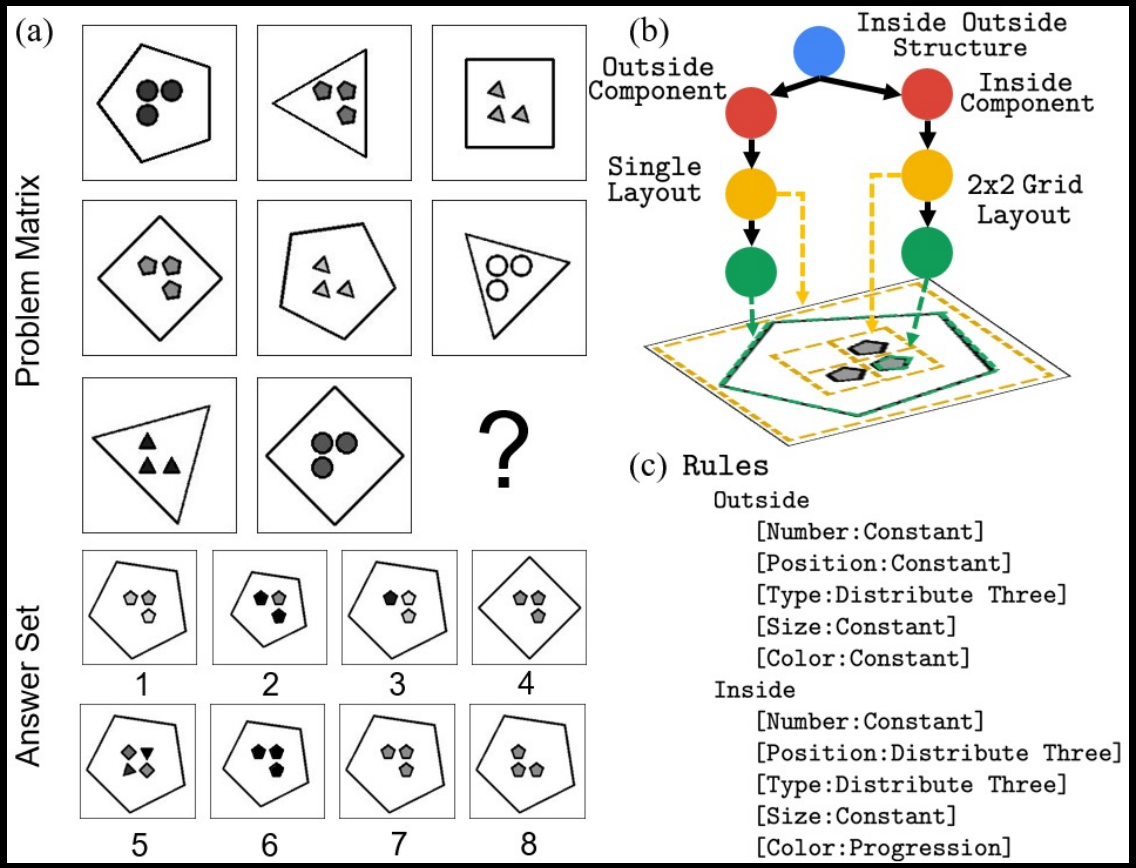


Topic 6: Theory of Mind





Topic 7: Abstract Reasoning

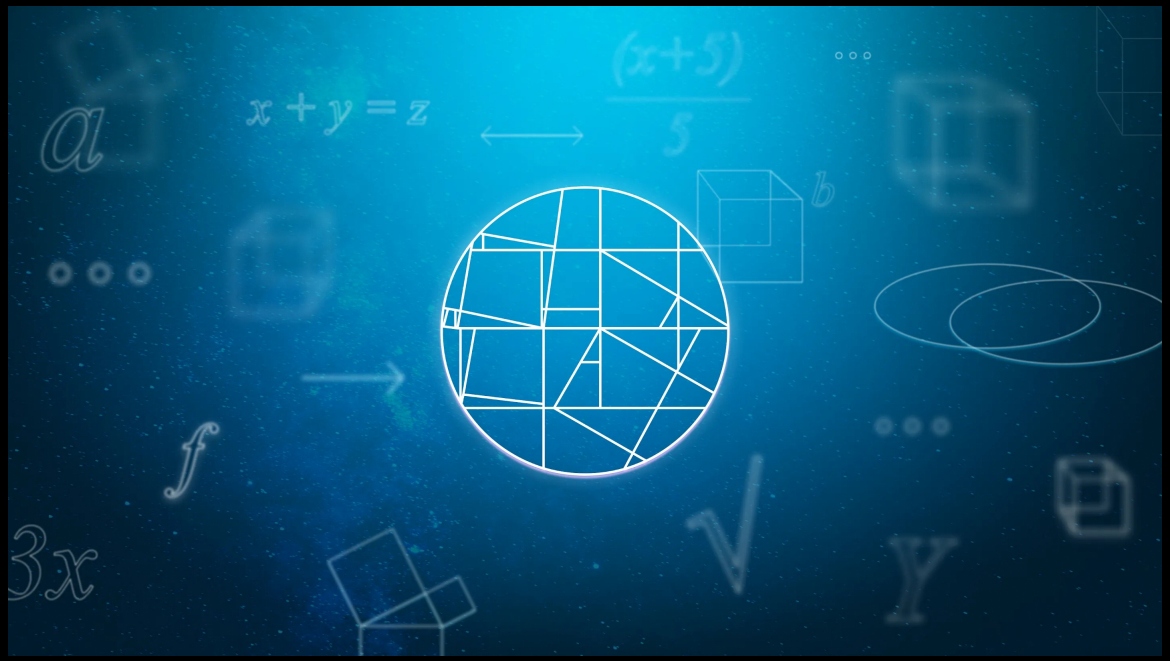


Case Study

- AlphaGo Series

Case Study

- TongGeometry, AlphaGeometry, AlphaProof and Gemini Deep Think



Final Report

- English
- Top-tier AI conference quality
  - Introduction
  - Related work
  - Method
  - Experiments
  - Conclusion
  - Code
- Topics related to AI and reasoning
- ~5 students as a team

Oral/Written Presentation Rubrics

- Soundness
  - Logical consistency and coherence of arguments
  - Accuracy of facts, data, and references
  - Valid methodology and sound reasoning
  - Credible sources and evidence to support claims
  - Absence of logical fallacies or contradictions

Oral/Written Presentation Rubrics

- Soundness
- Novelty
  - Original ideas or unique perspectives presented
  - Innovative approaches/perspective to existing problems
  - Fresh insights or unexpected connections
  - Creative solutions or methodologies
  - Contribution of new knowledge to the field

Oral/Written Presentation Rubrics

- Soundness
- Novelty
- Depth
  - Comprehensive coverage of the topic
  - Thorough analysis and detailed examination
  - Complex understanding demonstrated beyond surface level
  - Multiple perspectives or dimensions explored
  - Substantial research and investigation evident

Oral/Written Presentation Rubrics

- Soundness
- Novelty
- Depth
- Presentation
  - Clear and engaging delivery style
  - Well-organized structure and logical flow
  - Effective use of visual aids
  - Appropriate timing and pace
  - Strong audience engagement and communication skills

Oral/Written Presentation Rubrics

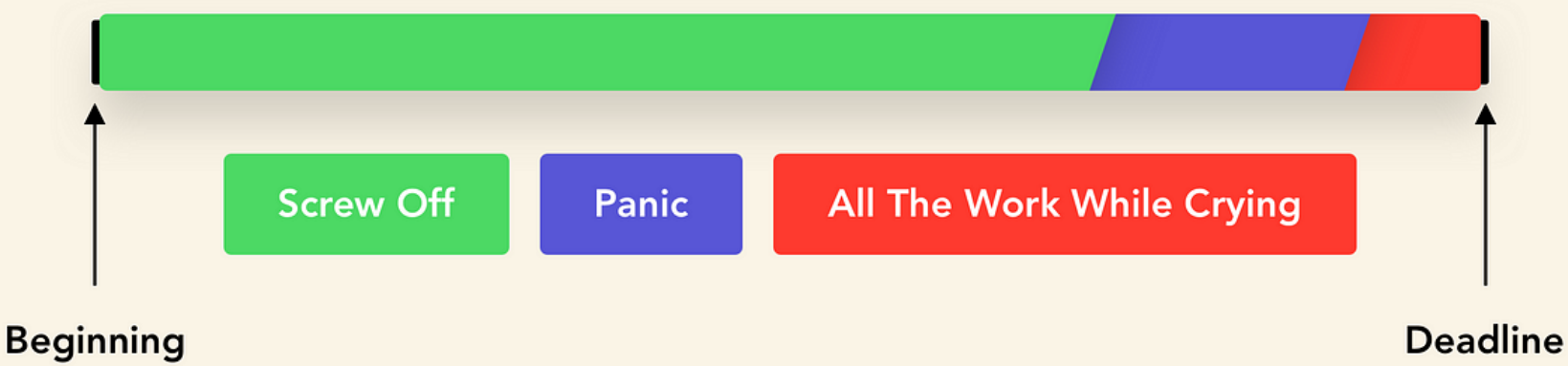
- Soundness
- Novelty
- Depth
- Presentation
- Technical Details
  - Accuracy of technical information and specifications
  - Proper use of terminology and concepts
  - Clear explanation of methods and processes
  - Sufficient detail for understanding and replication
  - Appropriate level of technical complexity for audience

More Tips

- Rule of Thumbs
  - You'll always be busy, so start early!
  - You'll have other deadlines, so start early!
  - You'll have trouble using software to make figures, record videos, etc., so start early!
  - You'll always encounter all kinds of problems during submission, so start early!
- Things to be Prepared
  - Start early on writing. Do this while reviewing the literature, designing models, and running experiments.
  - Do NOT wait till the last minute to write everything.
  - Start early on making figures. A pretty illustration is a key to impressing reviews and delivering essential ideas.

These are also crucial for submitting an actual academic research paper!

The Most Important Slide Today!



TODO: Before the end of Week 4

- Find potential teammates for projects
- Choose a team lead
  - The team lead will report individual's contributions
- Brainstorm an interesting topic
  - LLM, Agent
  - Tasks of value
  - Fun tools

GPUs

- AutoDL
- Kaggle Free GPU
- Google Colab GPU
- 阿里巴巴云公开物
- Consider inference only work (agentic / tool use)

2. What is Plagiarism?

Plagiarism

University Policy

Once suspected, the case will be documented and reported to the dean's office. If confirmed by the dean's office, the student's final grade will be 0. **NO EXCEPTION**.

- 北京大学学生违纪处分办法
- 北京大学研究生基本学术规范
- 北京大学考试纪律
- 清华大学学生纪律处分管理规定
- 清华大学预防与处理学术不端行为办法
- 清华大学考试违规行为的处理办法
- definition
- how to avoid
- sample cases
- FAQ

一经发现，按挂科严肃处理，并提交教务处

往年均发现过学生文章、代码抄袭，不要以身试法

On LLMs

- I assume everyone uses LLMs and encourage everyone to uses LLMs for various stages of your work
- However, you should be solely responsible for everything your model produces
- Your model is an executor, but ideas should come from you

According to the Merriam-Webster online dictionary, to "plagiarize" means:

to steal and pass off (the ideas or words of another) as one's own

to use (another's production) without crediting the source

to commit literary theft

to present as new and original an idea or product derived from an existing source

“Plagiarize.” Merriam-Webster.com. Merriam-Webster. Accessed May 20, 2020. https://www.merriam-webster.com/dictionary/plagiarize.

Plagiarism does not always mean word-for-word copying

#1 Clone

Submitting another's work, word-for-word, as one's own

Source text

A Natural Setting: A History of Exploration and Settlement in Yosemite Valley

Since its first discovery by non-indigenous people in the mid-nineteenth century, Yosemite Valley has held a special, even religious, hold on the American conscience because its beauty makes it an incomparable valley and one of the grandest of all special temples of Nature. While Yosemite holds a special grip on the western mind, perceptions about the Valley have evolved over time due to changing politics, migration patterns and environmental concerns as man has become more attuned to his relationship and impact on nature.

Student work

A Natural Setting: A History of Exploration and Settlement in Yosemite Valley

Since its first discovery by non-indigenous people in the mid-nineteenth century, Yosemite Valley has held a special, even religious, hold on the American conscience because its beauty makes it an incomparable valley and one of the grandest of all special temples of Nature. While Yosemite holds a special grip on the western mind, perceptions about the Valley have evolved over time due to changing politics, migration patterns and environmental concerns as man has become more attuned to his relationship and impact on nature.

#2 Ctrl+C

Contains significant proportions of text from a single source without alterations

Source text

A Natural Setting: A History of Exploration and Settlement in Yosemite Valley

Since its first discovery by non-indigenous people in the mid-nineteenth century, Yosemite Valley has held a special, even religious, hold on the American conscience because its beauty makes it an incomparable valley and one of the grandest of all special temples of Nature. While Yosemite holds a special grip on the western mind, perceptions about the Valley have evolved over time due to changing politics, migration patterns and environmental concerns as man has become more attuned to his relationship and impact on nature.

Student work

The Beautiful Yosemite Valley

From the time of its first discovery by non-indigenous people in the mid-nineteenth century Yosemite Valley has held a special, even religious, possession on the American conscience because its beauty makes it an incomparable valley and one of the grandest of all special temples of Nature. And Yosemite holds a special grip on the western mind, while perceptions about the Valley have evolved over time due to changing politics, migration patterns and environmental concern as man has become more attuned to his relationship and impact or nature.

#3 Find - Replace

Changing key words and phrases but retaining the essential content of the source

Source text

A Natural Setting: A History of Exploration and Settlement in Yosemite Valley

Since its first discovery by non-indigenous people in the mid-nineteenth century, Yosemite Valley has held a special, even religious, hold on the American conscience because its beauty makes it an incomparable valley and one of the grandest of all special temples of Nature. While Yosemite holds a special grip on the western mind, perceptions about the Valley have evolved over time due to changing politics, migration patterns and environmental concerns as man has become more attuned to his relationship and impact on nature.

Student work

A Beautiful Setting in Yosemite

Since its first discovery by non-native people in the mid-19th century Yosemite Valley has held a special, even sacred, hold on the American psyche because its beauty makes it an incomparable valley and one of the grandest of all special temples of Nature. While Yosemite holds a special grip on the western mindset, perceptions about the Valley have evolved over time due to changing political movements, migration patterns and environmental issues as man has become more attuned to their relationship and impact on nature.

#4 Remix

Paraphrases from multiple sources, made to fit together

Source text

A Natural Setting: A History of Exploration and Settlement in Yosemite Valley

Since its first discovery by non-indigenous people in the mid-nineteenth century, Yosemite Valley has held a special, even religious, hold on the American conscience because its beauty makes it an incomparable valley and one of the grandest of all special temples of Nature. While Yosemite holds a special grip on the western mind, perceptions about the Valley have evolved over time due to changing politics, migration patterns and environmental concerns as man has become more attuned to his relationship and impact on nature.

Student work

An Untouched View A History of Settlement and Change in Yosemite Valley

The first non-natives to see Yosemite Valley were probably members of the mid-nineteenth century Joseph Walker Party, who crossed the Sierra Nevada from East to West. The first descriptions of Yosemite came almost 20 years later. From the start, the valley has been renowned for its natural beauty, and highly regarded as he center-piece of Yosemite National Park, attracting visitors from around the world. While Yosemite holds a special grip on the western mind, perceptions about the Valley have evolved over time due to changing politics, migration patterns and environmental concerns as man has become more attuned to their relationship and impact on nature.

#5 Recycle

Borrows generously from the writer's previous work without citation

Source text

A Natural Setting: A History of Exploration and Settlement in Yosemite Valley

Since its first discovery by non-indigenous people in the mid-nineteenth century, Yosemite Valley has held a special, even religious, hold on the American conscience because its beauty makes it an incomparable valley and one of the grandest of all special temples of Nature. While Yosemite holds a special grip on the western mind, perceptions about the Valley have evolved over time due to changing politics, migration patterns and environmental concerns as man has become more attuned to his relationship and impact on nature.

Student work

Ecology 201
An Unnatural Expectation: The Impact of Exploration and Settlement in Yosemite Valley

From its first discovery by the Ahwahnechee in the mid-nineteenth century Yosemite Valley has held a unique, even religious, hold on the American conscience because its beauty makes it an incomparable valley and one of the grandest of all special temples of Nature. While Yosemite holds a special grip on the western mind, perceptions about the Valley have evolved over time reflecting changing politics, human migration patterns and the rise of environmental concerns as man has become more attuned to his relationship and impact on nature.

#6 Hybrid

Combines perfectly cited sources with copied passages without citation

Source text

A Natural Setting: A History of Exploration and Settlement in Yosemite Valley

Since its first discovery by non-indigenous people in the mid-nineteenth century, Yosemite Valley has held a special, even religious, hold on the American conscience because its beauty
makes it an incomparable valley and one of the grandest of all
special temples of Nature. While Yosemite holds a special grip
on the western mind, perceptions about the Valley have evolved
over time due to changing politics, migration patterns and environmental concerns as man has become more attuned to his relationship and impact on nature.

Student work

A Natural Setting A History of Exploration and Settlement in Yosemite Valley

Since its first discovery by non-indigenous people in the mid-nineteenth century, Yosemite Valley has held a special, even religious, hold on the American conscience because its beauty makes it an incomparable valley and one of the grandest of all special temples of Nature. "Native Americans have lived in the Yosemite region for as long as 8,000 years. The first people that we have record of was a band of Native Americans that called the Valley "Ah-wah-nee" and themselves the Ahwahnechee."[1] While Yosemite holds a special grip on the western mind, perceptions about the Valley have evolved over time due to changing politics, migration patterns and environmental concerns as man has become more attuned to his relationship and impact on nature.

1 "Yosemite Valley." Wikipedia. Wikipedia. 20 Apr. 2012. <http://en.wikipedia.org/wiki/Yosemite_Valley>

#7 Mashup

Mixes copied material from multiple sources

Source text

A Natural Setting: A History of Exploration and Settlement in Yosemite Valley

Since its first discovery by non-indigenous people in the mid-nineteenth century, Yosemite Valley has held a special, even religious, hold on the American conscience because its beauty makes it an incomparable valley and one of the grandest of all special temples of Nature. While Yosemite holds a special grip on the western mind, perceptions about the Valley have evolved over time due to changing politics, migration patterns and environmental concerns as man has become more attuned to his relationship and impact on nature.

Student work

A Natural Setting A History of Exploration and Settlement in Yosemite Valley

Since its first discovery by non-indigenous people in the mid-nineteenth century Yosemite Valley has held a special, even religious, hold on the American conscience. The work of Ayres gave easterners an appreciation for Yosemite Valley and started a move-ment to preserve it. While Yosemite holds a special grip on the western mind, perceptions about the Valley have evolved over time due to changing politics, migration patterns and environmental concerns as man has become more attuned to their relationship and impact on nature. The 1849 California Gold Rush led to conflicts between miners and natives, and the volunteer Mariposa Battalion was formed by the state of California as a punitive expedition against natives in the Yosemite area.

#8 404 Error

Includes citations to non-existent or inaccurate information about sources

Source text

A Natural Setting: A History of Exploration and Settlement in Yosemite Valley

Since its first discovery by non-indigenous people in the mid-nineteenth century, Yosemite Valley has held a special, even religious, hold on the American conscience because its beauty makes it an incomparable valley and one of the grandest of all special temples of Nature. While Yosemite holds a special grip on the western mind, perceptions about the Valley have evolved over time due to changing politics, migration patterns and environmental concerns as man has become more attuned to his relationship and impact on nature.

Student work

Perceived Beauty: Man's Nature
Exploration and Settlement in Yosemite Valley

"Since its first discovery by the Joseph Walker Party in the mid-nineteenth century" [1] "Yosemite Valley has held a special, even religious, hold on the American conscience because its beauty makes it an incomparable valley and one of the grandest of all 'special temples of Nature." [2] "While Yosemite holds a special grip on the western mind, perceptions about the Valley have evolved over time due to changing politics, migration patterns and environmental concerns as man has become more attuned to their relationship and impact on nature." [3]

1 For an insightful discussion of Walker's expedition, including an extended description of their first sight of Yosemite Valley, see Walker's original piece in The Golden Times. May 5, 1833: p. 4. Print.

2 Bloom, H. "Aesthetics of the Found: The 'Nature' of the Natural." Aesthetics and the Found. November 2000: 18-22. Print.

3 Huang et al. "Politics, Policy, and the Environment." Nature and Society.

25.3 (1997) : 127-53. Print.

#9 Aggregator

Includes proper citation to sources but the paper contains almost no original work

Source text

A Natural Setting: A History of Exploration and Settlement in Yosemite Valley

Since its first discovery by non-indigenous people in the mid-nineteenth century, Yosemite Valley has held a special, even religious, hold on the American conscience because its beauty makes it an incomparable valley and one of the grandest of all special temples of Nature. While Yosemite holds a special grip on the western mind, perceptions about the Valley have evolved over time due to changing politics, migration patterns and environmental concerns as man has become more attuned to his relationship and impact on nature.

Student work

A Natural Setting: A History of Exploration and Settlement in Yosemite Valley [1]

Yosemite Valley was first sighted by non-Indians in 1833 by Joseph Rutherford Walker and his group of explorers. [2] Yosemite has a long and diverse cultural history that helped to shape the region, country, and even the world. [3] While Yosemite holds a special grip on the western mind, perceptions about the Valley have evolved over time due to changing politics, migration patterns and environmental concerns as man has become more attuned to their relationship and impact on nature. [4]

1 Harrick, C. "A Natural Setting." Nature and its Discontents 23.1 (1996) : 41-50. Print.

2 "History of Yosemite," Yosemitegold.com Ace & Friends, n.d. Web. 24 Apr. 2012. <http://www.yosemitegold.com/yosemite/history.html>

3 "Yosemite National Park Cultural History," Yosemitepark.com DNC Parks and Resorts at Yosemite, Inc., n.d. Web. 24 Apr. 2012. <http://www.yosemitepark.com/cultural-history.aspx>

4 Harrick, C. "A Natural Setting," p. 41.

#10 Re-Tweet

Includes proper citation, but relies too closely on the text's original wording and/or structure

Source text

A Natural Setting: A History of Exploration and Settlement in Yosemite Valley

Since its first discovery by non-indigenous people in the mid-nineteenth century, Yosemite Valley has held a special, even religious, hold on the American conscience because its beauty makes it an incomparable valley and one of the grandest of all special temples of Nature. While Yosemite holds a special grip on the western mind, perceptions about the Valley have evolved over time due to changing politics, migration patterns and environmental concerns as man has become more attuned to his relationship and impact on nature.

Student work

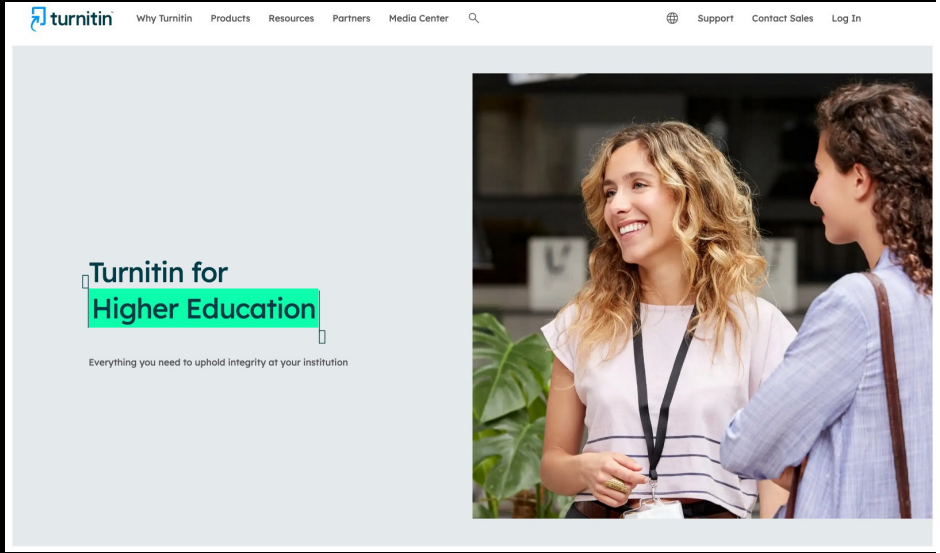
A History of Exploration and Settlement in Yosemite Valley

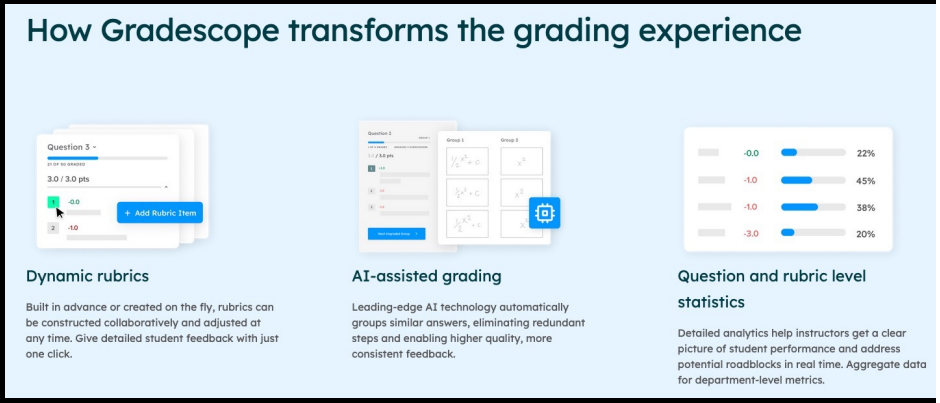
From its earliest discovery by non-indigenous people in the mid-nineteenth century Yosemite Valley has held a spiritual hold on the American conscience; its beauty is incomparable and has been held up as one of the grandest of all special temples of Nature. While Yosemite has a special grip on the western mind, impressions about the Valley have evolved over time, shaped by changing politics, migration patterns and environmental concerns as mankind has become more attuned to its relationship and impact on nature. [1]

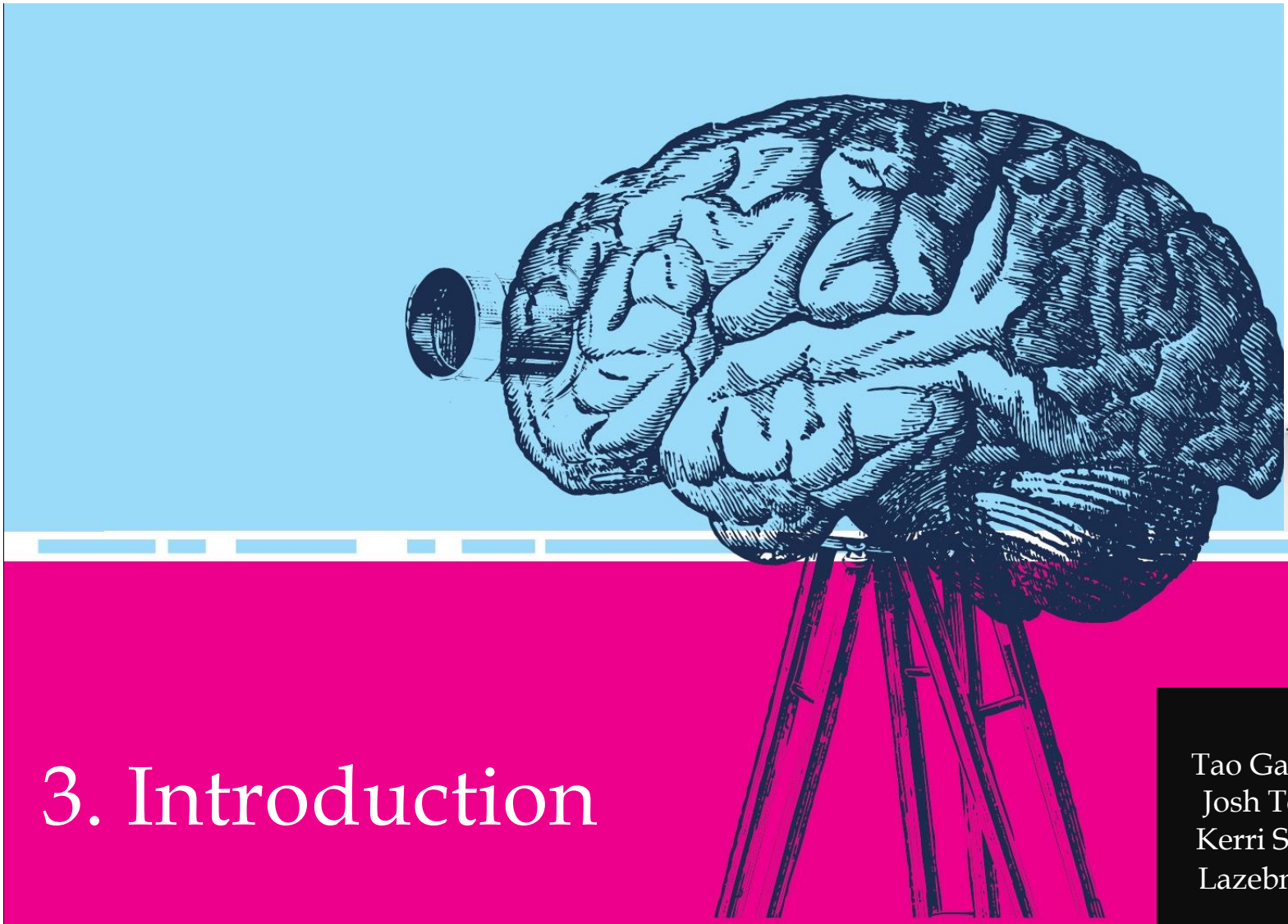
1 Harrick, C. "A Natural Setting." Nature and its Discontents 23.1 (1996) : 41-50. Print.

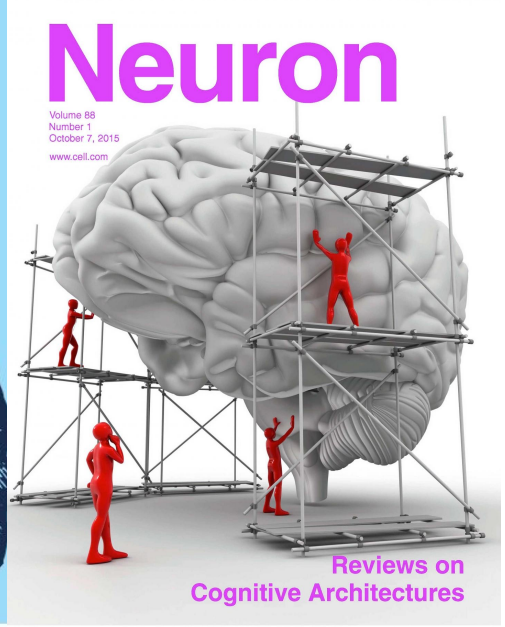
Avoiding Plagiarism

- Manage your time during the researching and writing process wisely.
- Employ effective research strategies.
- Choose when to summarize, paraphrase, or quote information from your sources.
- Provide proper citations for your paraphrased, quoted, or summarized information.
- Own your academic integrity and ask questions when you need help.







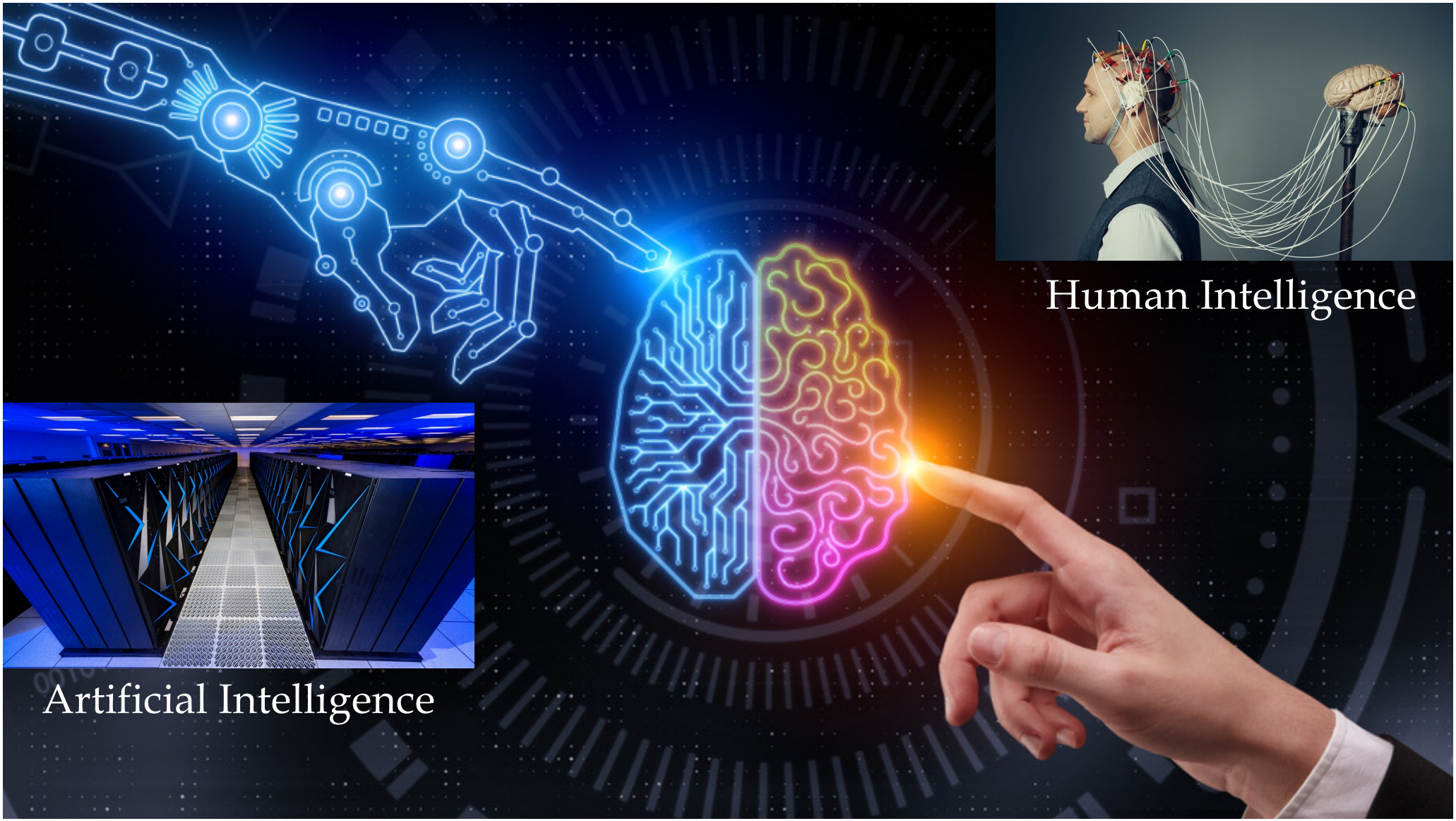


3. Introduction

Slide credits

Tao Gao, Hongjing Lu, Qian Wang, Josh Tenenbaum, Hyun Soo Park, Kerri Smith, Rick Szeliski, Svetlana Lazebnik, Song-Chun Zhu, 小红书的博主们

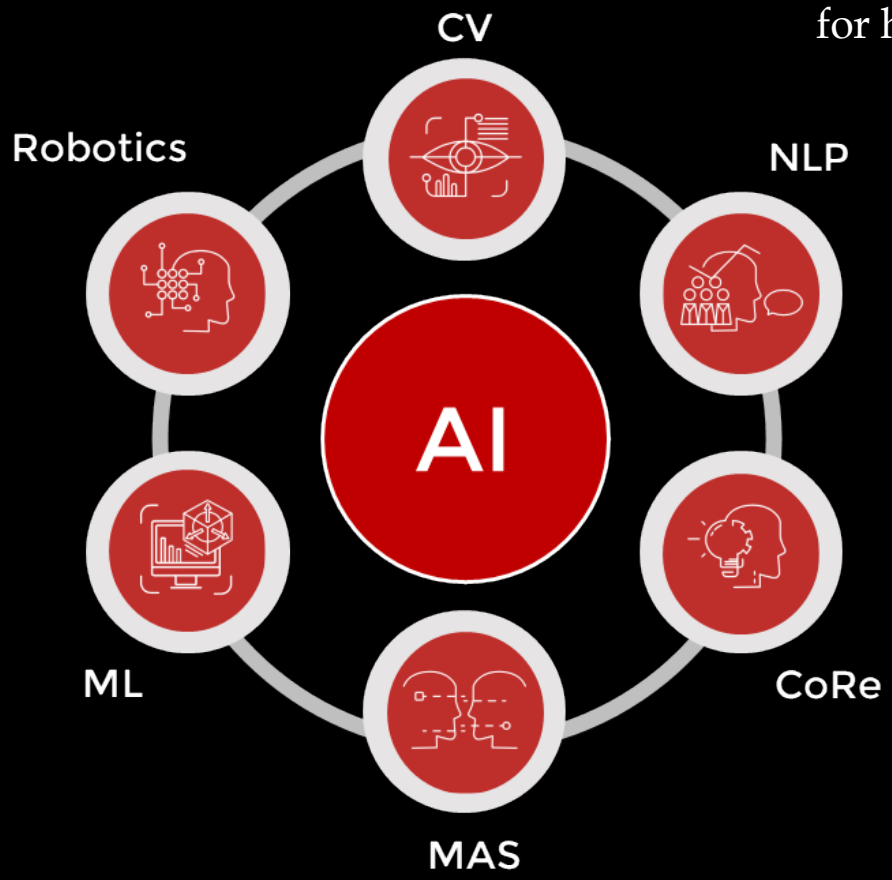
Chapter 1
Intelligence from scaling up vs growing up

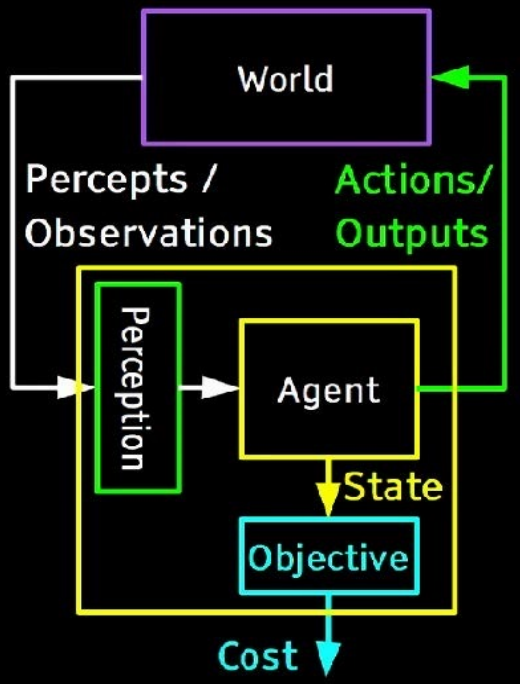

Human Intelligence
Artificial Intelligence

Artificial Intelligence

An attempt will be made to find how to make machines use language, form abstractions and concepts, solve kinds of problems now reserved for humans, and improve themselves.

John McCarthy



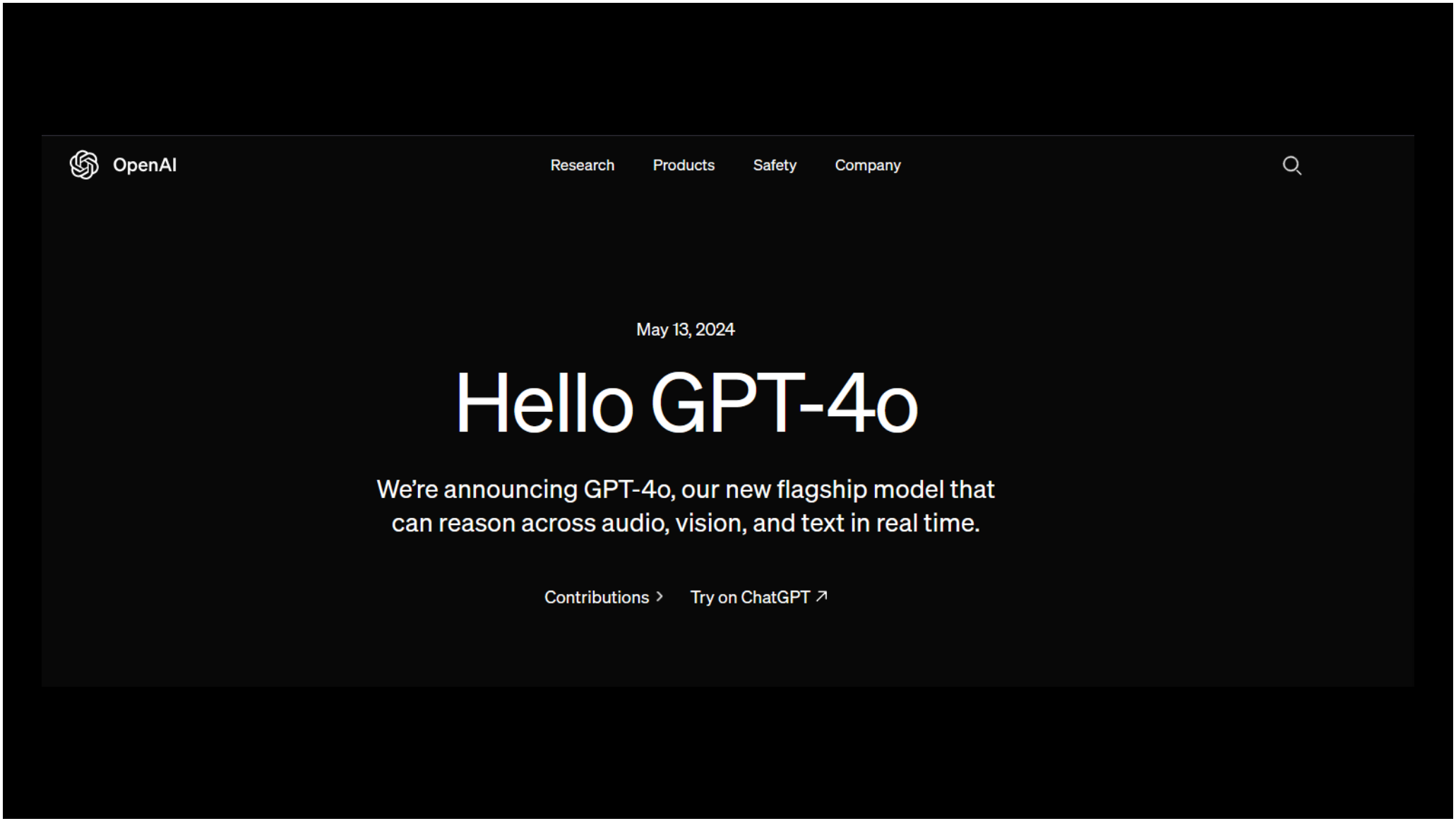


TECHNOLOGY

AlphaGo

AlphaGo mastered the ancient game of Go, defeated a Go world champion, and inspired a new era of AI systems.

Share

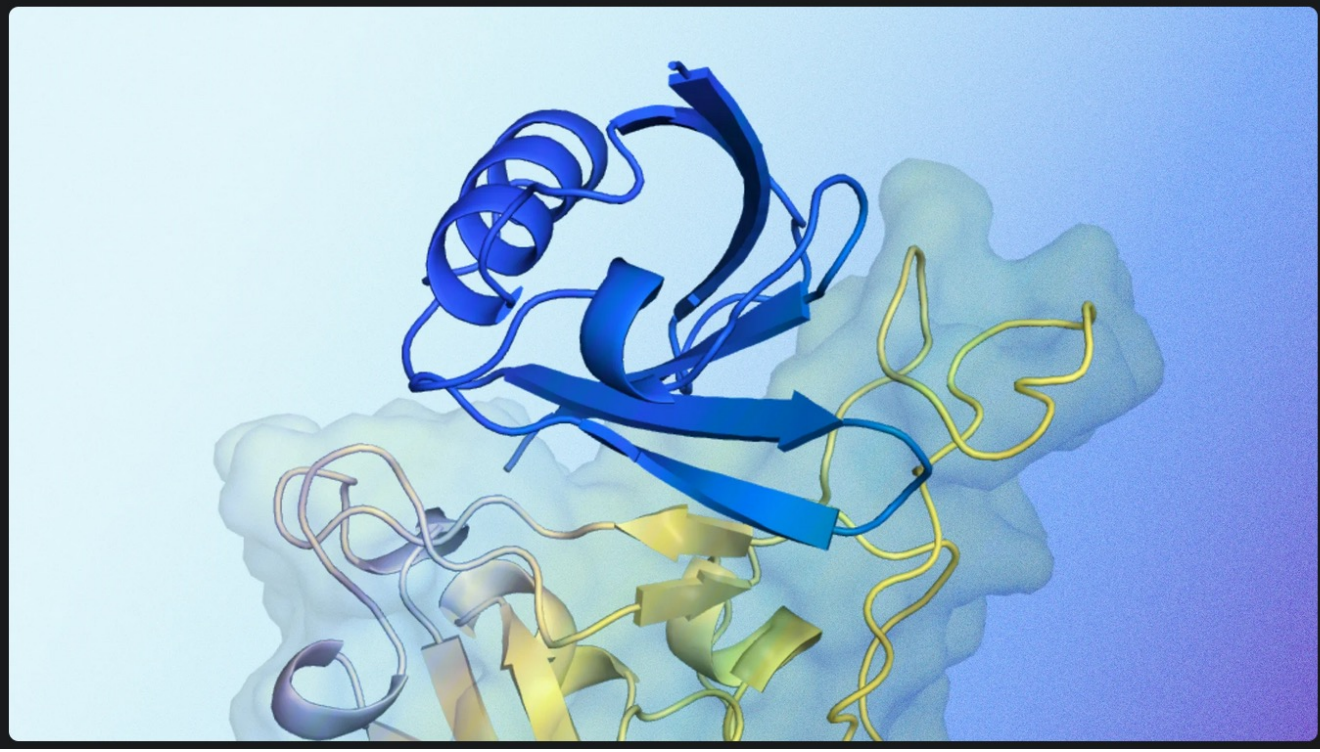

OpenAI
Research
Products
Safety
Company
May 13, 2024
Hello GPT-4o
We're announcing GPT-4o, our new flagship model that can reason across audio, vision, and text in real time.
Contributions
Try on ChatGPT

AlphaProteo generates novel proteins for biology and health research

5 SEPTEMBER 2024

Protein Design and Wet Lab teams

Share



OpenAI

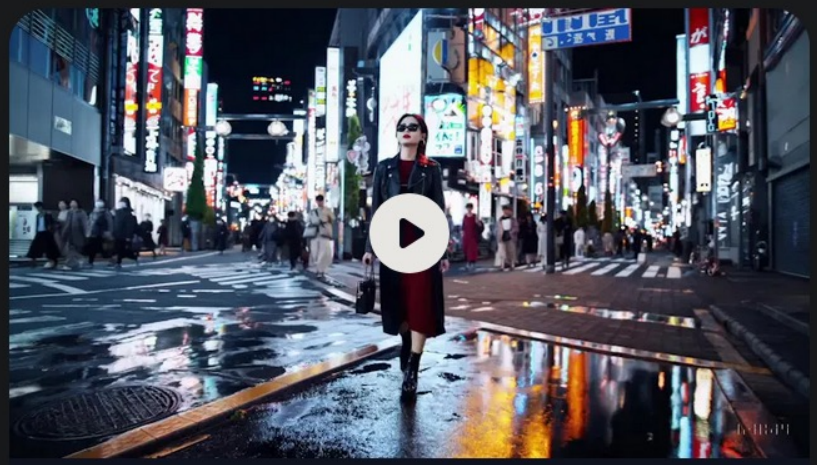
Research Products Safety Company

January 5, 2021

DALL·E: Creating images from text

We've trained a neural network called DALL·E that creates images from text captions for a wide range of concepts expressible in natural language.

English



Prompt: A stylish woman walks down a Tokyo street filled with warm glowing neon and animated city signage. She wears a black leather jacket, a long red dress, and black boots, and carries a black purse. She wears sunglasses and red lipstick. She walks confidently and casually. The street is damp and reflective, creating a mirror effect of the colorful lights. Many pedestrians walk about.

Join our alert list for Sora's upcoming launch details!

you@example.com

Join WaitList

Generate Your Ideas into Videos with Sora

Unlock the power of AI storytelling with Sora, OpenAI's revolutionary text-to-video tool.

Sora empowers creators, educators, and businesses alike to effortlessly bring their narratives to life. With just a few lines of text, you can generate stunning videos that captivate your audience, enhance your message, and elevate your digital presence.

Experience the seamless fusion of creativity and technology, and let Sora transform your imaginative concepts into visual masterpieces. Whether you're crafting educational content, marketing materials, or personal stories, Sora is your gateway to a world where your ideas are not just heard, but seen and felt.

PlayGround →

未来交通看
Waymo。

Ride with Waymo

全球首家自动驾驶约车服务。震撼推出，全天无休。

App Store 下载 | 去商店下载 Google Play

菲尼克斯

Anyone in the Metro Phoenix area can download and ride today. Just open the Waymo app and hail a car.

在菲尼克斯使用约车服务

旧金山

Anyone in San Francisco can download and ride today. Just open the Waymo app and hail a car.

在旧金山使用约车服务

洛杉矶

Anyone in LA County can ride today. Just download the Waymo app and hail a car.

在洛杉矶使用约车服务

奥斯汀 RIDE ON UBER

We've partnered with Uber to bring the magic of Waymo to Austin, only on the Uber app.

Learn more

Tesla Full Self Driving requires human intervention every 13 miles

It gave pedestrians room but ran red lights and crossed into oncoming traffic.

JONATHAN M. GITLIN – SEP 25, 2024 11:31 PM | 324

User

User

What does this image show?

Gemini 1.5 Pro 002
Nov, 2024

User

User

What does this image show?

Model

The image shows the interior of a Tesla vehicle, specifically the driver's seat area, with the dashboard, steering wheel, and central touchscreen display visible. The touchscreen displays a map/navigation screen, while the view through the windshield shows a highway. The accompanying text is a headline and subheading from an Ars Technica article discussing the limitations of Tesla's Full Self-Driving system.

Gemini 1.5 Pro 002
Nov, 2024

User

What does this image show?

Model 2.1s ⚠

The image shows the CAPTCHA "52gCsIh5Zb2". CAPTCHAs are used to verify that a user is human and not a bot.

User

What does this image show?

Model 2.1s ⚠

The image shows the CAPTCHA "52gCsIh5Zb2". CAPTCHAs are used to verify that a user is human and not a bot.

User

How confident are you that you are reading it correctly, on a scale of 0-100?

Model 3.1s

I'd say around 70-80%. The "g", "I", and "2" (at the end) are the ones I'm least certain about due to the distortion. While I *think* I'm reading them correctly, the noise makes it hard to be completely sure. The other characters seem relatively clear.

User

What does this image show?

Model 2.3s

The image appears to be a very low-resolution and noisy rendition of the word "CAPTCHA". It simulates the visual challenge aspect of a CAPTCHA test, designed to differentiate humans from bots.

User

What does this image show?

Model 2.3s

The image appears to be a very low-resolution and noisy rendition of the word "CAPTCHA". It simulates the visual challenge aspect of a CAPTCHA test, designed to differentiate humans from bots.

User

How confident are you that you are reading it correctly, on a scale of 0-100?

Model 3.1s

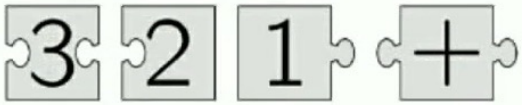
I'd say around 75-80. While the image is very noisy, the general shapes of the letters C, A, P, T, C, H, and A are discernible. The "P", "T", and second "C" are the clearest. There's definitely room for error given the extreme pixelation and noise, but the word "CAPTCHA" is the most likely interpretation.

Evaluating Large Vision-and-Language Models on Children's Mathematical Olympiads

MathKangaroo

Anoop Cherian[1] **Kuan-Chuan Peng**[1] **Suhas Lohit**[1] **Joanna Matthiesen**[2]
Kevin Smith[3] **Joshua B. Tenenbaum**[3]
[1]Mitsubishi Electric Research Labs, Cambridge, MA, [2]Math Kangaroo USA NFP
[3]Massachusetts Institute of Technology, Cambribdge, MA

Q: When you put the 4 puzzle pieces together correctly, they form a rectangle with a calculation on it. What is the result of this calculation?
A: 6 B: 15 C: 18 D: 24 E: 33



GPT-4o: To solve this puzzle, we need to arrange the 4 pieces (3, 2, 1, and +) to form a correct mathematical equation. Let's try arranging them and checking the validity of each possible equation: 1. 3 + 2 = 1 2. 3 + 1 = 2 3. 2 + 1 = 3 Let's check each possibility: 1. "3 + 2 = 1" is incorrect because 3 + 2 equals 5, not 1. 2. "3 + 1 = 2" is incorrect because 3 + 1 equals 4, not 2. 3. "2 + 1 = 3" is correct because 2 + 1 equals 3. Thus, we have: 2 + 1 = 3 So the correct equation formed by the puzzle pieces is: "2 + 1 = 3" Hence, the result of this calculation is: C: 18

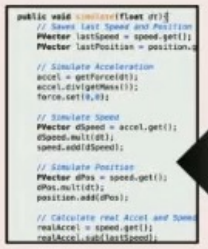
Gemini-Pro: 1. **Identify the Pieces:** The image shows four puzzle pieces, each with a part of a mathematical equation. 2. **Visualize the Rectangle:** Imagine fitting these pieces together to form a rectangle. The edges will help us connect the parts of the equation. 3. **Mentally Arrange:** Piece by piece, try to mentally arrange them. Notice that some pieces have parts of numbers, and others have operators (+, -). 4. **Form the Equation:** By carefully arranging the pieces, you can form the equation: ``` 18 + 6 ``` 5. **Calculate:** The result of the calculation 18 + 6 is 24. **D: 24**

Today's industry approach to building AI

EXTERNAL WORLD

Sense data

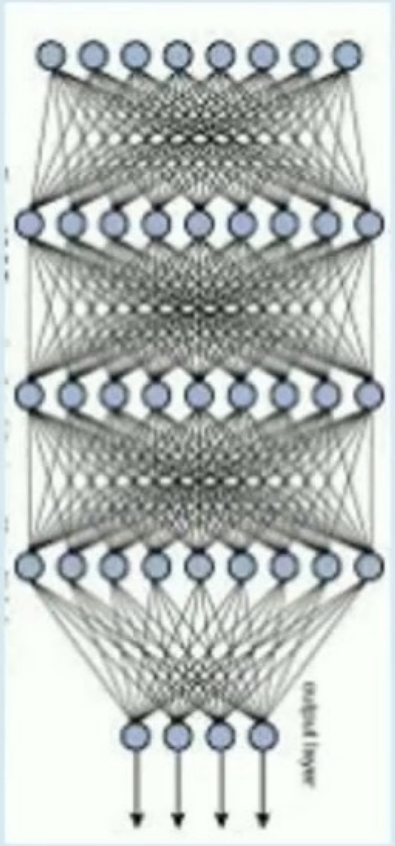
Underlying state



Perception

Action

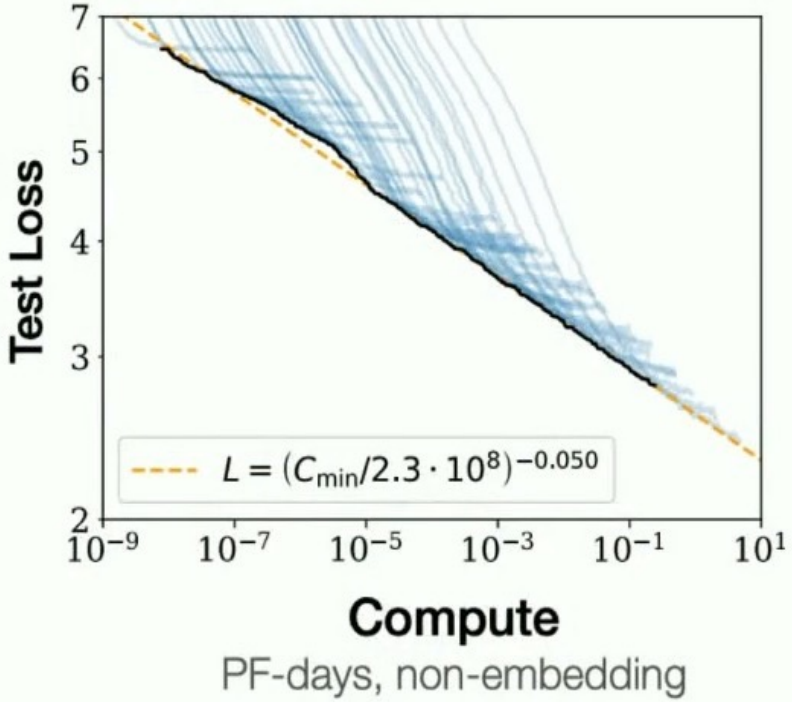
END-TO-END NEURAL NETWORK STACK



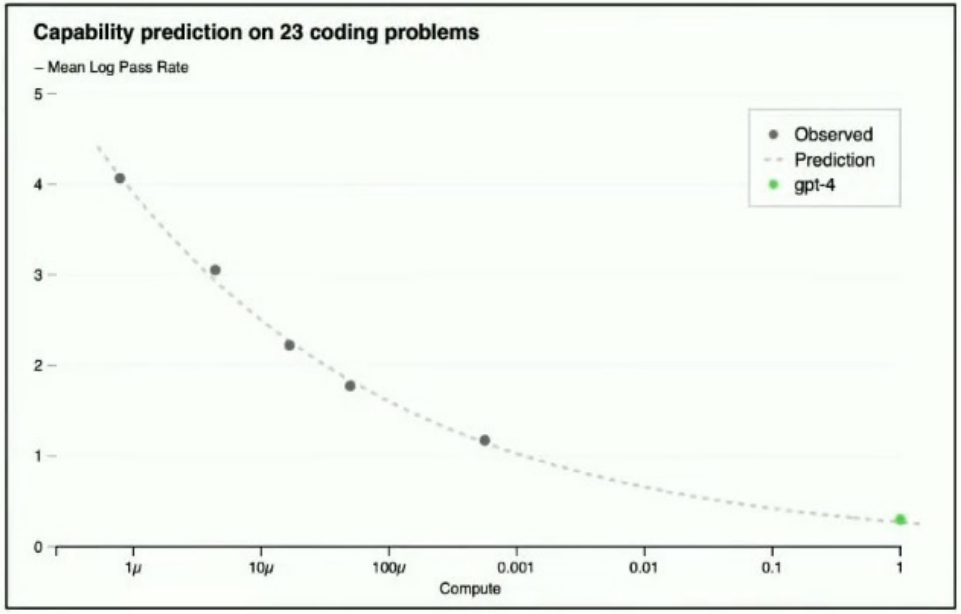
Scaling laws

"The point about this approach is that it scales beautifully. Basically you just need to keep making it bigger and faster, and it will get better. There's no looking back now."

Geoff Hinton, NY Times, 2012

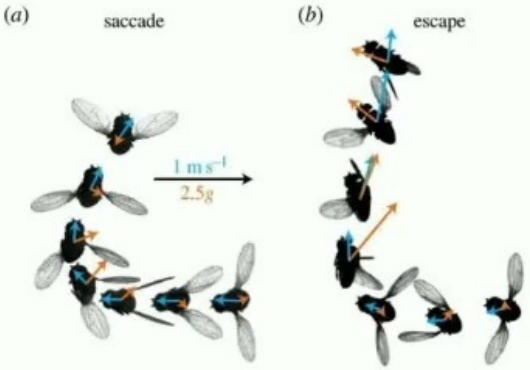


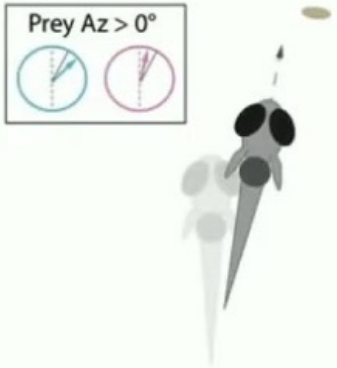
Scaling laws for neural language models (Kaplan et al., 2020)

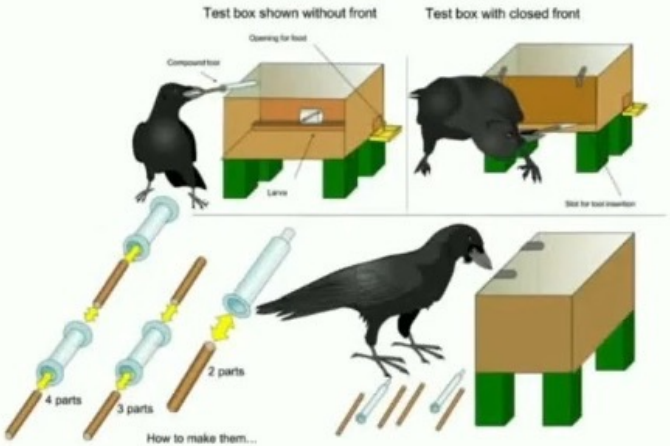


GPT4 technical report (OpenAI 2023)

Brains are world modeling machines, from the start

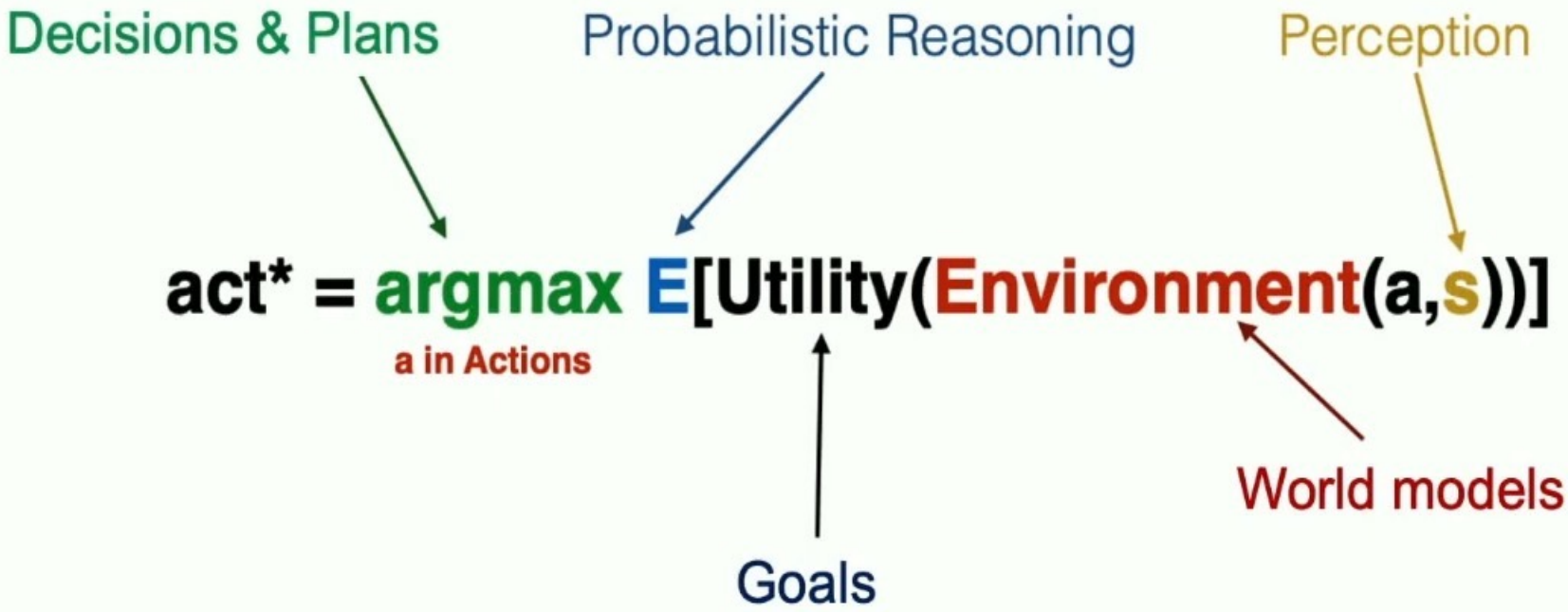






Intelligence starts with *Rationality*:

Using a model of the world, and your self in it, to make good guesses and good bets



Intelligence starts with *Rationality*:

Using a model of the world, and your self in it, to make good guesses and good bets

Decisions & Plans

Milliseconds
Seconds, Minutes,
Days, Years

Probabilistic Reasoning

Perception

$$act^* = \underset{a \text{ in Actions}}{\text{argmax}}\ E[Utility(Environment(a,s))]$$

Goals

Survival needs
Pleasures, Preferences
Values, Norms, Morality

World models

Intuitive physics
+ Models of other agents
and their world models...

The Developmental Scaling Route: A Roadmap

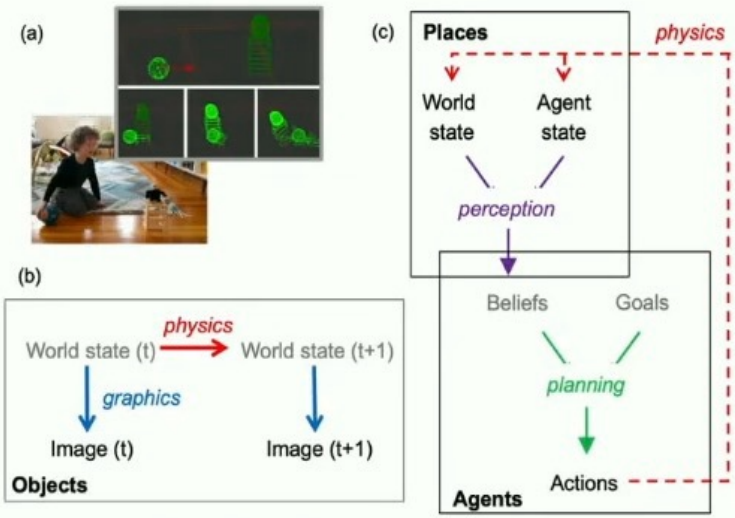
What is built in? And how do you learn the rest?

Built in:
The Core
Architecture

Representations:
function approximators,
probabilistic programs,
symbolic world models (physical, social) ...

Algorithms:
optimization
inference
decision making ...

Goals:
hedonic
epistemic
social ...



Learning:
Three Major
Stages of Human
Development

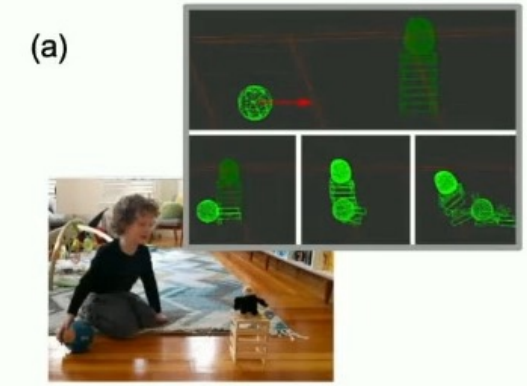
~ 0-1.5 years: learning within the core architecture, prior to language

~ 1.5-3 years: learning language, on top of the core architecture

~ 3 years and up: using language to learn everything else

The "Game engine in your head"

Mind Games: Game Engines as an Architecture for Intuitive Physics
(Ullman et al., *Trends in Cognitive Sciences* 2017)



The Naïve utility calculus: Computational principles
(Jara-Ettinger et al., *Trends in Cognitive Sciences* 2016)

Intelligence from scaling up versus *growing* up

Deep learning & Today's AI

- Intelligence is the **end result** of learning, and a surprising **emergent phenomenon** of simple "dumb" mechanisms operating on big data.
- Generalization depends on **similarity to training data**, and extends only **weakly and unpredictably beyond the data** distribution.
- Thinking – reasoning, planning, problem solving, judgment, social and moral cognition – **derives from language** and requires **carefully cleaned and curated language data**.

Human minds & Tomorrow's AI

- Intelligence is the **foundation** of learning, what's **present from the start**, and a **core built-in property** that lets us learn so much, from such small data.
- Generalization depends on **mental models of the world**, designed to extend **far beyond the data** distribution.
- Thinking **precedes** and **enables language,** and provides the basis for constructing language so robustly and resiliently.

Chapter 2

Why building a human-like AI is hard and the big questions

Why is AI hard?

What this image can tell us about the scene?

Rainy Day
Street Market
People/role
Interaction
Car/fruit/shelter
Distance
Height of the camera

Extremely difficult
# of pixels: 8.2M

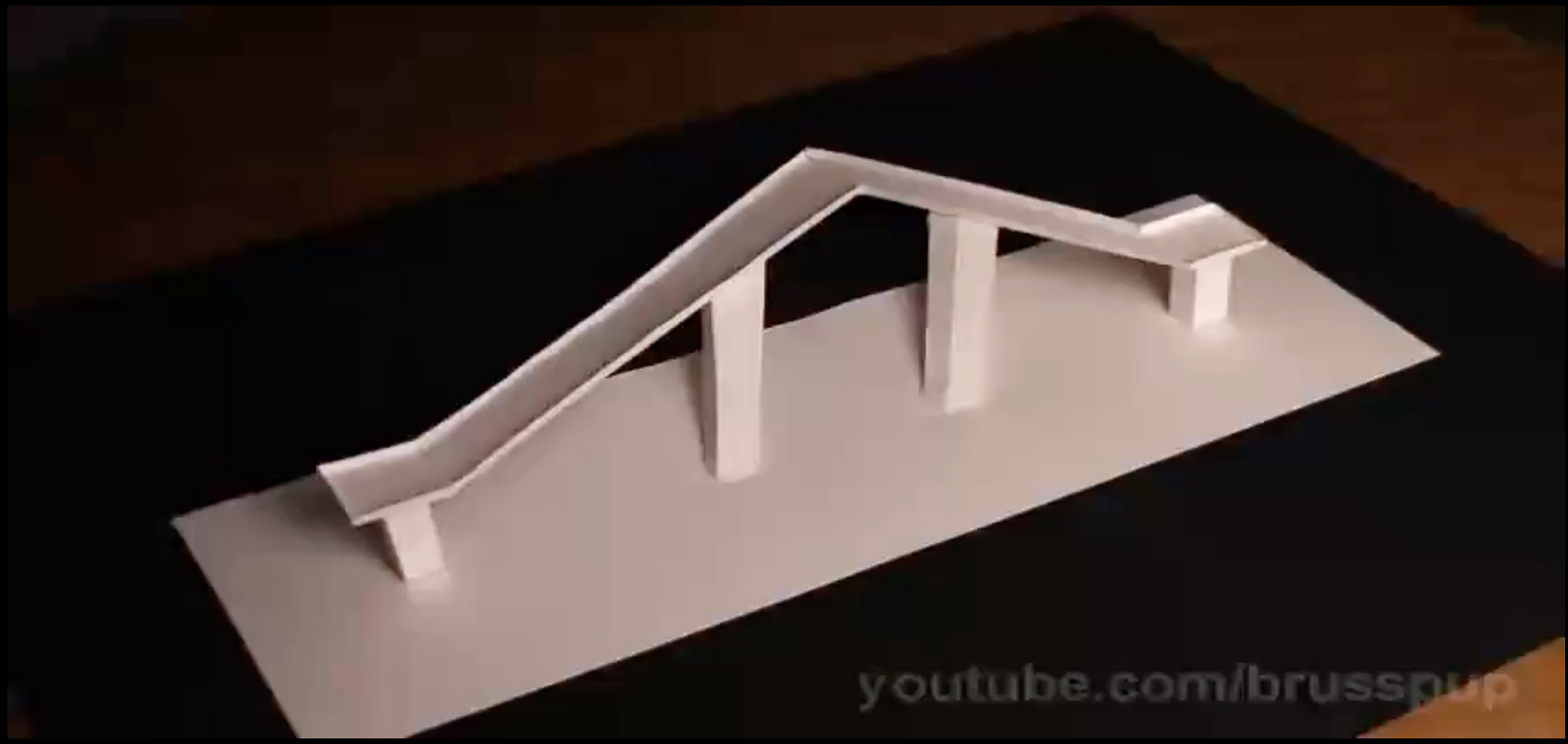
这张图片展示了一个户外市场或集市的场景，看起来可能是在热带或亚热带地区。以下是图片中的主要内容：

1. 左侧是一个水果摊位，展示着各种鲜艳的热带水果，如木瓜、芒果和西瓜等。
2. 摊位前站着两个人，一位穿着鲜艳的橙红色上衣，另一位穿着灰色的衣服。
3. 右侧有一个遮蓬下的区域，那里有几个小摊位和一些坐着的顾客。
4. 整个市场被绿树环绕，给人一种郁郁葱葱的热带感觉。
5. 天气似乎在下雨，可以看到雨水从屋顶上滴落。
6. 地面是碎石铺就的，显得有些潮湿。
7. 远处可以看到一些停泊的汽车。
8. 整体氛围显得悠闲而友好，是一个典型的社区市场场景，人们在此购物、社交和放松。

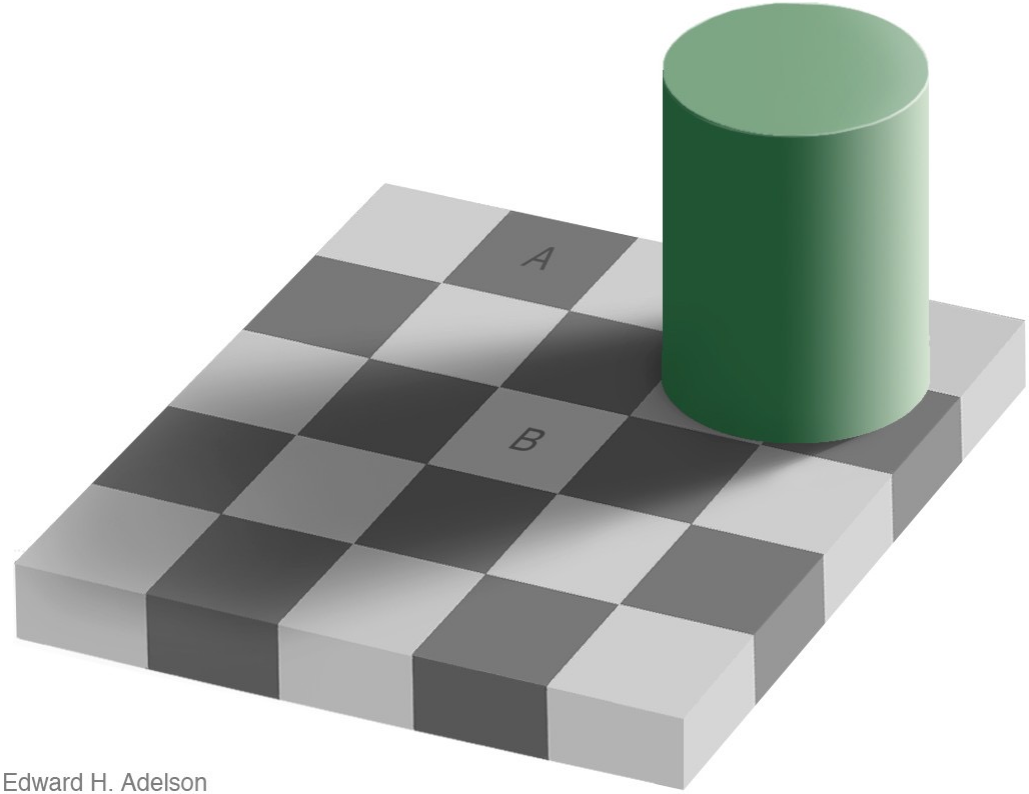
这个市场似乎融合了当地的特色和社区氛围，提供了新鲜的农产品和其他商品，同时也是人们聚会交流的场所。

Why is AI hard?

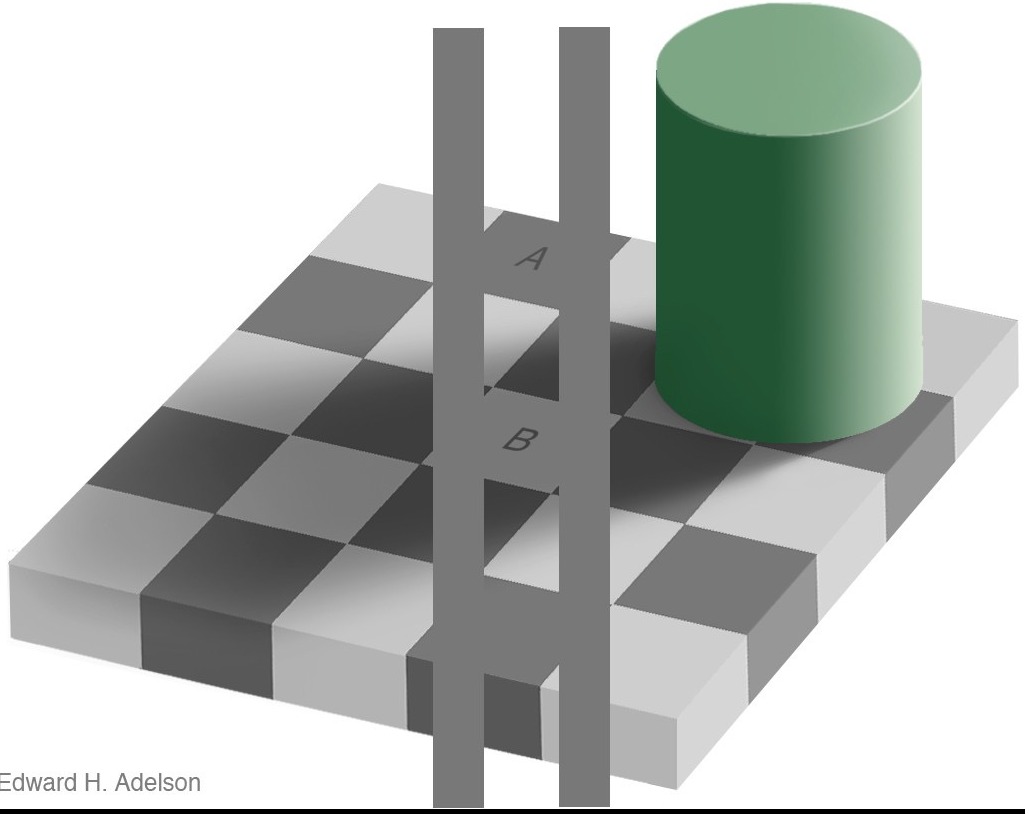
Perspective Illusion



Why is AI hard?



Edward H. Adelson



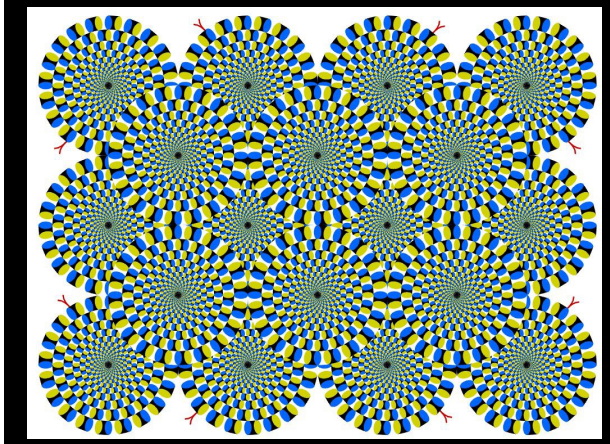
Edward H. Adelson

http://persci.mit.edu/gallery/checkershadow

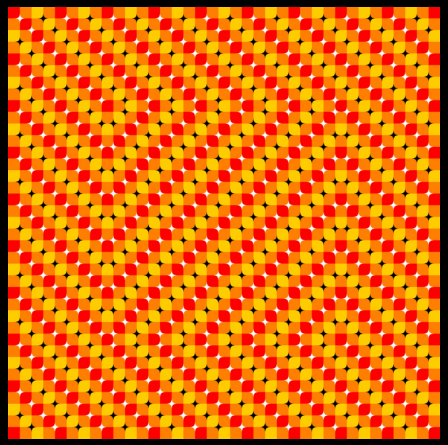
Perception vs. measurement

Why is AI hard?

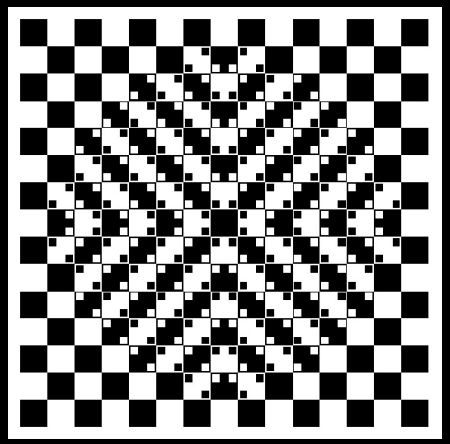
http://www.ritsumei.ac.jp/~akitaoka/index-e.html



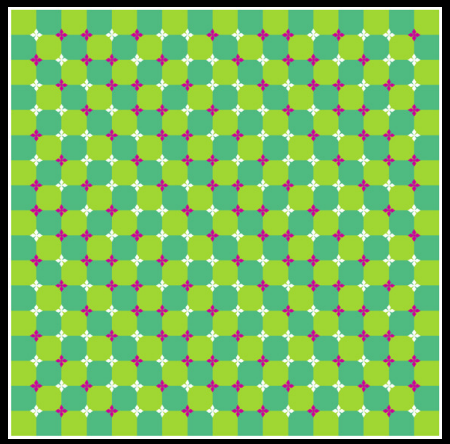
"Rotating snakes"
Circular snakes appear to rotate 'spontaneously'



"The autumn color swamp"
The inset appears to move.

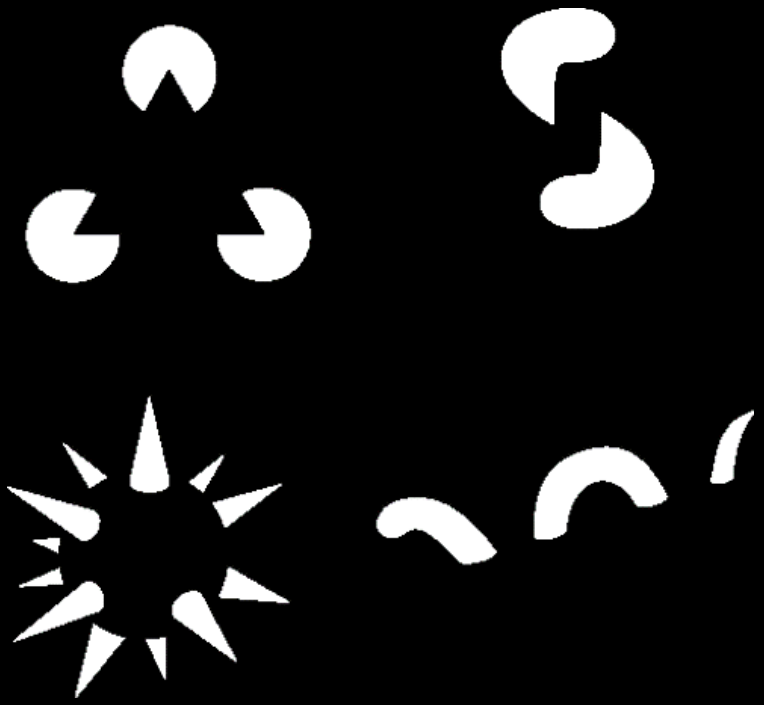


"A bulge"
The floor appears to bulge out, though this image consists of only squares.



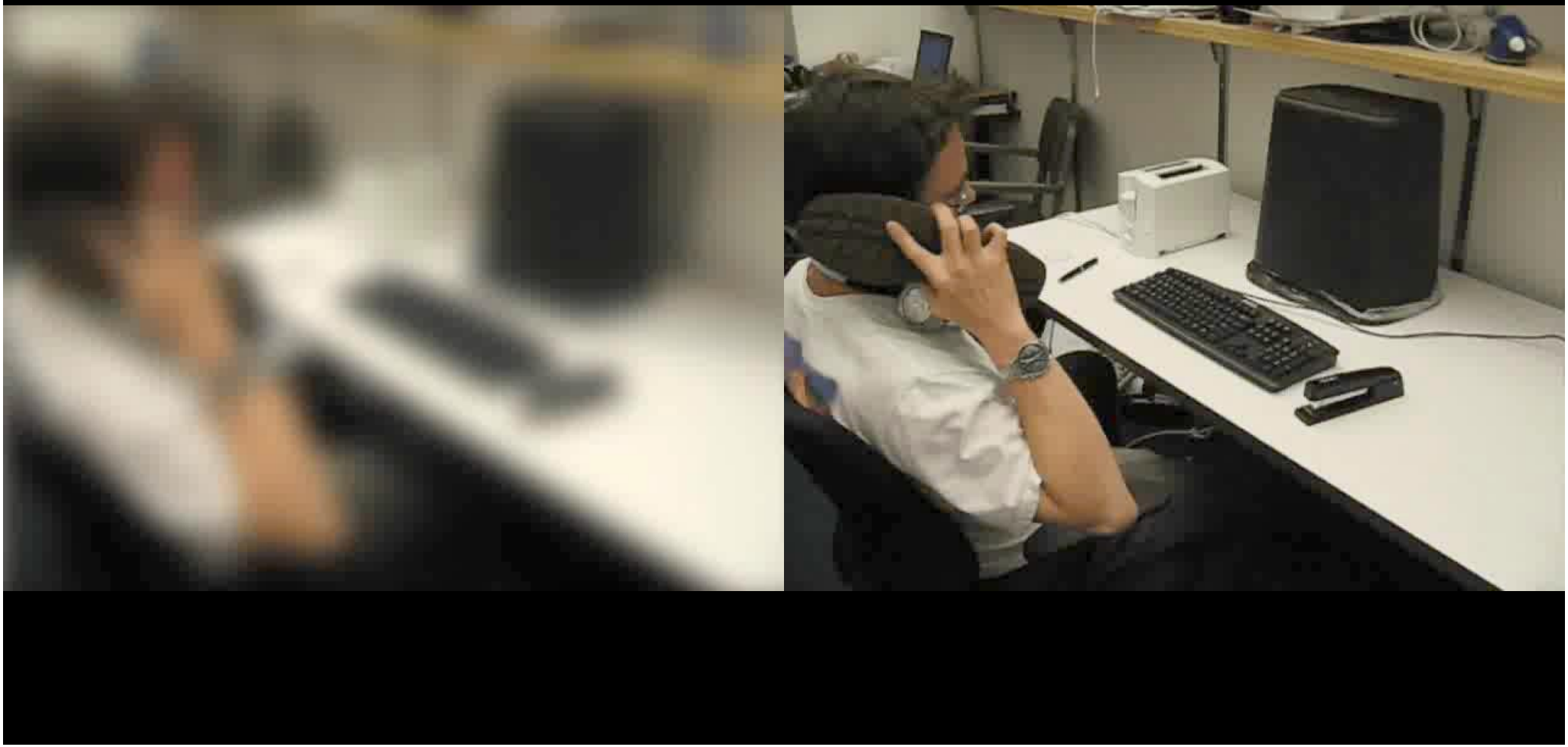
"Primrose's field"
This checkered background consists of squares but appears to wave. In addition, this figure also shows a waving motion illusion.

Why is AI hard?



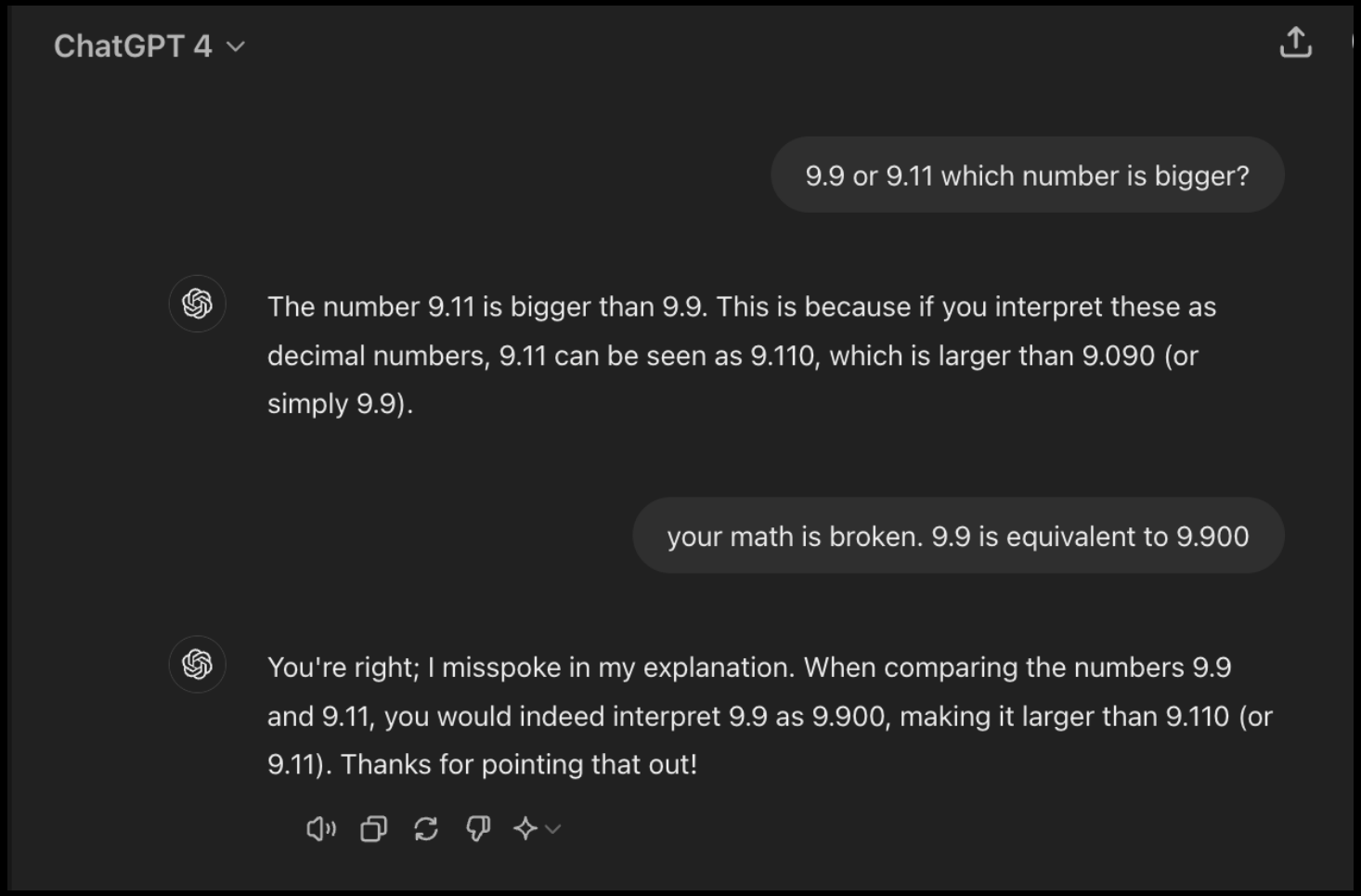
Kanizsa, Gaetano. "Subjective contours." Scientific American 234.4 (1976): 48-53.

Why is AI hard?

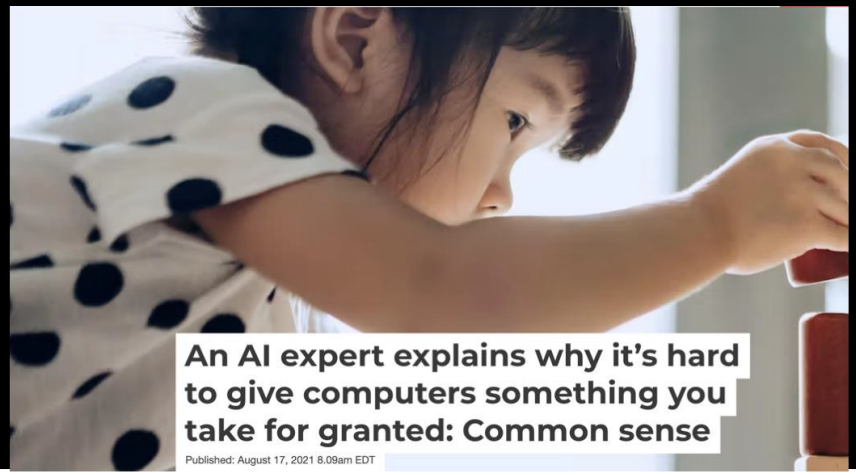


Why is AI hard?

Why is AI hard?



GPT 3 in 2020

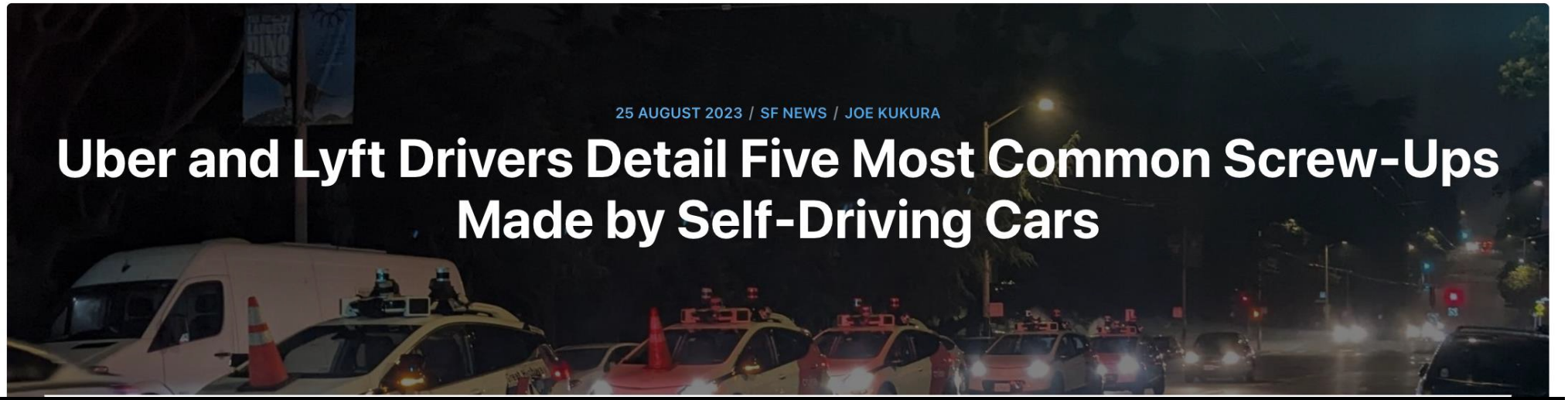


Modern AI is designed to tackle highly specific problems, in contrast to common sense, which is vague and can't be defined by a set of rules. Even the latest models make absurd errors at times, suggesting that something fundamental is missing in the AI's world model. For example, given the following text:

"You poured yourself a glass of cranberry, but then absentmindedly, you poured about a teaspoon of grape juice into it. It looks OK. You try sniffing it, but you have a bad cold, so you can't smell anything. You are very thirsty. So you"

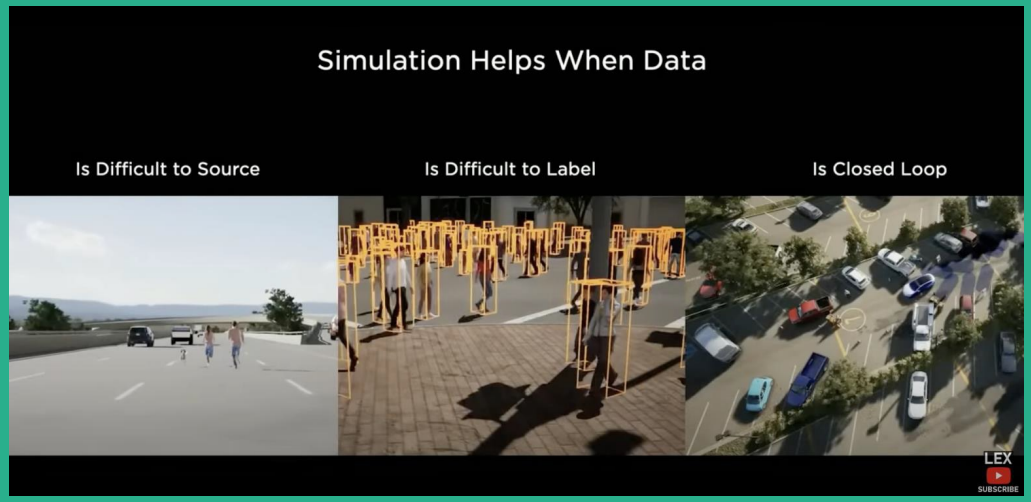
the highly touted AI text generator GPT-3 supplied

"drink it. You are now dead."



25 AUGUST 2023 / SF NEWS / JOE KUKURA

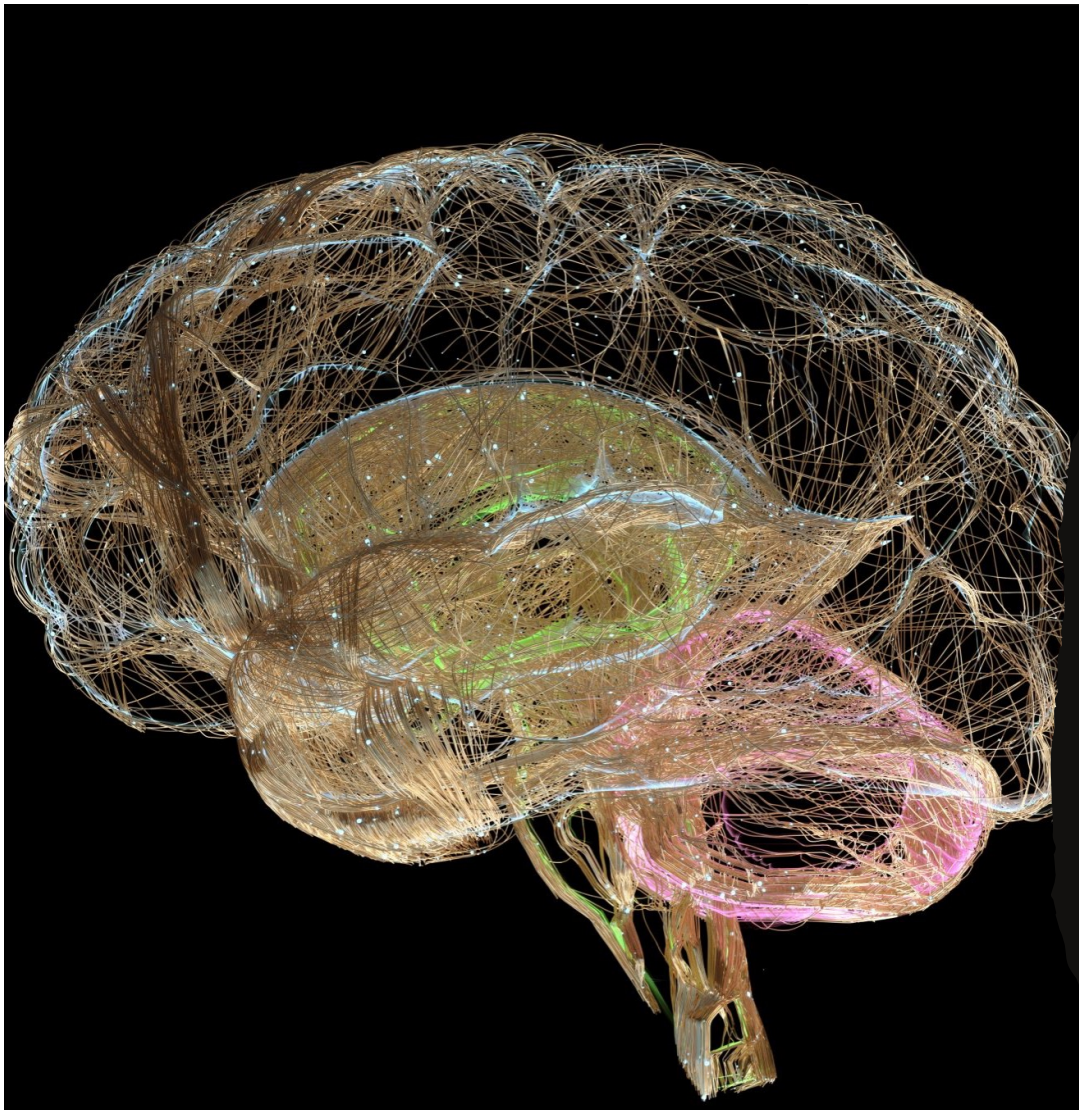
Uber and Lyft Drivers Detail Five Most Common Screw-Ups Made by Self-Driving Cars


Simulation Helps When Data
Is Difficult to Source
Is Difficult to Label
Is Closed Loop
LEX
SUBSCRIBE

The Big Questions

- **Human Intelligence:**
  - What makes human intelligence so robust?
  - What kind of computer is the human brain/mind?
- **Learning Mechanisms:**
  - How does the human mind learn so much from so little?
  - What is built in, and how do you learn the rest?
- **Generalization & Meta-Learning:**
  - How can the same intelligent system learn to do so many things so well, without having to be specially engineered or extensively trained?
  - Can you learn to be intelligent (in general, not just for a specific task)? Can you learn to learn?
- **Machine Intelligence Implications:**
  - How could machines achieve similar capabilities?

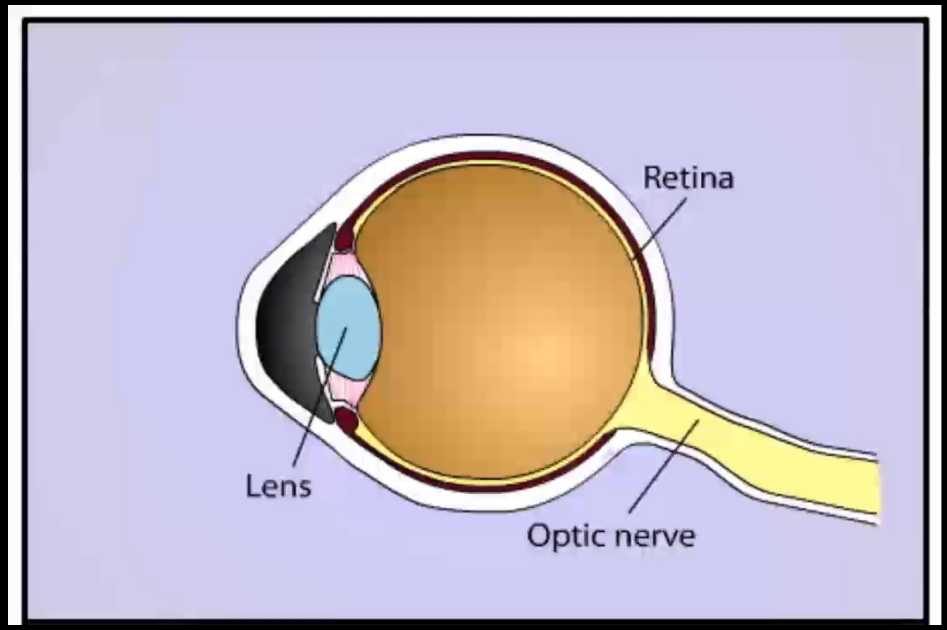
Chapter 3
Let the nature speak



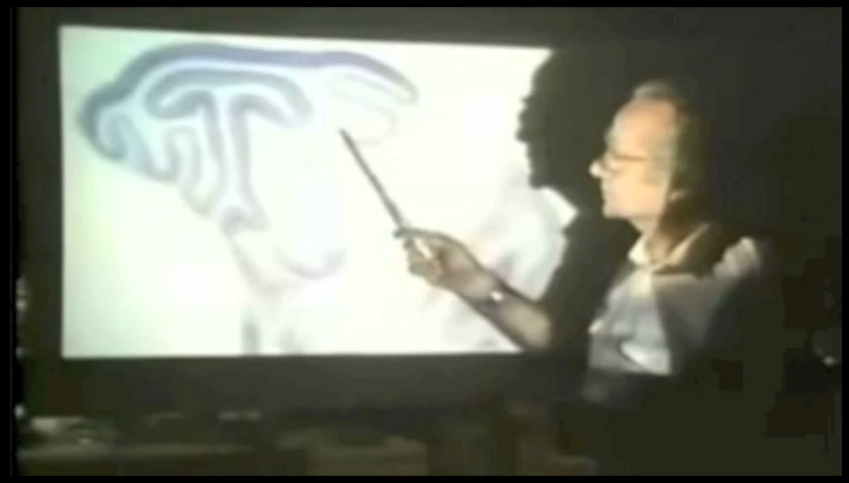
AI and Neuroscience

- Human intelligence is based on the brain.
- Brain consists of neurons.
- Hence, neural networks will rule.

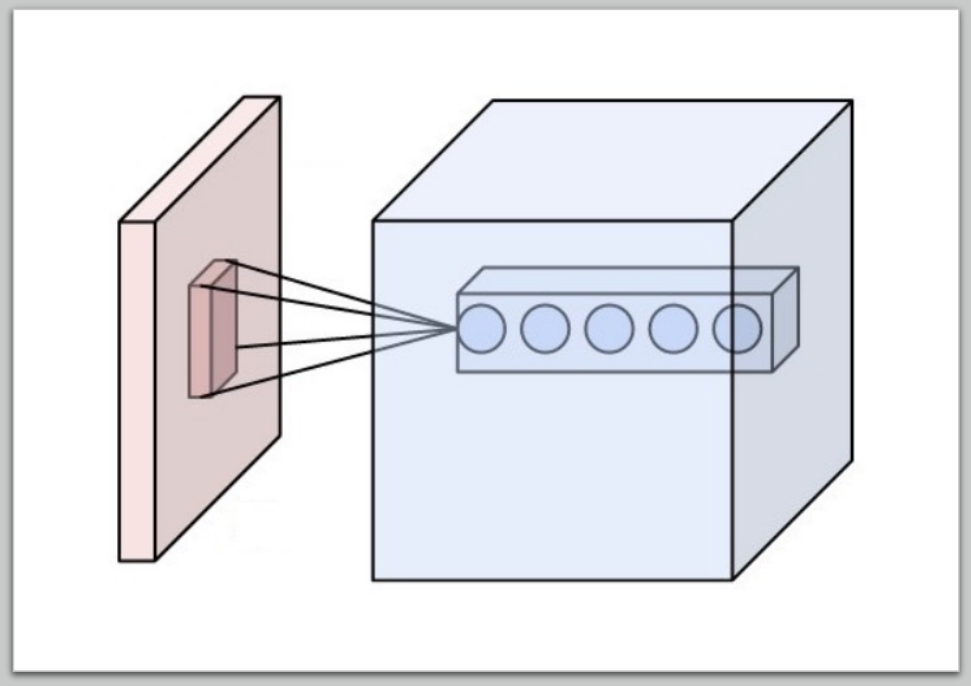
Receptive Field: Respond Locally

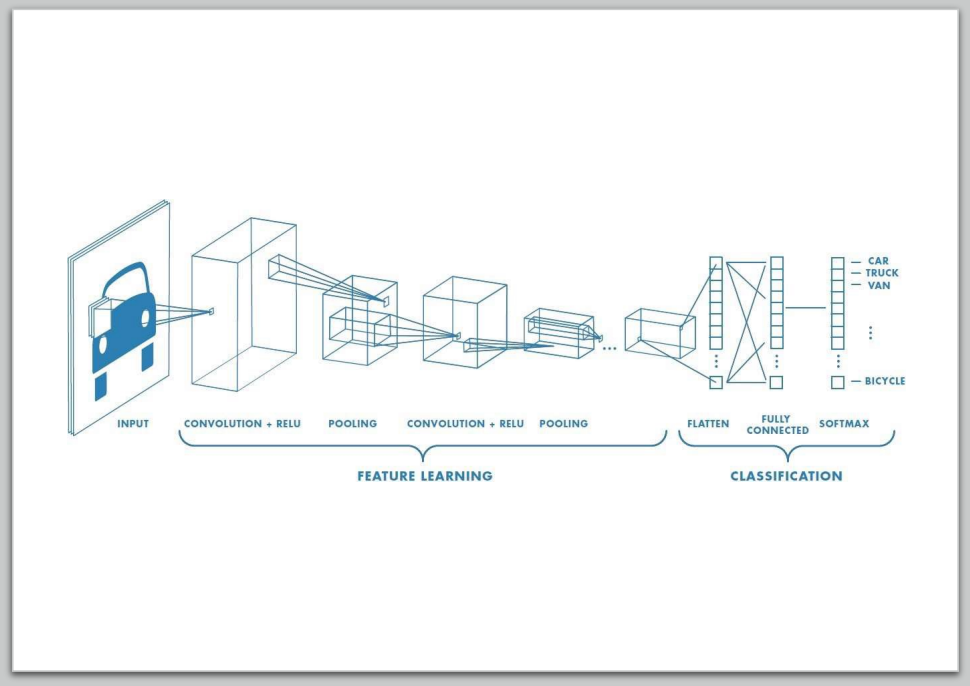


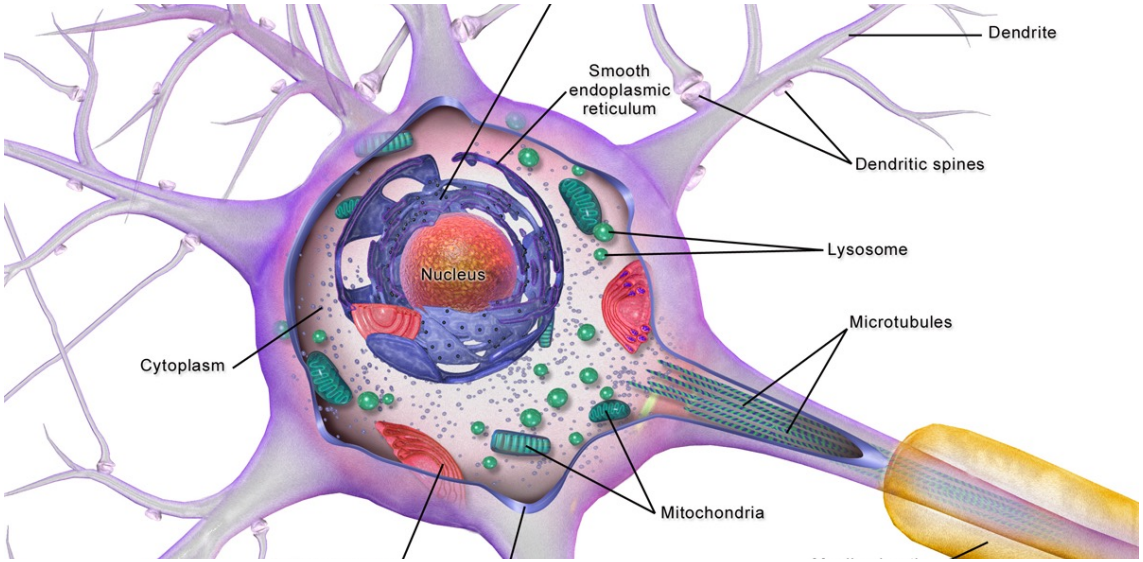
Application: Edge Detection

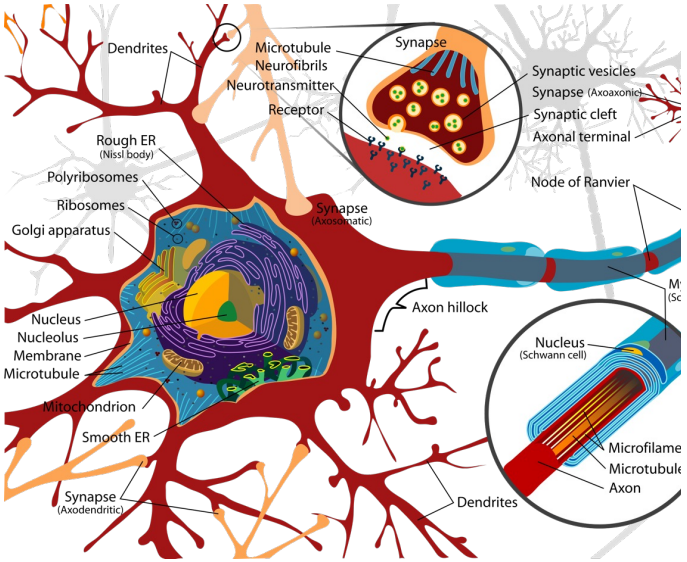


Neurons of a convolutional layer (blue), connected to their receptive field (red)

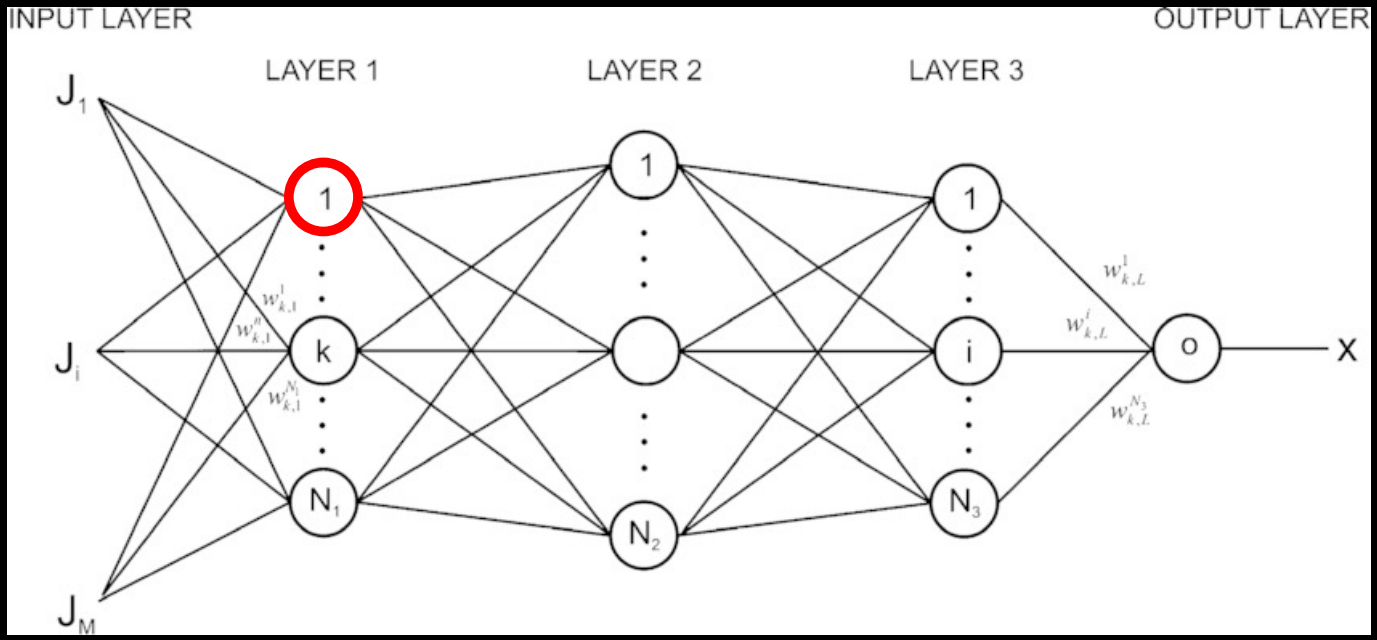


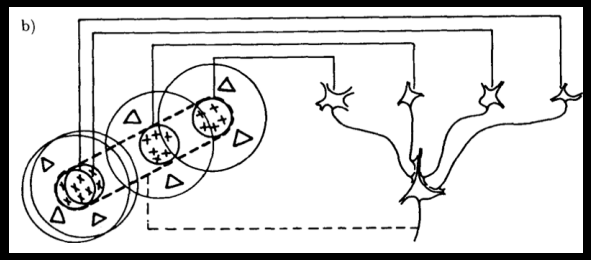


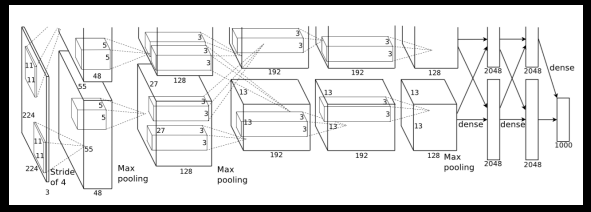


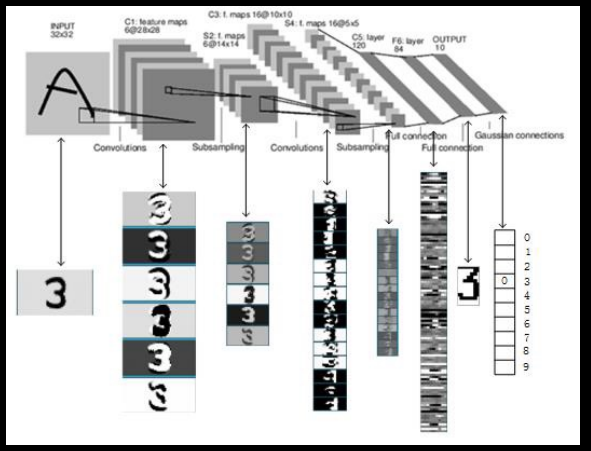


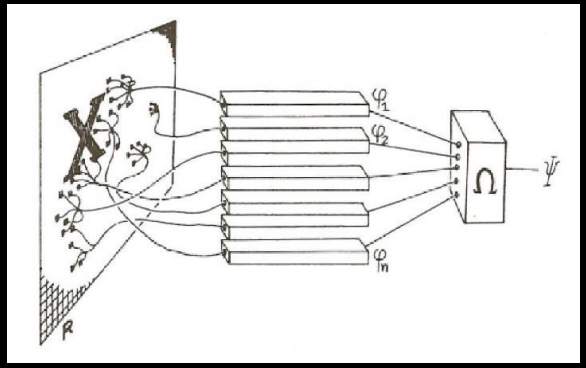
A single neuron


INPUT LAYER
OUTPUT LAYER
LAYER 1
LAYER 2
LAYER 3
J1
Ji
JM
1
k
N1
1
N2
1
i
N3
o
X









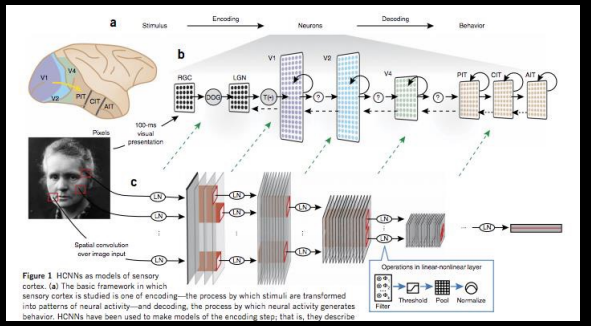


Figure 1 HCNNs as models of sensory cortex. (**a**) The basic framework in which sensory cortex is studied is one of encoding—the process by which stimuli are transformed into patterns of neural activity—and decoding, the process by which neural activity generates behavior. HCNNs have been used to make models of the encoding step; that is, they describe

How neurons solve problems?

The Challenge:

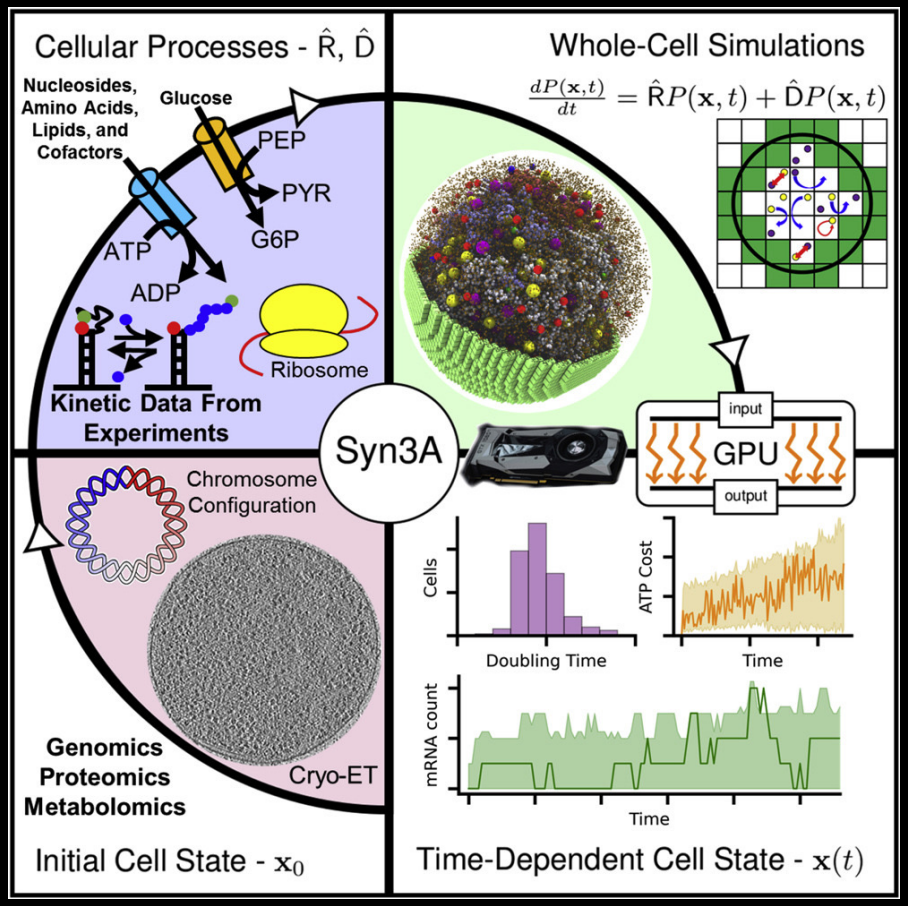
- We know surprisingly little about how neurons actually solve problems

A New Perspective:

- Think of a neuron as a micro-computer
- Focus on computational principles, not just biological details

Research Approach:

- Don't be constrained by traditional neuroscience boundaries
- Evaluate models based on function, not biological plausibility alone



Fundamental behaviors emerge from simulations of a living minimal cell

Zane R. Thornburg [1] · David M. Bianchi [1] · Troy A. Brier [1] · … · Hamilton O. Smith [3] · John I. Glass [3] · Zaida Luthey-Schulten [1,4,5,6] … Show more

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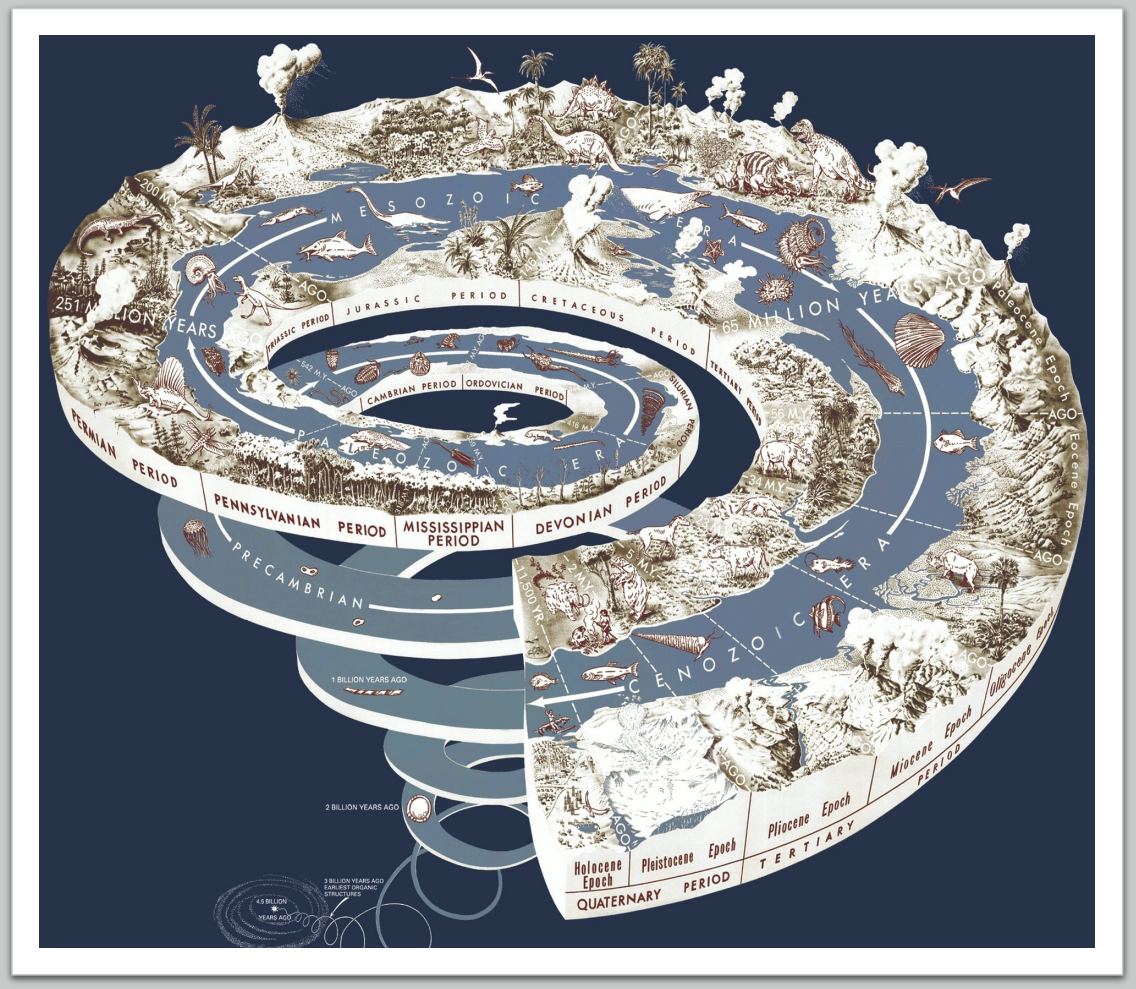
Show Outline

Highlights

- 3D spatial resolution of a fully dynamical whole-cell kinetic model
- Detailed single-reaction, single-cell accounting of time-dependent ATP costs
- Genome-wide mRNA half-lives emerge from length-dependent kinetics and diffusion
- Connections among metabolism, genetic information, and cell growth are revealed

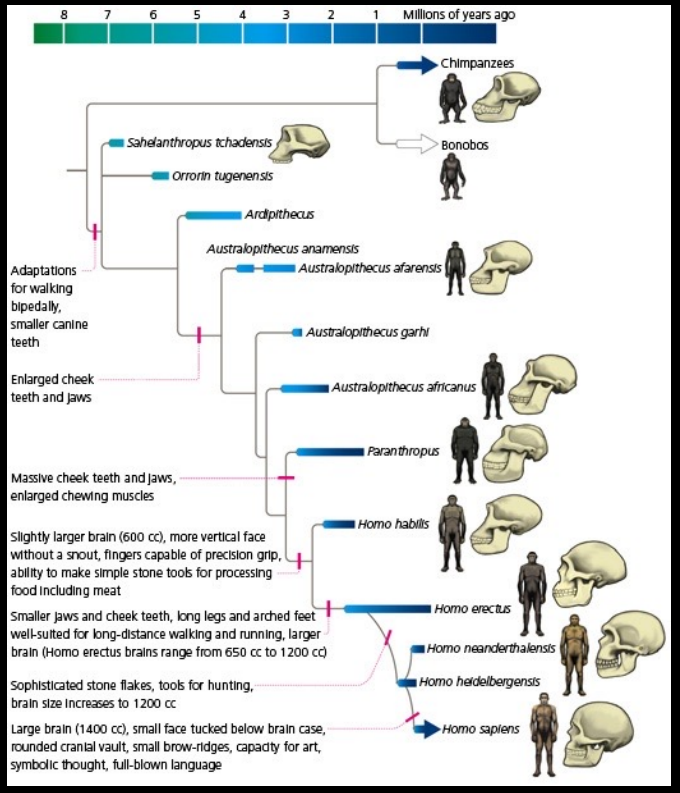
Summary

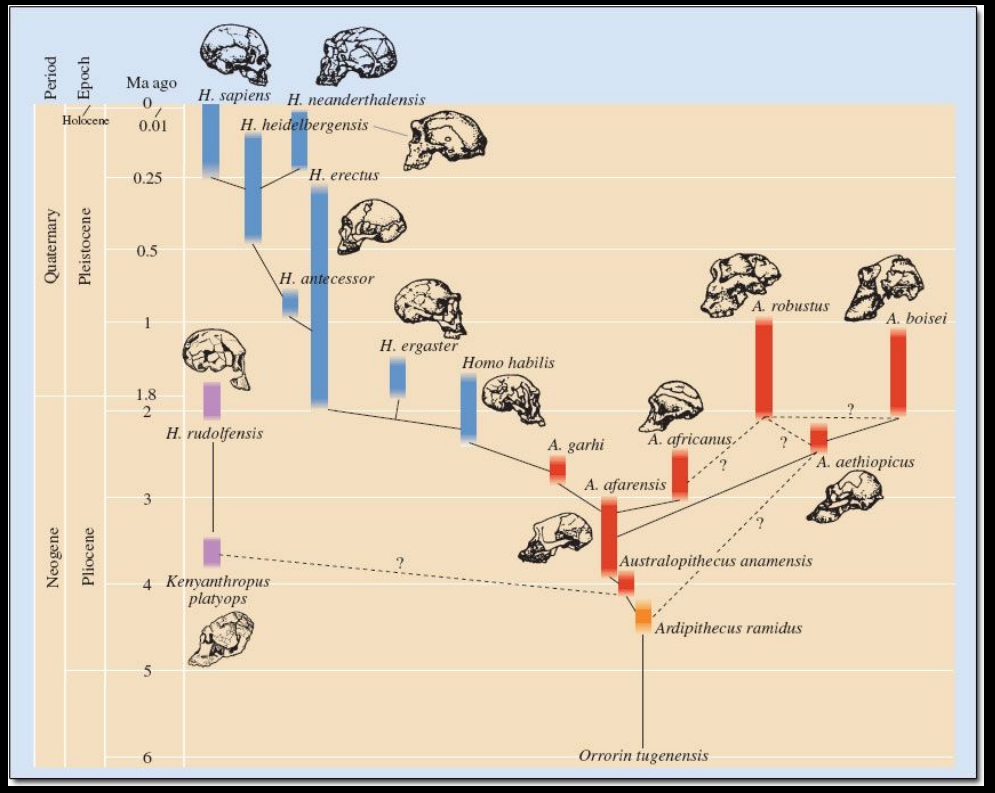
We present a whole-cell fully dynamical kinetic model (WCM) of JCVI-syn3A, a minimal cell with a reduced genome of 493 genes that has retained few regulatory proteins or small RNAs. Cryo-electron tomograms provide the cell geometry and ribosome distributions. Time-dependent behaviors of concentrations and reaction fluxes from stochastic-deterministic simulations over a cell cycle reveal how the cell balances demands of its metabolism, genetic information processes, and growth, and offer insight into the principles of life for this minimal cell. The energy economy of each process including active transport of amino acids, nucleosides, and ions is analyzed. WCM reveals how emergent imbalances lead to slowdowns in the rates of transcription and translation. Integration of experimental data is critical in building a kinetic model from which emerges a genome-wide distribution of mRNA half-lives, multiple DNA replication events that can be compared to qPCR results, and the experimentally observed doubling behavior.



AI and Evolution

- Life is the outcome of evolution.
- Evolution is essentially a type of learning and optimization
- Hence, learning from scratch, end-to-end.





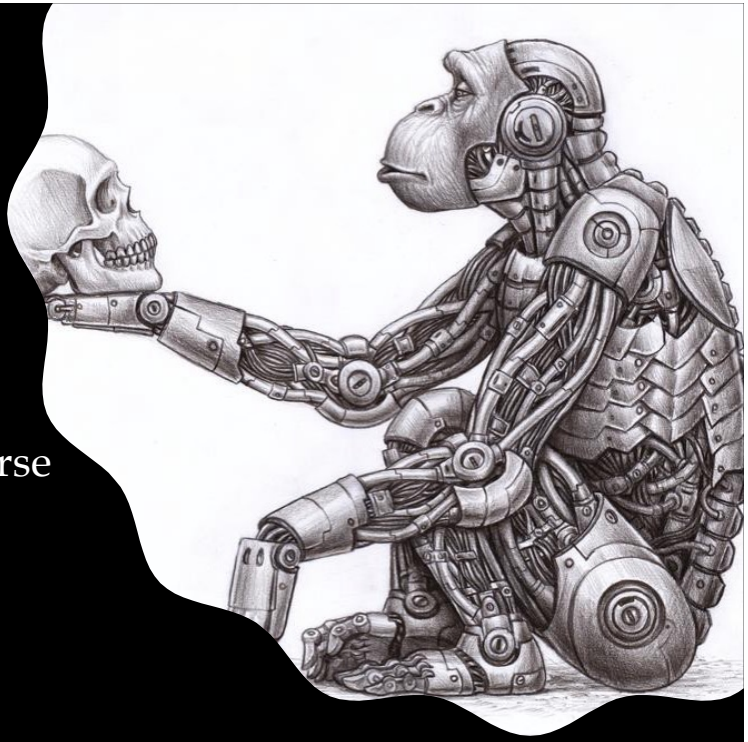
Evolution with many dead ends

due to end-to-end

Evolved Virtual Creatures
Karl Sims

Just let the nature speak

- Reinforcement Learning:
  - Attempts within-lifetime learning
  - But suffers from state explosion and dimensionality curse
  - Over-engineered for simple problems
- Genetic Algorithms:
  - Mimic evolution but operate "between lives”
  - Rely heavily on engineered heuristics
  - Miss the key: learning within a lifetime
- The Common Problem:
  - Both approaches over-constrain the learning process
  - Nature works with fewer assumptions and more flexibility



Summary: What else do we need?

- Intelligence is not fundamentally about recognizing patterns, approximating functions and reinforce based on rewards
- **True Intelligence = World Modeling:**
  - Core capability: Modeling the world and ourselves in it
  - Purpose: Using models to guide actions and change the world
- **The Complete Intelligence Stack:**
  - Level 1 - Understanding: Explaining what we observe VLM
  - Level 2 - Imagination: Envisioning unseen possibilities Video Generation
  - Level 3 - Planning: Making plans to realize these possibilities VLA, Humanoid
  - Level 4 - Learning: Building new models from few experience and generalize to new problems Age: 0-3
  - Level 5 - Collaboration: our minds and sharing our models , to cooperate on joint goals, learn socially, and grow knowledge culturally across generations. Age: 3-7

Chapter 4
Different computing problems

Cognitive Science

- The study of intelligent systems
- Cognition as information processing

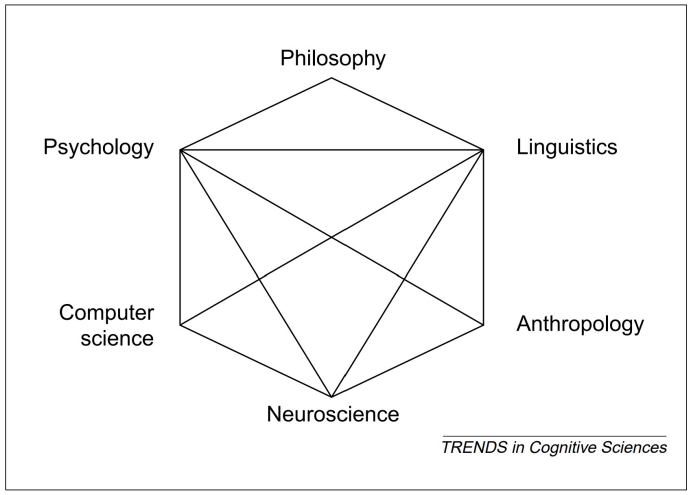
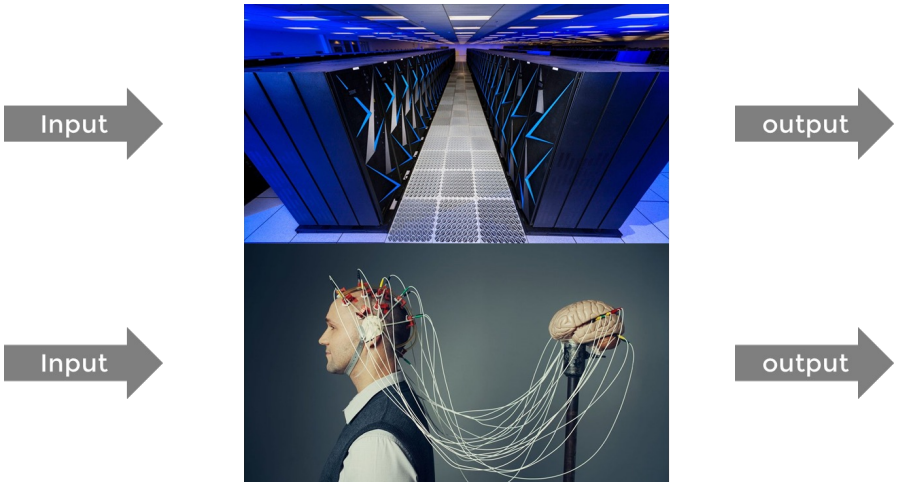
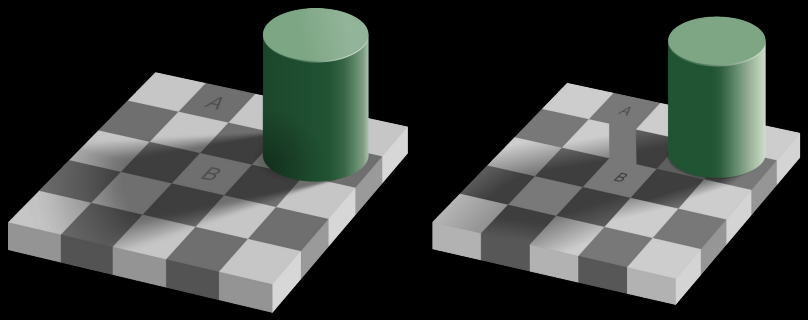


Fig. 1. Cognitive science in 1978. Each line joining two disciplines represents interdisciplinary inquiry that already existed in 1978.



Generative Perception and Cognition

Perception and cognition are active construct and reasoning of the external world



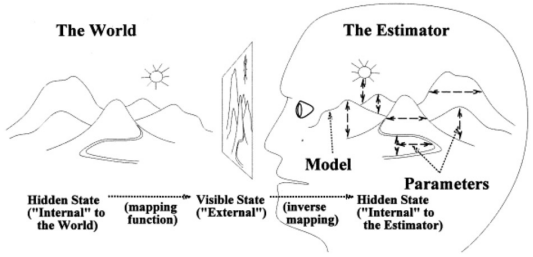
Perception is unconscious inference

Hermann von Helmholtz
(1821–1894)

$$S \rightarrow X \rightarrow \hat{S}$$

posterior likelihood prior

$$P(S|X) = \frac{P(X|S)P(S)}{P(X)}$$



Thomas Bayes
(1702-1761)

Goals

Cognition

Explain human cognition (and behavior) in terms of the underlying computation

Computation

Gain insight into how to solve some challenging computational problems

Easy (relatively)

World chess champion Garry Kasparov (left) playing against IBM's supercomputer Deep Blue in 1996 during the ACM Chess Challenge in Philadelphia.

Easy (relatively)

Vladimir Kramnik of Russia, the chess world champion, playing his sixth and last match today against Deep Fritz, a computer program, in Bonn, Germany.

The New York Times

Once Again, Machine Beats Human Champion at Chess

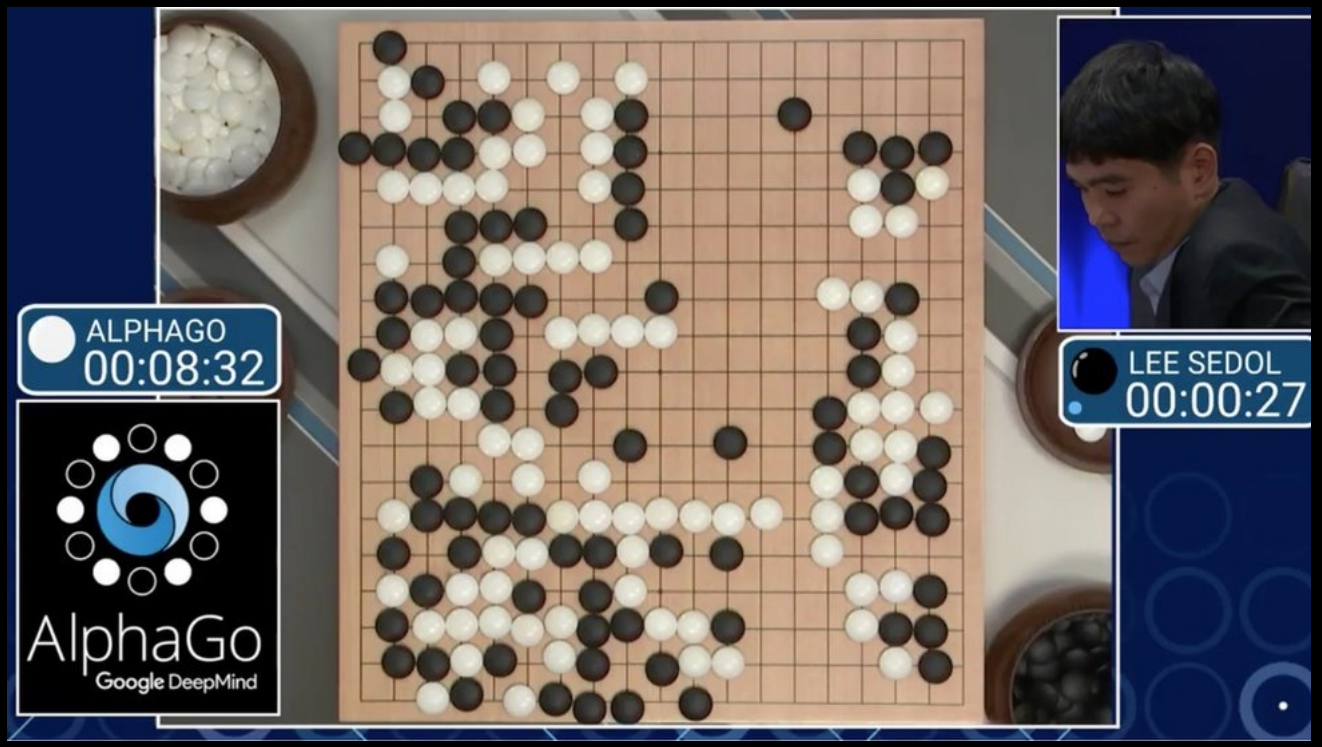
Give this article

By Dylan Loeb McClain

Dec. 5, 2006

In the continuing quest to see if humans can outpace their electronic creations, the humans have lost another, perhaps decisive, round.

Easy (relatively)



Google's AlphaGo beats Go master Lee Se-dol

Easy (relatively)

No limit: AI poker bot is first to beat professionals at multiplayer game

Computational Problems

arithmetic, algebra, chess

learning and using language

sophisticated senses: vision, hearing

similarity and categorization

representing the structure of the world

scientific investigation

Computational Problems

- **Easy** computational problems: deduction and search
  - Arithmetic, algebra, chess
- We know what the underlying formal system should be for these problems
  - We know how computers can solve these problems (at least in principle)
  - In many cases, computers can solve these problems better than humans

Computational Problems

- **Difficult** computational problems:
  - Cognition as inductive inference
- Drawing conclusions that are not fully justified by the available data: detective work

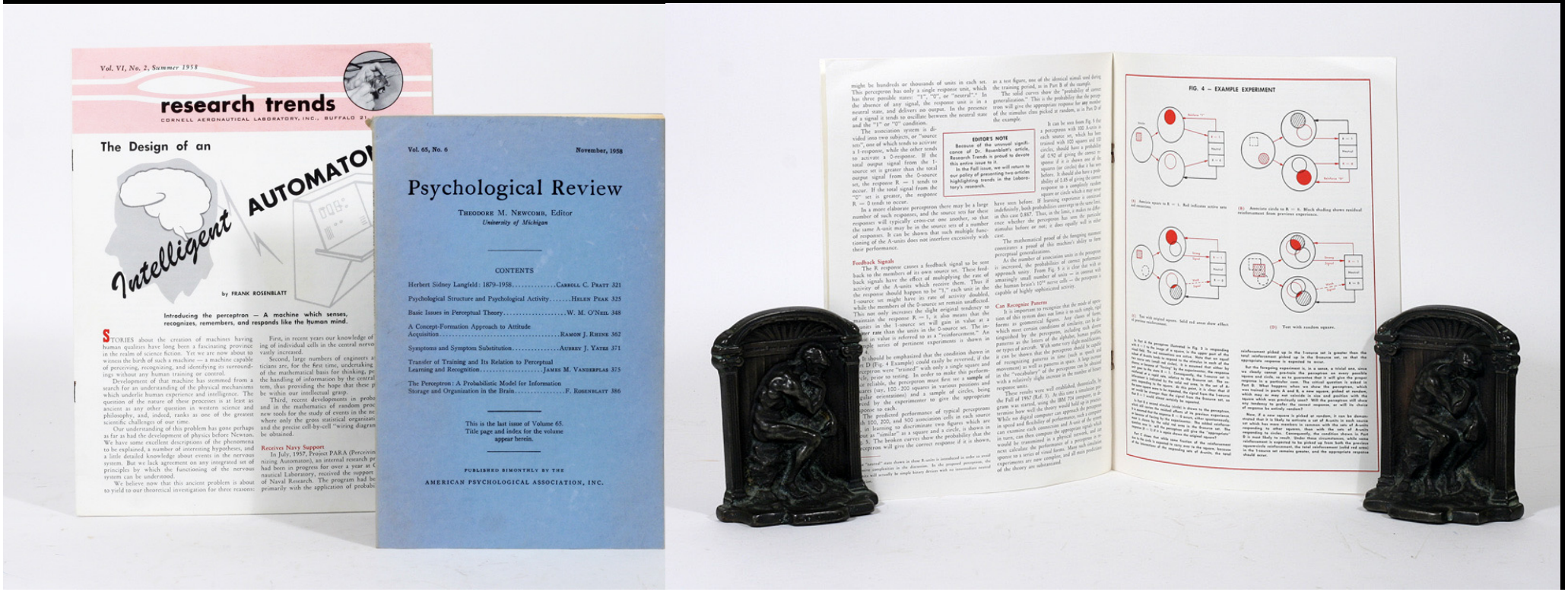
Sir Arthur Conan Doyle: A Study in Scarlet
PART II. The Country of the Saints.
7. CHAPTER VII. THE CONCLUSION.

"In solving a problem of this sort, the grand thing is to be able to reason backwards.

That is a very useful accomplishment, and a very easy one, but people do not practise it much."

Chapter 5
A (missing) brief history and key ingredients of AI

The Perceptron: A Probabilistic Model for Information Storage and Organization in the Brain



Psychological Review
Vol. 65, No. 6, 1958

THE PERCEPTRON: A PROBABILISTIC MODEL FOR INFORMATION STORAGE AND ORGANIZATION IN THE BRAIN[1]

F. ROSENBLATT

Cornell Aeronautical Laboratory

If we are eventually to understand the capability of higher organisms for perceptual recognition, generalization, recall, and thinking, we must first have answers to three fundamental questions:

1. How is information about the physical world sensed, or detected, by the biological system?
2. In what form is information stored, or remembered?
3. How does information contained in storage, or in memory, influence recognition and behavior?

and the stored pattern. According to this hypothesis, if one understood the code or "wiring diagram" of the nervous system, one should, in principle, be able to discover exactly what an organism remembers by reconstructing the original sensory patterns from the "memory traces" which they have left, much as we might develop a photographic negative, or translate the pattern of electrical charges in the "memory" of a digital computer. This hypothesis is appealing in its simplicity and ready intelligibility, and a large family of theoretical brain

Learning representations by back-propagating errors

David E. Rumelhart*, Geoffrey E. Hinton† & Ronald J. Williams*

* Institute for Cognitive Science, C-015, University of California, San Diego, La Jolla, California 92093, USA
† Department of Computer Science, Carnegie-Mellon University, Pittsburgh, Philadelphia 15213, USA

We describe a new learning procedure, back-propagation, for networks of neurone-like units. The procedure repeatedly adjusts

COGNITIVE SCIENCE 9, 147–169 (1985)

A Learning Algorithm for Boltzmann Machines*

DAVID H. ACKLEY
GEOFFREY E. HINTON
Computer Science Department
Carnegie-Mellon University
TERRENCE J. SEJNOWSKI
Biophysics Department
The Johns Hopkins University

Psychological Review
1981, Vol. 88, No. 2, 135–170

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0033-295X/81/8802-0135$00.75

Toward a Modern Theory of Adaptive Networks: Expectation and Prediction

Richard S. Sutton and Andrew G. Barto
Computer and Information Science Department
University of Massachusetts—Amherst

COGNITIVE SCIENCE, **14**, 179-211 (1990).

Finding Structure in Time

JEFFREY L. ELMAN
University of California, San Diego

Two approaches to intelligence

Pattern recognition

- Discriminative
- Comprehension

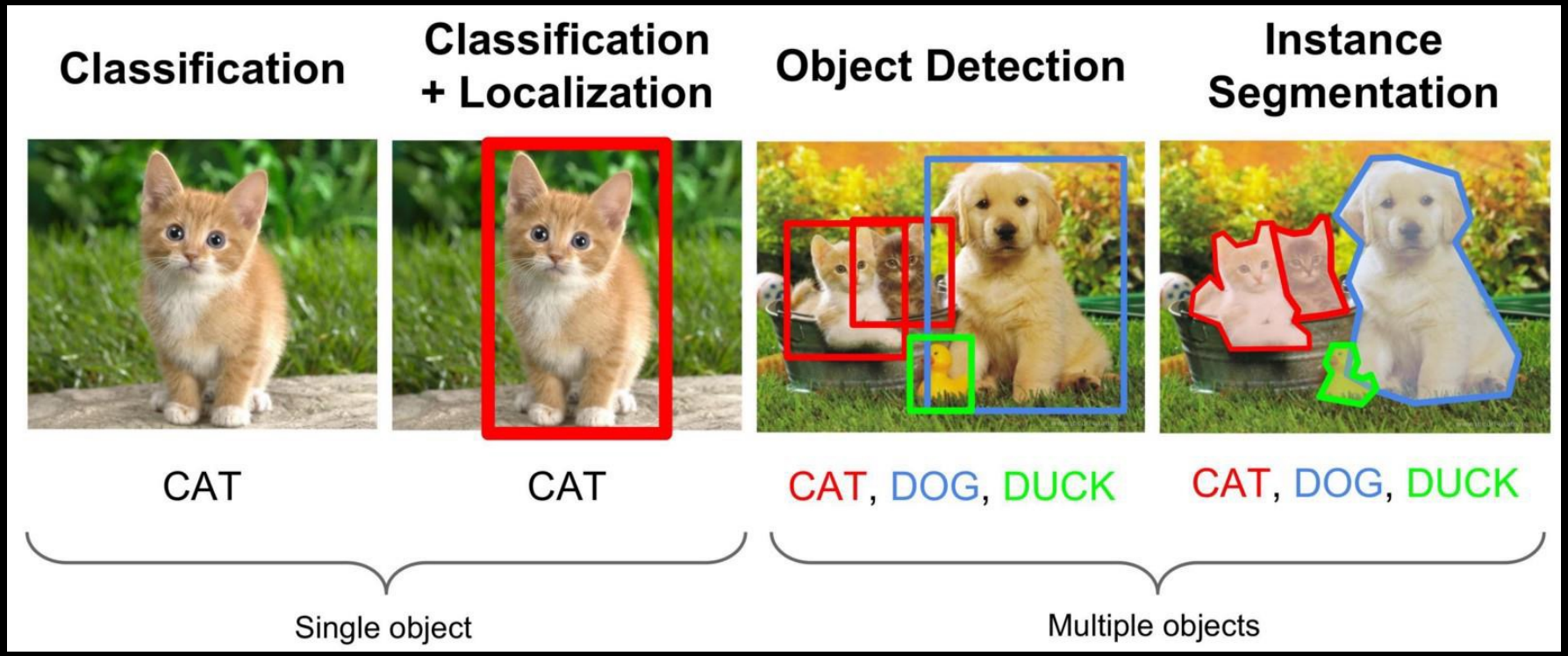
Model building

- Generative
- Production

Origin

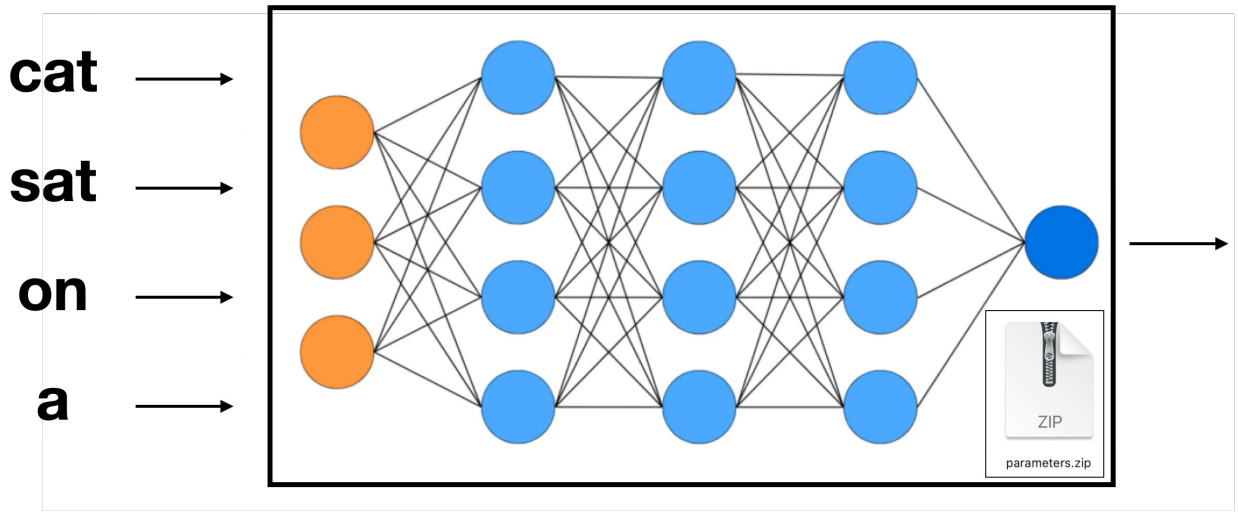
- Build “child-machine” (Turing, 1950)
- “General Problem Solver” (Newell and Simon, 1961)
- “I draw no boundary between a theory of human thinking and a scheme for making an intelligent machine” (Minsky, 1974)

Pattern Recognition



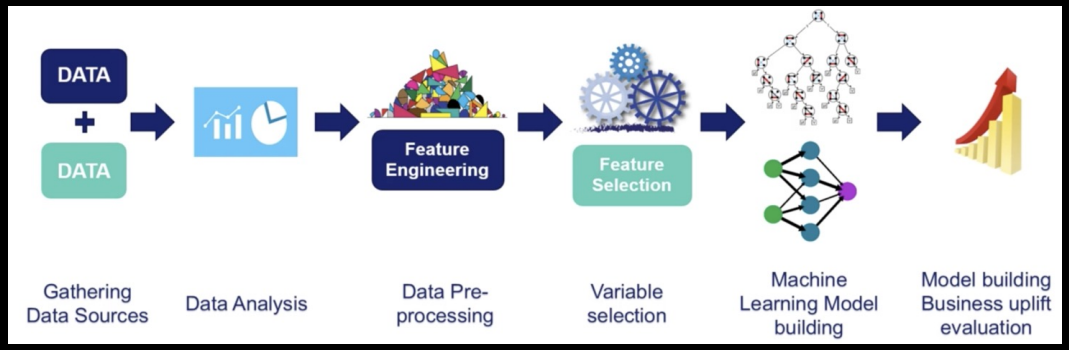
Neural Network

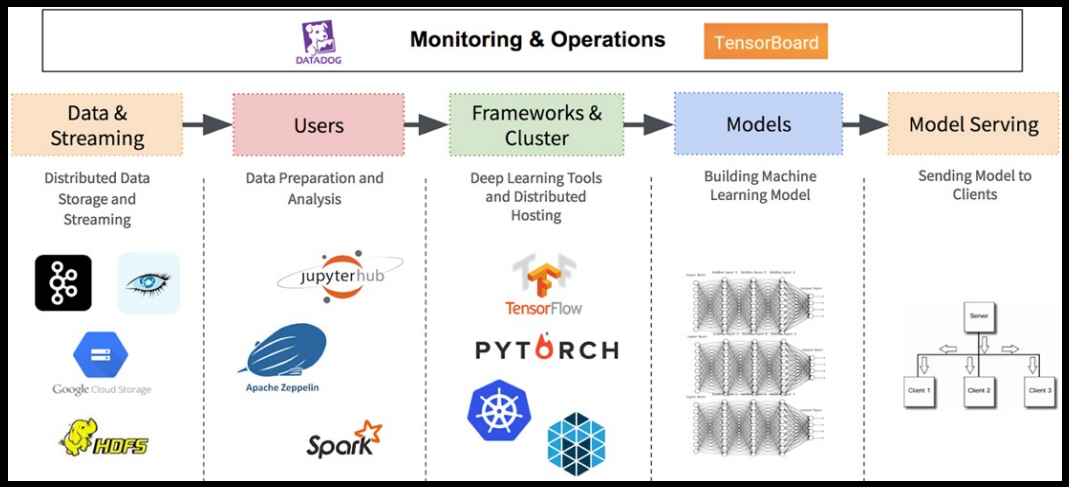
Predicts the next word in the sequence.



e.g. context of 4 words

predict next word





General Pipeline

Model Building

- Understanding and explaining what we see

Obama surreptitiously tips the scales as White House trip director Marvin Nicholson weighs himself on a tour of a locker room at the University of Texas: a president with a genuine sense of humor

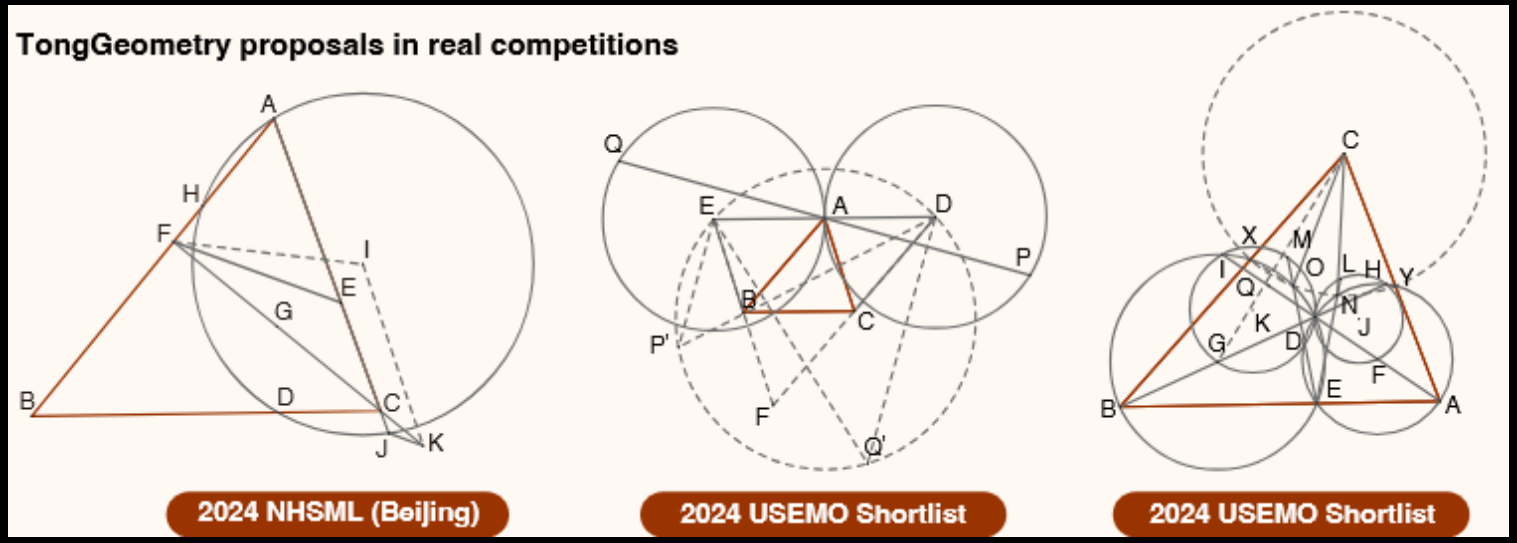
Photograph: Pete Souza/The White House

Model Building

- Understanding and explaining what we see
- Imaging things we could see but haven't yet

Model Building

- Understanding and explaining what we see
- Imaging things we could see but haven't yet
- Problem solving and planning to make these things real



Model Building

- Understanding and explaining what we see
- Imaging things we could see but haven't yet
- Problem solving and planning to make these things real
- Building more models as we learn more about the world

Model Building

- Understanding and explaining what we see
- Imaging things we could see but haven't yet
- Problem solving and planning to make these things real
- Building more models as we learn more about the world
- Sharing our models, communicating to others, understanding their models, and learning from them and with them, and about them.

I think it's a person sitting in a living room.

Key Ingredients

- Build causal models of the world that support explanation and understanding, rather than merely solving pattern recognition problems

Berkeley Blicket Experiment

Key Ingredients

- Build causal models of the world that support explanation and understanding, rather than merely solving pattern recognition problems
- Harness compositionality and learning to learn to rapidly acquire and generalize knowledge to new tasks and situations

Human-level concept learning through probabilistic program induction

Brenden M. Lake

Ruslan Salakhutdinov

Joshua B. Tenenbaum

Key Ingredients

- Build causal models of the world that support explanation and understanding, rather than merely solving pattern recognition problems
- Harness compositionality and learning to learn to rapidly acquire and generalize knowledge to new tasks and situations
- Ground learning in intuitive theories of physics and psychology to support and enrich the knowledge that is learned

100x
MICKEY MOUSE

Key Ingredients

- Build causal models of the world that support explanation and understanding, rather than merely solving pattern recognition problems
- Harness compositionality and learning to learn to rapidly acquire and generalize knowledge to new tasks and situations
- Ground learning in intuitive theories of physics and psychology to support and enrich the knowledge that is learned
- Build cognitive infrastructure, motivation and skills for collaboration, communication and pro social norms

Summary

1. Intelligence from scaling up vs growing Up
2. Why building a human-like AI is hard and the big questions
3. Let the nature speak
4. Different computing problems
5. A (missing) brief history and key ingredients of AI

Goal: Born like a baby, and learn like a child

Thanks for your listening and sleeping